KEYENCE

N49GB

Chapter 1 Configuration and Specifications

Chapter 2 Wiring of EtherCAT Communication

Chapter 3 Setting of EtherCAT Communication

Chapter 4 Communication Function of EtherCAT

• Error list
• System variable list

Programmable Controller
KV-X500/X300 Series

EtherCAT Communication Function User's Manual

Read this manual before using the product.
Keep this manual in a safe place for later reference.

Frequently Asked Questions

FAQ

www.keyence.com/controlfaq

Supported CPU Units
- KV-X550
- KV-X530
- KV-X520
- KV-X310

Preface

This manual describes the overview, features, and how to use the EtherCAT™ communication function built into the “KV-X500/X300 series”.
Be sure to thoroughly read and fully understand this manual before installing. In addition, store this manual in a safe place so that you can retrieve it whenever necessary.

Related Manuals

When using the EtherCAT communication function with the “KV-X500/X300 series”, read also the following manuals.
All the following PDF manuals can be found and opened in the help file of “KV STUDIO”. The latest version of PDF manuals can be downloaded from the Keyence web site.

Name	Description
KV-X500/X300 Series EtherCAT Communication Function User's Manual	This manual. Explains connections/specifications, and methods for creating programs for the EtherCAT communication function built into the CPU unit.
KV-X500 Series CPU Unit User’s Manual”	Explains the system configuration, specifications, and methods for creating programs for the "KV-X500 series”.
KV-X300 Series CPU Unit User’s Manual”	Explains the system configuration, specifications, and methods for creating programs for the "KV-X300 series”.
KV-X500/X300 Series Motion Control Function User’s Manual	Explains specifications and methods for creating programs for the motion control function built into the CPU unit.
KV-X500/X300 Series Command Reference Manual	Explains each instruction that can be used in programming. This manual does not describe the operation of each instruction. For the description of the detailed operation of each instruction, see “KV-X500/X300 Series Command Reference Manual”.
KV STUDIO User’s Manual	Explains the operation methods for the programming software "KV STUDIO".

Safety Information

The following describes notes and other information for users to know to correctly use the EtherCAT communication function.
Familiarize yourself with the following information before using the function.

Symbols

This document contains notices which you should observe to ensure your own personal safety, as well as to protect the product and connected equipment.

DANGER	**Indicates a hazardous situation which, if not avoided, will result in death or serious injury.**
WARNING	**Indicates a hazardous situation which, if not avoided, could result in death or serious injury.**
CAUTION	**Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.**
NOTICE	**Indicates a situation which, if not avoided, could result in product damage as well as property damage.**

Important	**Indicates cautions and limitations that must be followed during operation.**
Point	**Indicates additional information on proper operation.**
Reference	Indicates tips for better understanding or useful information.

Indicates a page to be referred to in this manual or other manuals.

General Precautions

DANGER	• **Do not use the device with the purpose of protecting human beings.** • **This device is not intended for use as an explosion-proof product. Do not use this device in a hazardous location or in a location that has a potentially explosive atmosphere.**
WARNING	• **Provide a safety circuit that bypasses the PLC to enable failsafe operation of the entire system in the event that the PLC fails.** • **Output circuit and internal circuit malfunctions sometimes prevent control from being performed normally. Be sure to provide a safety circuit in control systems where circuit malfunctions may lead to fire or other serious accidents.**
CAUTION	• **Before you use this device, verify its functionality and performance at startup and during operation.** • **If the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.**
NOTICE	• **Proceed with care when modifying the device, or when using it in a manner that falls outside of the ranges indicated in its specifications, since KEYENCE is unable to guarantee device functionality or performance in such situations.** • **Use this device in combination with other devices only after careful consideration, since the device may fail to satisfy its functionality and performance capabilities as a result of factors such as its usage conditions and the environment in which it is used.**

CE and UKCA Marking/UL Standards

For restrictions related to CE and UKCA Marking, and restrictions to comply with UL Standards, see the user's manual of the CPU unit in use.

How This Manual Is Organized

Chapter 1	Configuration and Specifications	This chapter describes the mechanism, names of the parts, and communication specifications of the EtherCAT communication function built into the CPU of “KV-X500/X300 series”.
Chapter 2	Wiring of EtherCAT Communication	This chapter describes the overview of supported network topologies, wiring pattern, and situations where errors occur.
Chapter 3	Setting of EtherCAT Communication	This chapter describes node information and a procedure for setting network configuration to establish EtherCAT communication.
Chapter 4	Communication Function of EtherCAT	This chapter describes the status transition of the EtherCAT communication function, refresh, PDO communication, SDO communication, and how to check the communication status.
—	Appendix	The appendix lists the errors of the EtherCAT communication function and system variables of the “KV-X500/X300 series”.

Terminology

The following terminologies are used for descriptions except for some contents in this manual.

Terminology	Description
Programmable logic controller	An electronic device that gives users complete control over equipment by changing programs. Also referred to as PLC (Programmable Logic Controller).
KV-X500 series	This refers to KEYENCE KV-X550/X530/X520 programmable controllers.
KV-X300 series	This refers to KEYENCE KV-X310 programmable controllers.
CPU unit	This refers to KEYENCE KV-X550/X530/X520/X310 programmable controllers.
KV STUDIO	This refers to the software that supports program creation for the KEYENCE programmable controllers.
Program	The program made with “KV STUDIO”. The “KV STUDIO” enables users to create programs using the ladder language and ST language.
EtherCAT setting	This is the setting tool “EtherCAT setting” for communicating with the EtherCAT that comes with the “KV STUDIO”.
Master	A master of EtherCAT communication.
Node	A device that is connected to the master in EtherCAT communication. This term is used as the same meaning as slave. The KV-X500/X300 series is an EtherCAT master. It cannot be used as an EtherCAT node (slave).

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Revision History

Chapter 1 Configuration and Specifications

This chapter describes the overview, features, names of the parts and functions, and specifications of the EtherCAT communication function of the “KV-X500/X300 series”.

1-1 Overview

The following describes the overview of the EtherCAT communication function.

What is EtherCAT?

The EtherCAT (Ethernet for Control Automation Technology) is one of the industrial open field network standards for real-time communication compatible with Ethernet. This standard provides high-speed and highly efficient communication using effectively the communication band.
The EtherCAT, a patented technology licensed by Beckhoff Automation GmbH, is a registered trademark of this company.
The standards of EtherCAT is managed by ETG (EtherCAT Technology Group).

Reference
- Beckhoff Automation GmbH: An FA manufacturer based in Germany
- ETG (EtherCAT Technology Group): An organization that stipulates the function requirements and other items of the EtherCAT.

Features of EtherCAT

EtherCAT has the following features.

Ultra high-speed communication at 100 Mbps

EtherCAT can achieve fast and highly efficient transmission of a variety of data by fully using the bandwidth of Ethernet data frames. The EtherCAT uses one data frame to communicates with all devices. Of all data within the data frame, each device only processes data allocated to own device (On-the-fly method). This method, compared with other methods that transmit data frames individually to each device, causes less communication delay due to data reading and writing, significantly increase I/O response at each device, which in turn leads to a shorter communication cycle.

High versatility based on Ethernet

While using own communication protocol, the EtherCAT uses the standard Ethernet technology on the physical layer, which delivers such an excellent versatility that Ethernet cables can be used. This versatility makes the function work fully effectively not only for large-scale control systems that requires high processing speed or system integrity but for small- and mid-scale control systems as well.

Real-time communication by highly accurate time synchronization

DC (distributed clock) is used for highly accurate time synchronization between devices. This achieves advanced real-time communication, enabling synchronization control required for motion control and robot control.

Support for flexible network topologies

EtherCAT supports a variety of network topologies, including the line type (daisy-chain type), star type (branched type), and ring type. Also, it enables duplication of lines for taking measures against disconnection and detection for disconnection.

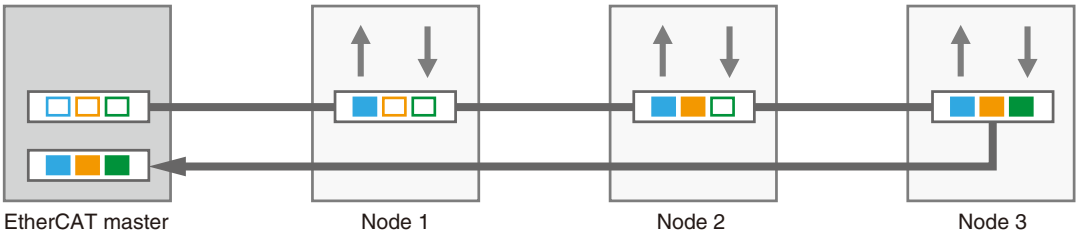
How EtherCAT works

The network of the EtherCAT communication consists of one master and multiple nodes. A data frame sent from the master passes through all nodes and returns to the master using the nature of the full-duplex communication of Ethernet. Data are read and written between the master and nodes by the on-the-fly method, which focuses on real-time property.

Data reading and writing by the on-the-fly method

A specific area is allocated in a data frame for each node. In each node, when a data frame is passing, data are read and written in the area allocated to the node.
Since data are exchanged between one data frame and all nodes, no data frame congestion occurs. Also, at each node, only the allocated area is processed, reducing processing delay within the node. This mechanism ensures high-speed and real-time data transmission.

The following illustrates an example of the EtherCAT communication in the case of a line type (daisy-chain type) topology.



PDO communication and SDO communication

EtherCAT uses two communication methods to exchange data between the master and nodes.

PDO communication

This communication uses process data objects (PDO).

Data are exchanged cyclically in a set communication cycle.

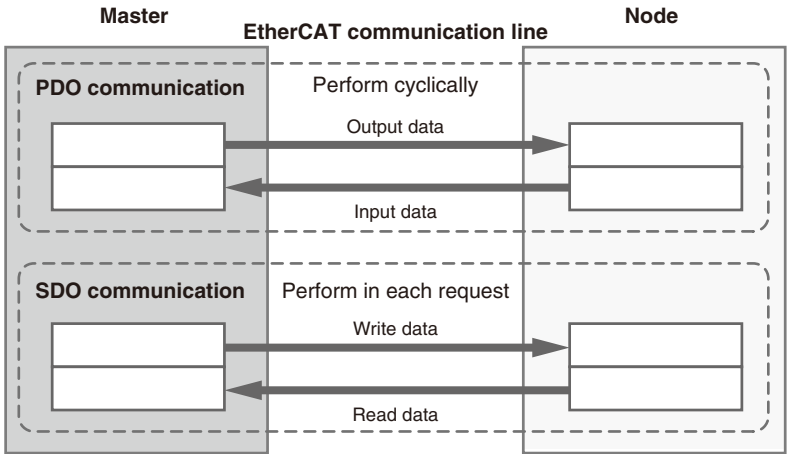
"PDO communication function"

SDO communication

This communication uses service data objects (SDO).

Data are exchanged at any time.

"SDO communication function"

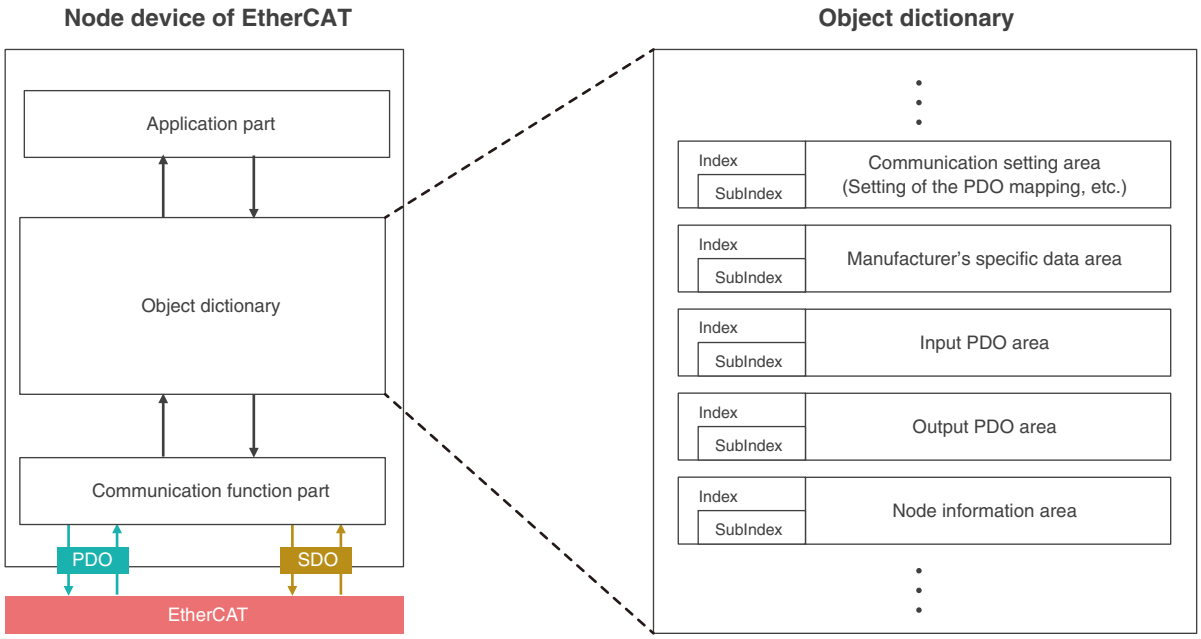


The EtherCAT communication function in the "KV-X500/X300 series" uses the PDO communication for instructions for updating I/O data in a certain cyclic period, such as I/O in the nodes and position control of the servo motor. Unlike this, the SDO communication is used for instructions for reading/writing data at the specified timing, such as parameter transfer.

The PDO communication constantly repeats in each communication cycle, during which the SDO communication is processed as instructed.

Object dictionary

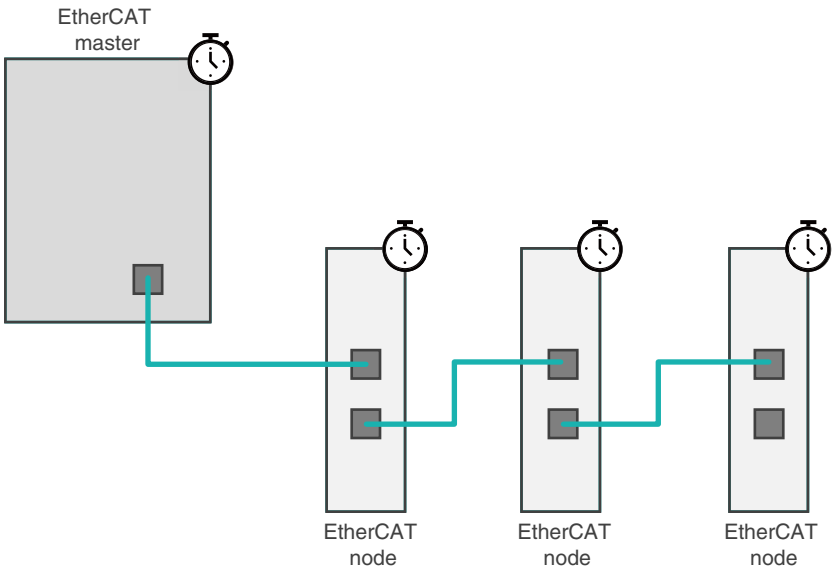
The object dictionary is the information table that mediates between the application part and the communication part in an EtherCAT node. The object dictionary includes information on the node, information on communication settings, such as PDO mapping, and information on data used in the PDO communication and SDO communication. Each of these pieces of information is called object. Each object has an index and sub-index, which are specified when object information is read or written. In the PDO communication, the function reads and writes data in the input PDO area and the output PDO area according to the information in the communication setting area. In the SDO communication, the function reads and writes data in the manufacturer's specific data area.



Highly accurate time synchronization using DC (distributed clock)

The DC (distributed clock) is an EtherCAT specific function to achieve highly accurate time synchronization between nodes within the network.

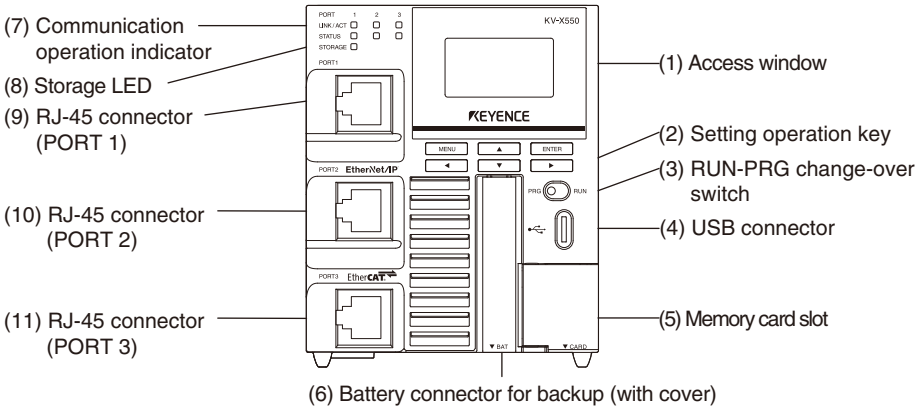
This function corrects time and synchronize the operation timing by obtaining the reference clock, which is used as the reference time within the network, from the master or a node having the DC function and supplying the obtained reference clock to each node.



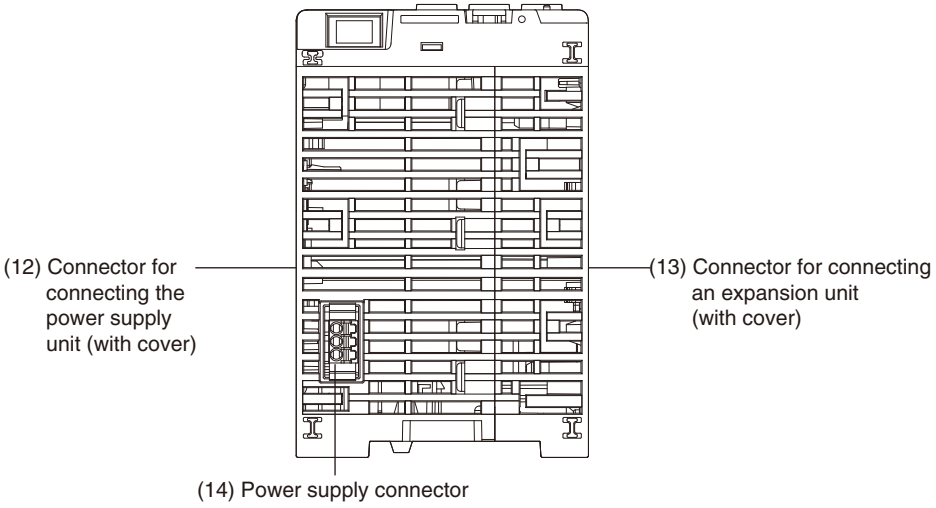
1-2 Names and functions of parts

KV-X500 series

Front side of the CPU unit



Underside of the CPU unit



No.	Name	Function
(1)	Access window	This window allows you to monitor the unit and make settings. The back light indicates the operation status. RUN: White PRG: Red
(2)	Setting operation key	Use this key for operating the access window.
(3)	RUN-PRG change-over switch	Change the mode between RUN and PROGRAM.
(4)	USB connector	This is the connector (USB Type-C) for USB connection
(5)	Memory card slot	This is the slot for inserting a memory card (microSD).
(6)	Battery connector for backup (with cover)	Use this when installing KV-B1.
(7)	Communication operation indicator	LINK/ACT (green): Indicates the communication status with the other device. Not lit: The port is set to “Not use”, or no link is established. Lit: Link status, STATUS (green/red): Indicates the communication operation status*2.
(8)	Storage LED	Indicates the access status to the SD card/CPU memory. Lit in red: Accessing*1 Not lit: No access

No.	Name	Function	
(9)	RJ-45 connector (PORT 1)	This is the slot for connecting the connector (RJ-45 modular jack) of a network cable.	General-purpose Ethernet communication can be used.
(10)	RJ-45 connector (PORT 2)		The EtherNet/IP™ communication can be used in addition to general-purpose Ethernet communication.
(11)	RJ-45 connector (PORT 3)		The EtherCAT communication can be used.
(12)	Connector for connecting the power supply unit (with cover)	This is the connector for connecting the KV-PU1 (power supply unit)	
(13)	Connector for connecting an expansion unit (with cover)	This is the connector for connecting an expansion unit.	
(14)	Power supply connector	This is the connector for power supply.	

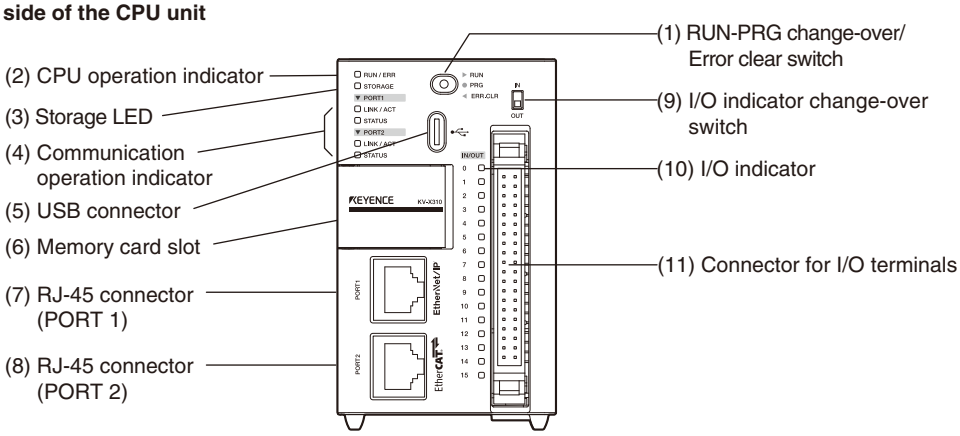
*1: For the CPU memory, the indicator is lit red only when the project data are being saved.

*2: The table below shows the communication operation statuses to be indicated.

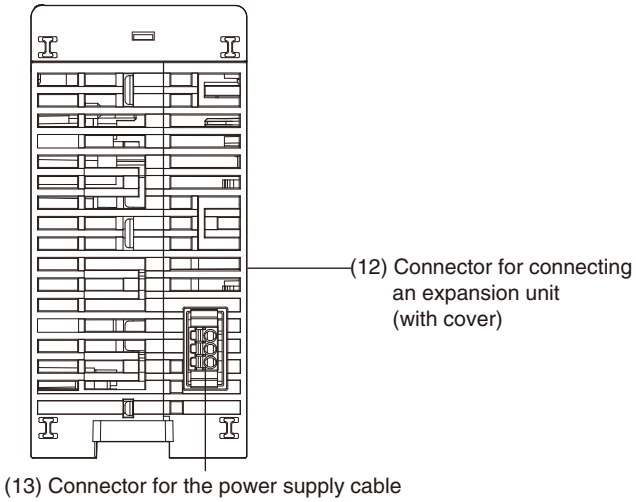
Status	PORT 1 (General-purpose Ethernet)	PORT 2 (General-purpose Ethernet / EtherNet/IP)	PORT 3 (EtherCAT)
Not lit	No IP address is set.	• When General-purpose Ethernet No IP address is set. • When Ethernet/IP No IP address is set.	No communication
■ Lit green	An IP address is set and is not duplicated.	• When General-purpose Ethernet An IP address is set and is not duplicated. • When Ethernet/IP One or more connections are established.	Communicating
□ Blinking green	–	• When General-purpose Ethernet – • When Ethernet/IP No connection is established.	Connecting, ring discon-nected
■ Lit red	The IP address is duplicated.	• When General-purpose Ethernet The IP address is duplicated. • When Ethernet/IP The IP address is duplicated. Remove the cause and restart the function to resolve the problem.	In error
□ Blinking red	–	• When General-purpose Ethernet – • When Ethernet/IP Time-out has occurred in one or more connections (except Class 3) to the CPU unit. Re-establish one or more connection or restart the function to resolve the problem.	–
■ Lit orange	–	–	Warning
□ Blinking orange	–	–	Restart of the power supply for the first time after all clear

KV-X300 series

Front side of the CPU unit



Underside of the CPU unit



No.	Name	Function	
(1)	RUN-PRG change-over/Error clear switch	Change the mode between RUN and PROGRAM, and clear an error.	
(2)	CPU operation indicator	Indicates the operation status*2 of the CPU unit.	
(3)	Storage LED	Indicates the access status to the SD card/CPU memory. Lit in red: Accessing*1 Not lit: No access	
(4)	Communication operation indicator	LINK/ACT (green): Indicates the communication status with the other device. Not lit: The port is set to “Not use”, or no link is established. Lit: Linked STATUS (green/red): Indicates the communication operation status*3.	
(5)	USB connector	This is the connector (USB Type-C) for USB connection	
(6)	Memory card slot	This is the slot for inserting a memory card (microSD).	
(7)	RJ-45 connector (PORT 1)	This is the slot for connecting the connector (RJ-45 modular jack) of a network cable.	The EtherNet/IP communication can be used in addition to general-purpose Ethernet communication.
(8)	RJ-45 connector (PORT 2)		The EtherCAT communication can be used.
(9)	I/O indicator change-over switch	Switch the indication of the I/O indicator LED between the status of input terminal and the status of output terminal. (Set to IN by factory default)	
(10)	I/O indicator	Indicates the conduction state of the I/O terminal. Lit red: Conduced Not lit: Not conducted	
(11)	Connector for I/O terminals	This is the connector used for I/O. Use this together with the MIL connector.	
(12)	Connector for connecting an expansion unit (with cover)	This is the connector for connecting an expansion unit.	
(13)	Connector for the power supply cable	This is the connector for connecting the power supply cable.	

*1: For the CPU memory, the indicator is lit red only when the project data are being saved.

*2: The table below shows the operation statuses to be indicated.

Status	Operation status
□ Blinking green	In the PRG mode, and no error
■ Lit red	In the PRG mode, and a severe error has occurred
□ Blinking orange	In the PRG mode and a minor error has occurred
■ Lit green	In the RUN mode, and no error
■/■ Lit green and orange alternately	In the RUN mode and a minor error has occurred
□ Blinking red	Initializing
□ Not lit	Power off

*3: The table below shows the communication operation statuses to be indicated.

Status	PORT 1 (General-purpose Ethernet / EtherNet/IP)	PORT 2 (EtherCAT)
Not lit	• When General-purpose Ethernet No IP address is set. • When Ethernet/IP No IP address is set.	No communication
■ Lit green	• When General-purpose Ethernet An IP address is set and is not duplicated. • When Ethernet/IP One or more connections are established.	Communicating
□ Blinking green	• When General-purpose Ethernet – • When Ethernet/IP No connection is established.	Connecting, ring discon-nected
■ Lit red	• When General-purpose Ethernet The IP address is duplicated. • When Ethernet/IP The IP address is duplicated. Remove the cause and restart the function to resolve the problem.	In error status
□ Blinking red	• When General-purpose Ethernet – • When Ethernet/IP Time-out has occurred in one or more connections (except Class 3) to the CPU unit. Re-establish one or more connection or restart the function to resolve the problem.	–
■ Lit orange	–	Warning
□ Blinking orange	–	Restart of the power supply for the first time after all clear

1-3 Specifications

EtherCAT communication specifications

Item	KV-X550	KV-X530	KV-X520	KV-X310
Support service	CoE (PDO communication*1, SDO communication)*6			
Synchronous communication	DC (Distributed Clock) *2			
Physical layer	100BASE-TX			
Transmission speed	100Mbps			
Duplex mode	Full duplex*3			
Topology	Line (Daisy-chain), star, ring (Junction redundancy)			
Transmission medium	Category 5e compliant STP cable*4			
Maximum distance between nodes	100m*5			
Maximum number of nodes	512	256	192	72
Communication cycle	KV-X550/X530/X520: 62.5μs to 10ms (Incremented by 1 ms after 2 ms) KV-X310: 125μs to 10ms (Incremented by 1 ms after 2 ms)			
Maximum process data size	Input: 11,472 bytes Output: 11,472 bytes			Input: 5,736 byte Output: 5,736 bytes
Maximum process data size per node	Input: 1,434 bytes Output: 1,434 bytes			
Maximum message size	1018Byte			

*1: Not supported for the PDO communication that uses multiple FMMU in the same direction (Tx/Rx).
NU-EC1, a Keyence network communication unit, cannot be connected. Use NU-EC1A.

*2: The reference clock is supplied from “KV-X500/X300 series” to each node.

*3: Supported for MDI/MDI-X automatic switching function.

*4: STP = shielded twisted pair cable It is recommended to use twisted pair cable with double shield of aluminium tape and braid shield.

*5: Set the distance between nodes according to a device to be connected.

*6: VoE, FoE, EoE, FSoE, AoE, and SoE are not supported.

Point **The KV-X500/X300 series is an EtherCAT master. It cannot be used as an EtherCAT node.**

Chapter 2 Wiring of EtherCAT Communication

2-1 Wiring of EtherCAT

The following describes how to wire the “KV-X500/X300 series” and EtherCAT nodes.

1 Turn off the PLC and all EtherCAT node devices.

2 Connect the modular jack on one side of the STP cable to the connector of the “KV-X500/X300 series”.

Insert it until a click sound is heard. The modular jack and the connector are locked.

3 Connect the modular jack on the other side of the STP cable to the connector of a node.

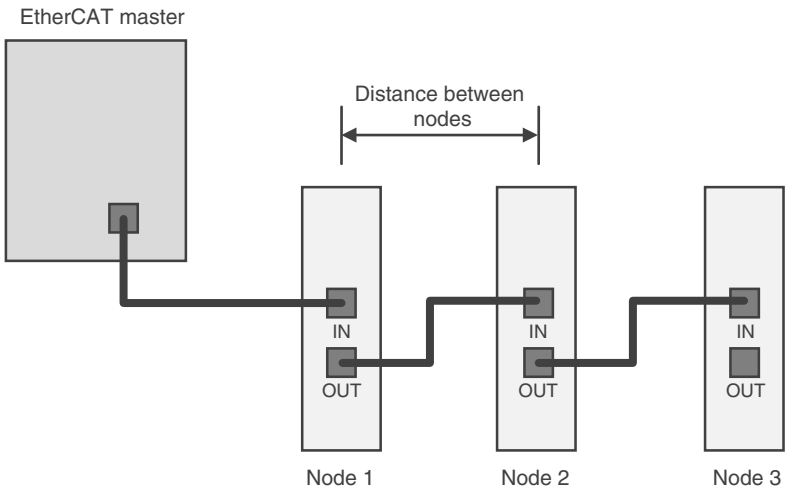
Insert it until a click sound is heard. The modular jack and the connector are locked.

4 Turn on the PLC and each node device, and check if the LINK/ACT lights or blinks green.

It takes about three seconds until the indicator lights.

Wiring

Connect the STP cable to respective EtherCAT ports. Connect the OUT port of each node to the IN port of the next node.
*STP = shielded twisted pair cable It is recommended to use twisted pair cable with double shield of aluminium tape and braid shield.



Item	Description
Maximum distance between nodes	100m*1

*1: The maximum distance between nodes may be limited by the specifications of the node. Check also the specifications of the node.

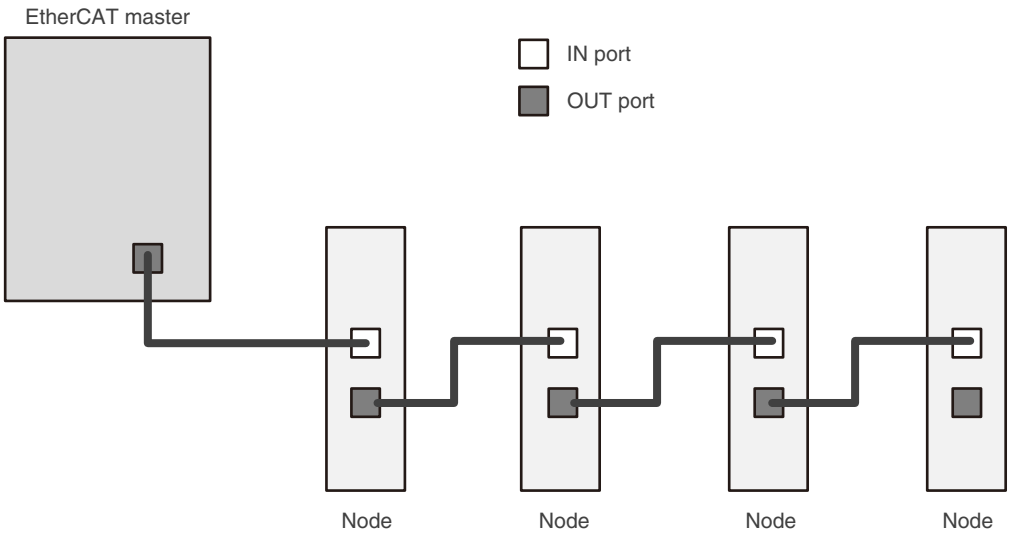
2-2 Supported network topologies

The EtherCAT communication function of the “KV-X500/X300 series” supports the following network topologies: line (daisy-chain), star (branched), and ring.

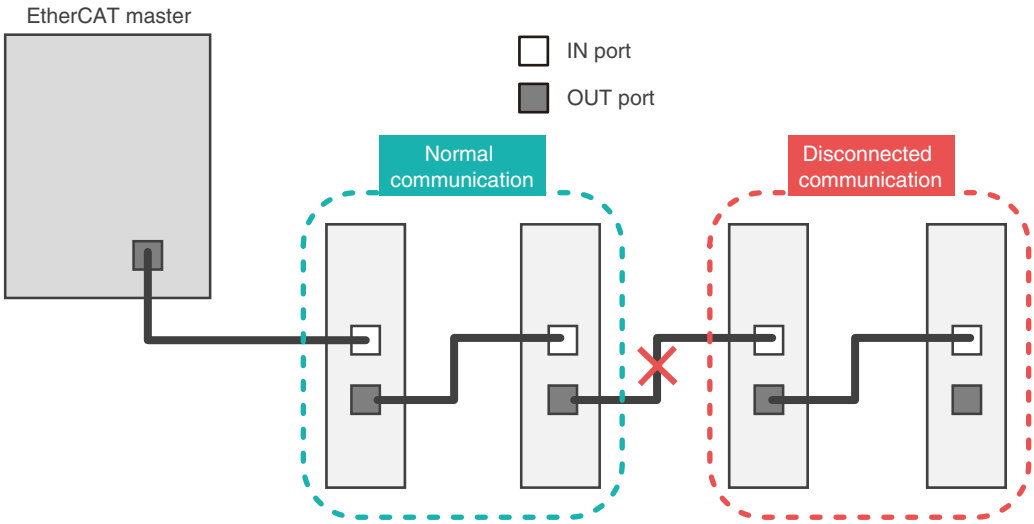
Line

Overview

In a line topology, the OUT port of a node is connected to the IN port of the next node, and like this, all ports are strung together in one line. This type of topology is also called daisy chain. This topology makes it easy to wire.



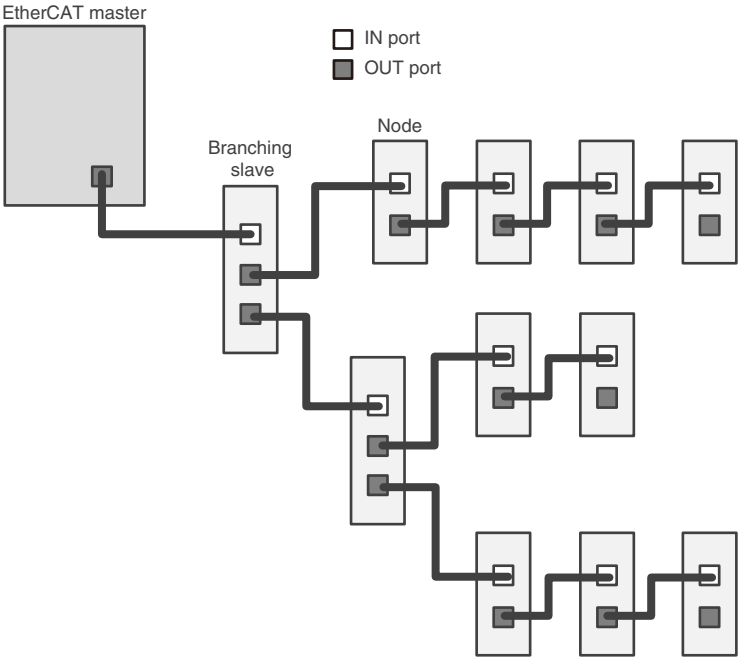
If a node is isolated because of cable disconnection or other reasons, the disconnected node and all the downstream nodes are disconnected from the network.



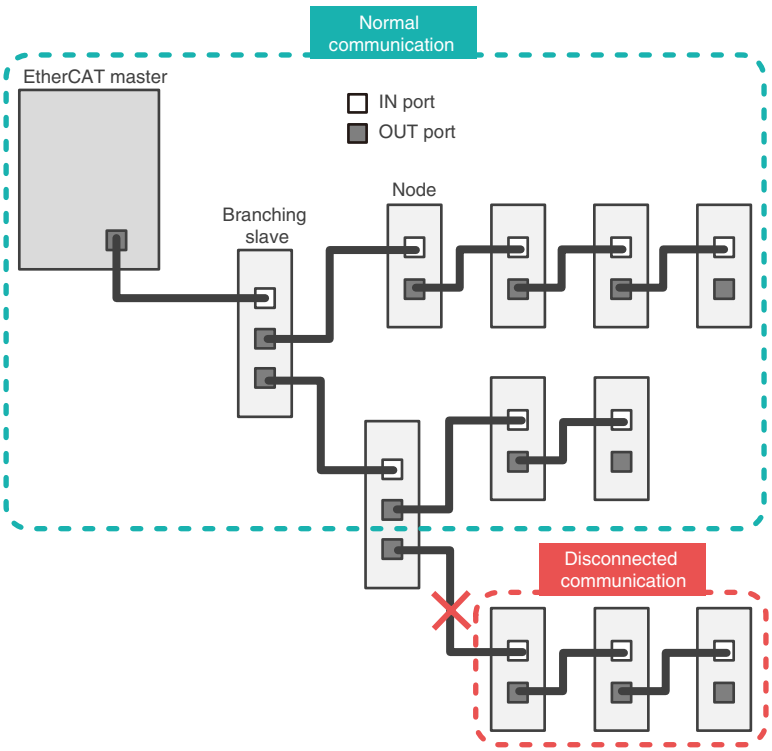
Star

Overview

The star topology uses a branching slave to configure a topology consisting of multiple lines. This type of topology is also called branched topology.

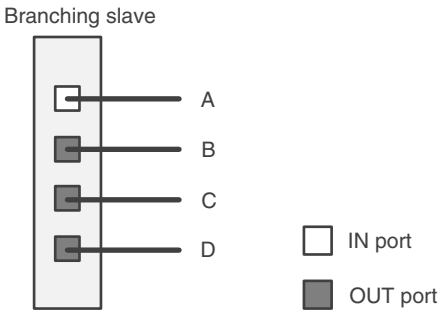


In a star topology, if a cable is disconnected, only the disconnected branched nodes are isolated from the network.



Notes and restrictions on the star topology

The EtherCAT communication function checks if the following configurations match: the configuration of node connected to each OUT port of the branching slave and the node configuration set in “EtherCAT setting”. Make the configuration consistent between the settings and those of the actual machine.



●Error example

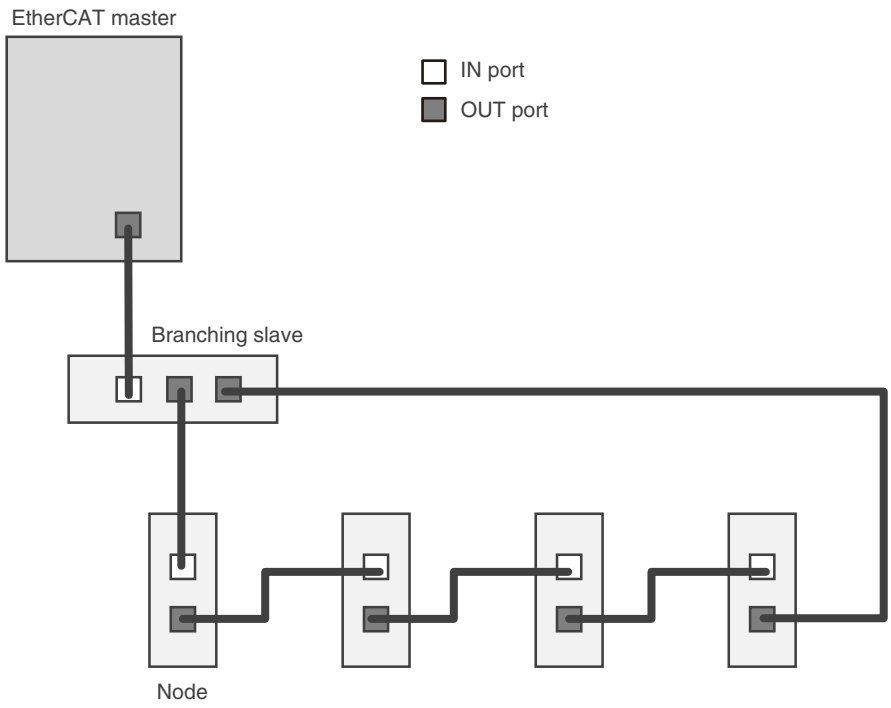
Setting: device connected to B = node 1, device connected to C = node 2

Actual machine: device connected to B = node 2, device connected to C = node 1

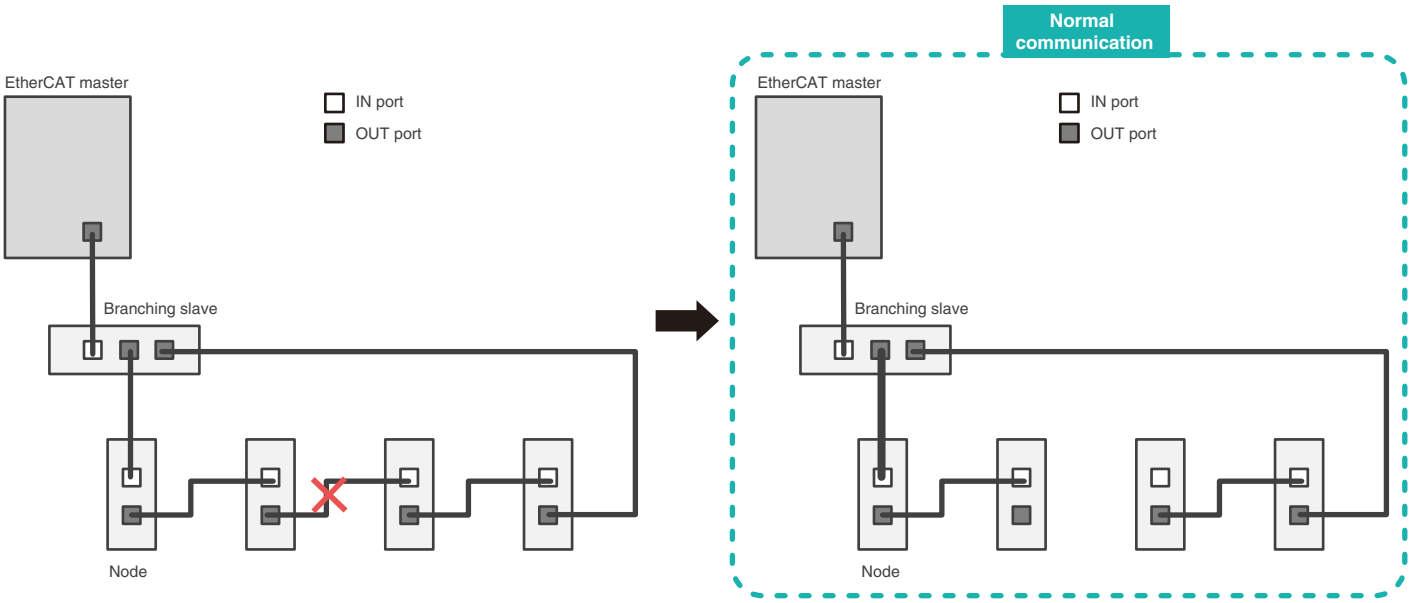
Ring

Overview

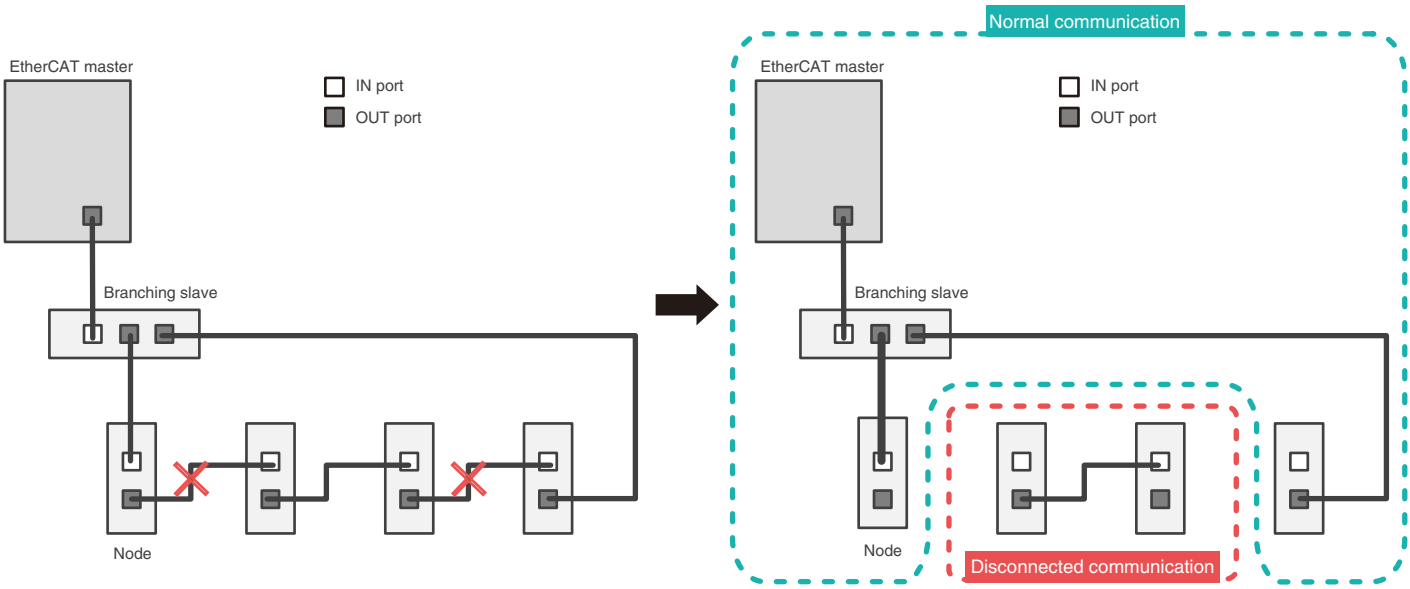
The ring topology uses a branching slave to connect nodes into a ring.



With disconnection of only one cable within the ring, communication between all nodes remains normal.



If a break occurs in two or more places, communication will be cut off only between the nodes that are sandwiched between the breaks, and communication will continue normally with the other nodes.



Reference

- If disconnection occurs within the ring, the system variable "_EC_MASTER.RingError" becomes TRUE.
- To find the disconnected cable, use the system variable "_EC_MASTER.RingErrorTbl". The status becomes TRUE for the "_EC_MASTER.RingErrorTbl" allocated to the downstream nodes from the disconnected node.
- If two or more nodes are disconnected within the ring, the communication stops between the branching slave and the disconnected node. The communication between the other nodes continues.

Ring configuration

The “KV-X500/X300 series” supports only the method (junction redundancy) that forms a ring from a branching slave. The following shows the components of the ring.

Components	Description
Branching slave	A branching slave used for configuring the ring.
Start port	The OUT port where the ring starts.
End port	The OUT port where the ring ends.

Reference

- The ports to be allocated to the start and end ports in the OUT port of the branching slave are fixed (the start port = D, end port = B). Incorrect wiring disables communication. For details on allocation of the start port and end port, see the manual of the branching slave.
- Some branching slave devices have multiple start points and end points. Incorrect wiring pairing disables communication. Refer to the manual of branching slave about port allocation to wire correctly.

For details about ring setting, see the following:

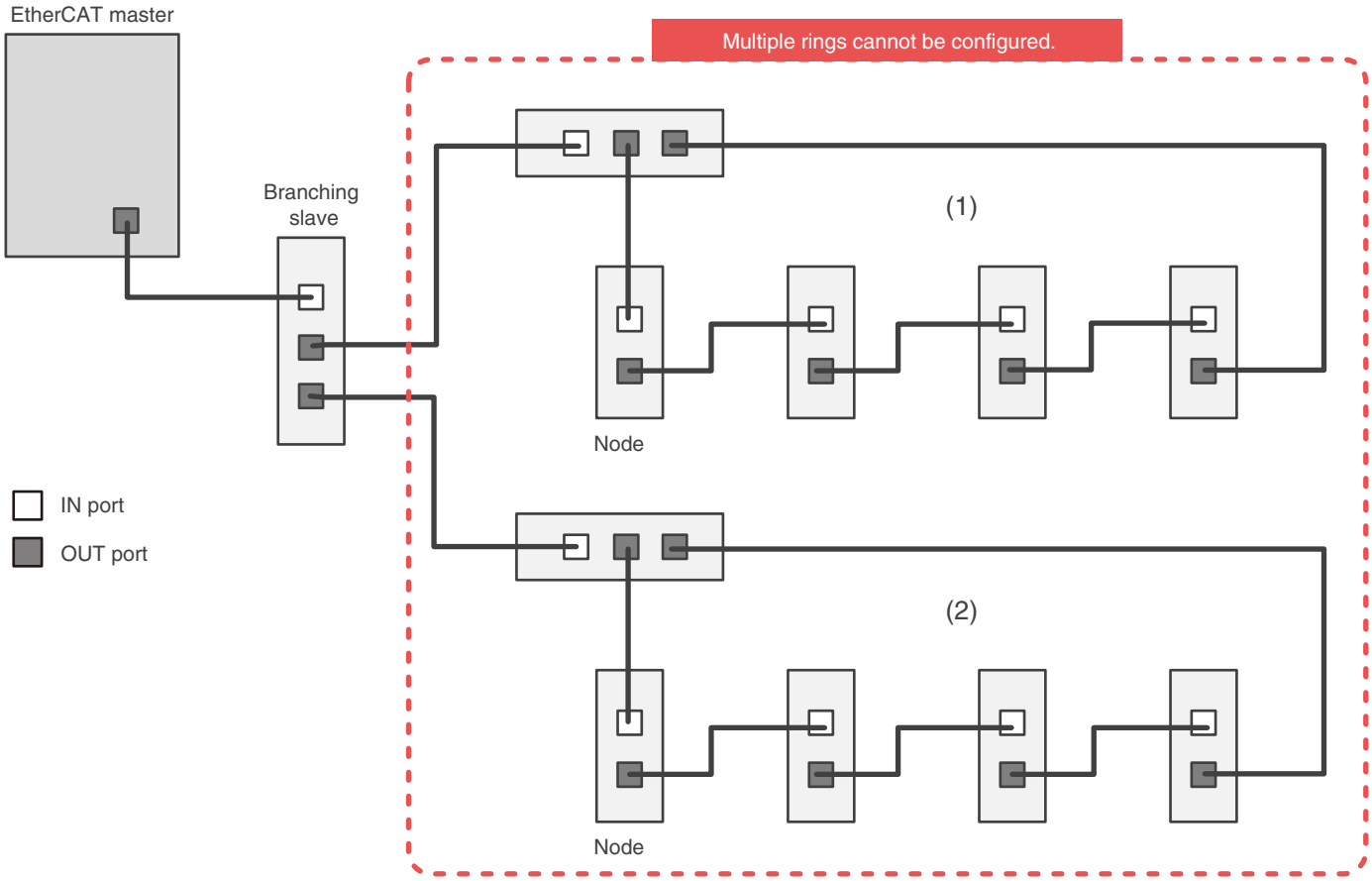
“Configuration setting of node”

Notes and restrictions on the ring topology

- When using the ring topology, set to “Used” for the Check product code of all ring components (the branching slave from where the ring starts and nodes making up the ring). If set to “Not used”, the ring cannot be used.

  “Check items when starting communication”
- When using a ring, set the Explicit Device ID check to "Yes" for all ring constituent devices (branch slaves that are the starting point of the ring, and nodes that make up the ring) and set Explicit Device IDs for the nodes.

  “Check items when starting communication”
- Incorrect wiring can cause communication across the entire network to stop. Check that the wiring is correct.
- Only one ring can be configured per network. Multiple rings cannot be configured within a network as shown below.



Reference You can also build a hybrid topology combining ring, line and star types.

Topology check

Topology check is performed each time an event occurs that may affect the topology, such as plugging or unplugging of a cable. A node checked as normal is considered valid and can participate in the PDO communication. If the checked node is abnormal, a node error occurs. If an unknown node is detected, a master error occurs.

Node configuration check

The function obtains information of actually connected nodes and checks if the information matches the node configuration set in “EtherCAT setting”. If an unmatched node is found, a master error “EtherCAT node device unmatched 2” occurs.

Unconnected node check

If a node to be checked is not connected, a mater error or node error occurs. Note that this does not apply to the following cases.

- Node that is requesting deactivation “Stop/resume”
- Node under reservation and has never participated in the PDO communication

Incorrect wiring check

The function checks for incorrect wiring, such as connection between OUT ports.
If incorrect wiring is found, the node error “EtherCAT port connected incorrectly” occurs.

Connected port check

The function checks if the following configurations match: the configuration of node connected to each OUT port of the branching slave and the node configuration set in “EtherCAT setting”. The function also checks if wiring is correctly implemented on the start port and end port.

“Notes and restrictions on the star topology”
“Ring configuration”

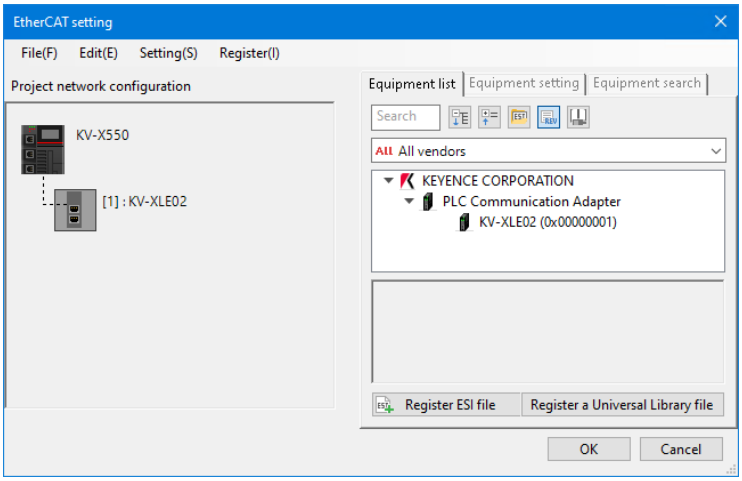
Chapter 3 Setting of EtherCAT Communication

3-1 EtherCAT setting

The following describes the setting of EtherCAT communication.

What is EtherCAT setting?

The “EtherCAT setting” is a setting tool for EtherCAT communication between the “KV-X500/X300 series” (EtherCAT master) and nodes. Start the “EtherCAT setting” from “KV STUDIO”, a ladder support software.

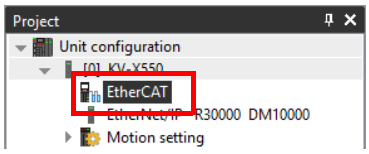


What you can do with "EtherCAT setting"

- Auto Configuration
- Equipment search
- Registration of an ESI file
- Configuration setting of nodes
- Master setting
- Node setting
- Advanced node setting (PDO mapping)

How to start EtherCAT setting

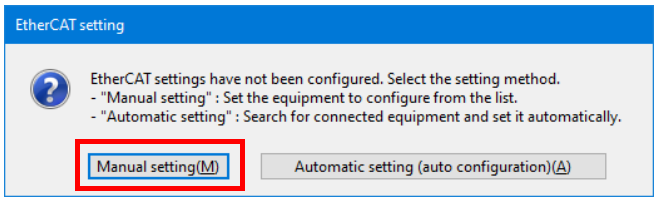
1 Double-click “EtherCAT” in the workspace.



If the EtherCAT has not yet set, the following dialog box appears.

If already set, no dialog box appears.

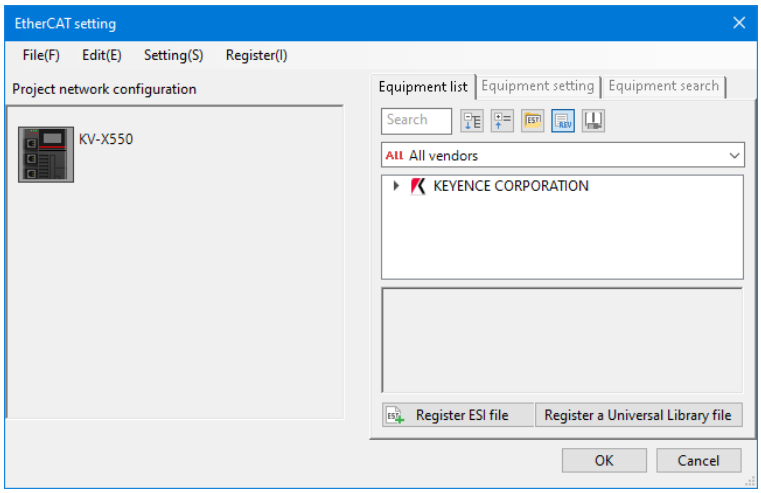
If appears, click [Manual setting].



Reference The dialog box allows you to select [Manual setting] or [Automatic setting (auto configuration)].

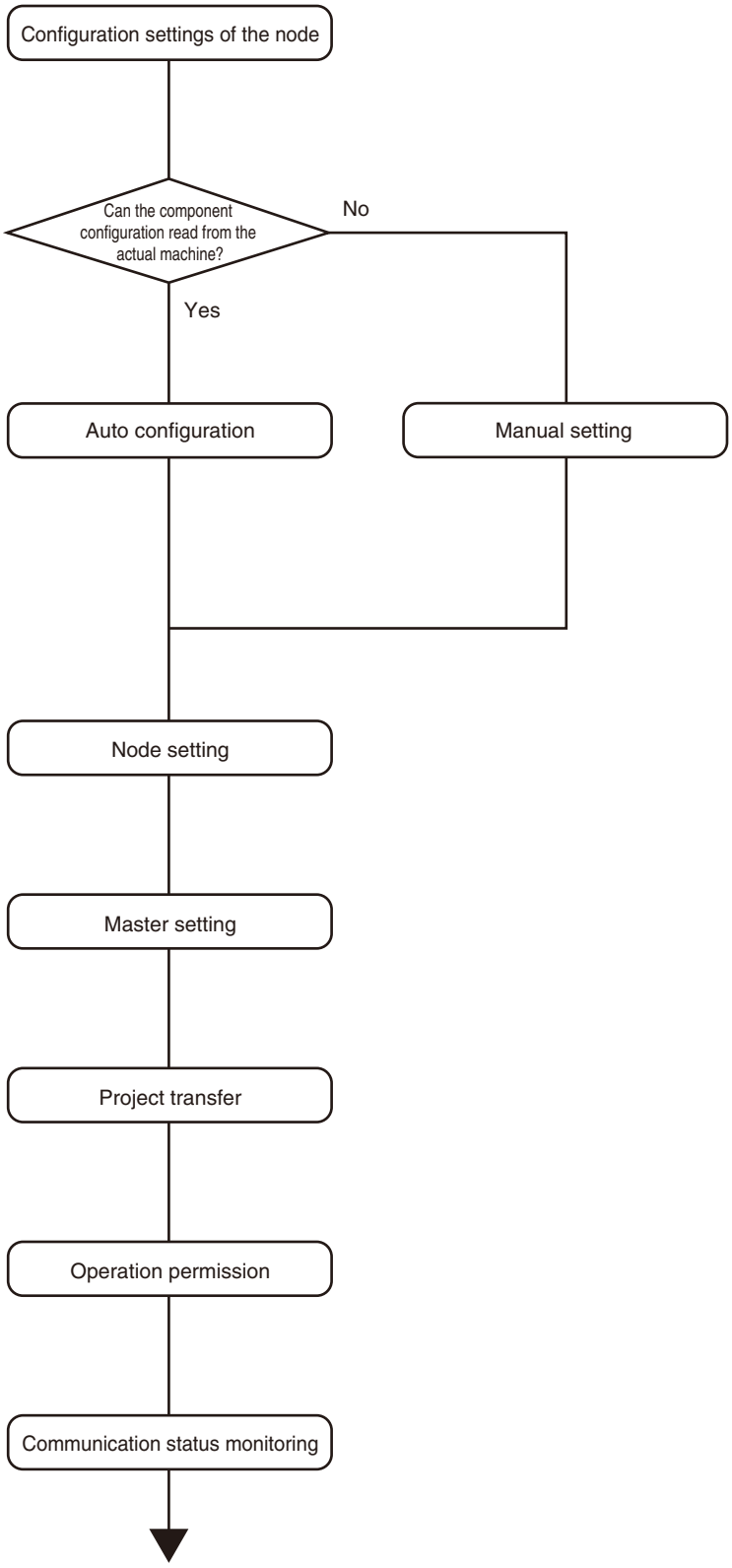
- Manual setting: Select and set a component from the components list (“Manual setting”).
- Auto configuration: The system searches for a connected component and automatically set it (“Auto configuration”).

2 The [EtherCAT setting] dialog box appears.



Setting flow of EtherCAT communication

The following shows the setting of EtherCAT communication.



Open the [EtherCAT setting] dialog box.

- "How to start EtherCAT setting"

Set the configuration of the node.

- "Auto configuration"
- "Manual setting"
  - "Registration of an ESI file"
  - "Configuration setting of node"

Set the parameters of the node.

- "Node setting"
- "Check items when starting communication"

Set the communication cycle and other items.

- "Master setting"

Transfer the "EtherCAT setting" and variable information.

Set the operation permission to on to reflect the settings.

- Start the EtherCAT communication.

Monitor the status of the EtherCAT communication.

- "How to check the communication status"

3-2 How to set EtherCAT setting

The following describes how to set the “EtherCAT setting”.

Auto configuration

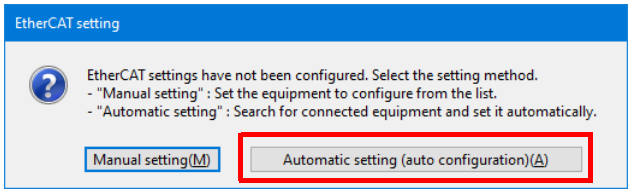
The Auto configuration is a function that automatically searches for a node connected to the “KV-X500/X300 series” and generates configuration settings of the node.

How to use

The Auto configuration can be used when the "KV STUDIO" is in the editor mode. The following two methods are available.

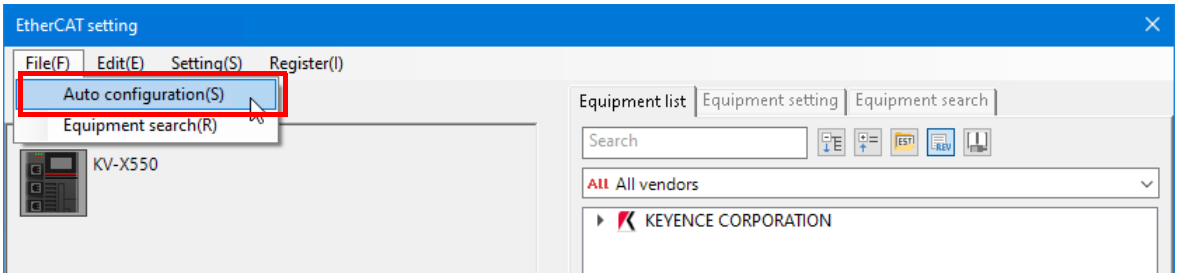
When the configuration of the node is not set

When the [EtherCAT setting] dialog box starts, the following dialog appears.
Click [Automatic setting (auto configuration)].



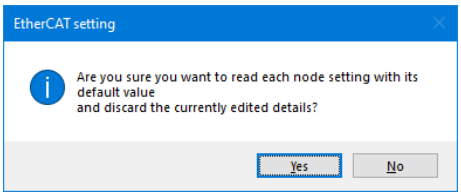
When executing from the [EtherCAT setting] dialog box

Select [File] > [Auto configuration] from the menu in the [EtherCAT setting] dialog box.



Reference

- If the ESI file of a connected node is not registered, the node is added as a general-purpose device to the configuration.
  In such a case, manually register the ESI file. “Registration of an ESI file”
  After registering the ESI file, right-click on the general-purpose device and select [Import ESI] to apply the ESI file and enable PDO mapping and other settings.
- When the node has already been configured, the following dialog appears.



- Even if the node configuration has already been set, you can perform a device search to select and add devices found in the search while retaining the existing settings. “Equipment search”

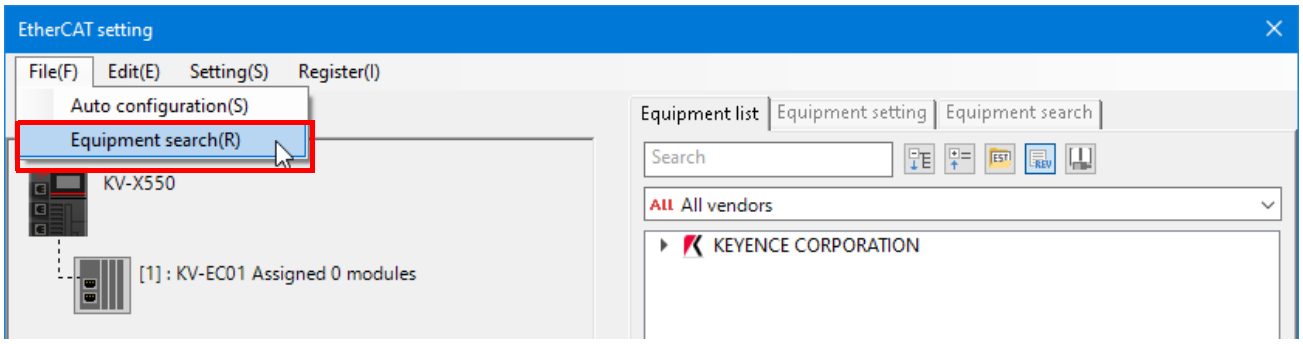
Equipment search

Equipment search is a function that searches for nodes connected to the “KV-X500/X300 series”. The detected nodes can be added to the EtherCAT network configuration. In Auto Configuration, the search results are directly applied to the network configuration, overwriting any existing settings. However, with Equipment search, only the selected nodes can be added, allowing new devices to be included without overwriting the existing network configuration.

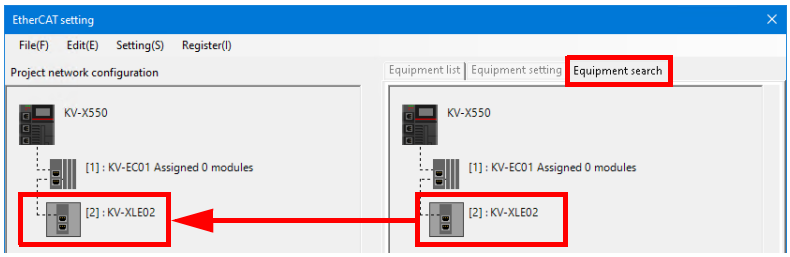
How to use

Equipment search can be used in “KV STUDIO” when in Editor Mode, Monitor Mode, or Online Edit Mode. However, nodes can only be added to the network configuration in Editor Mode.

In the [EtherCAT Settings] dialog, select [File] > [Equipment search] from the menu.

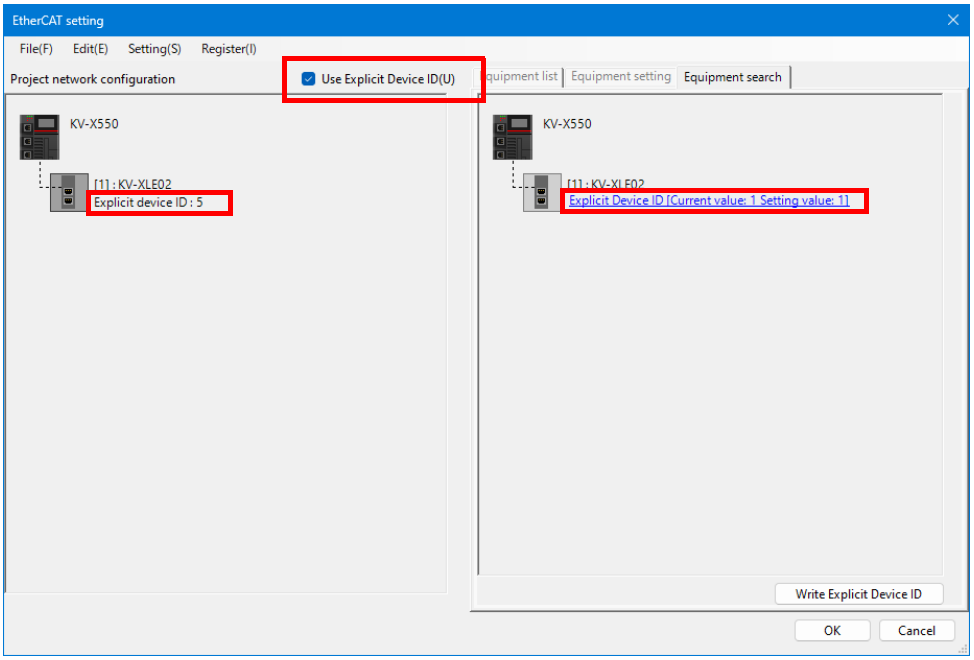


The search results will be displayed under the “Equipment search” tab. If a node matches the search, double - click it to add the node to the network configuration.

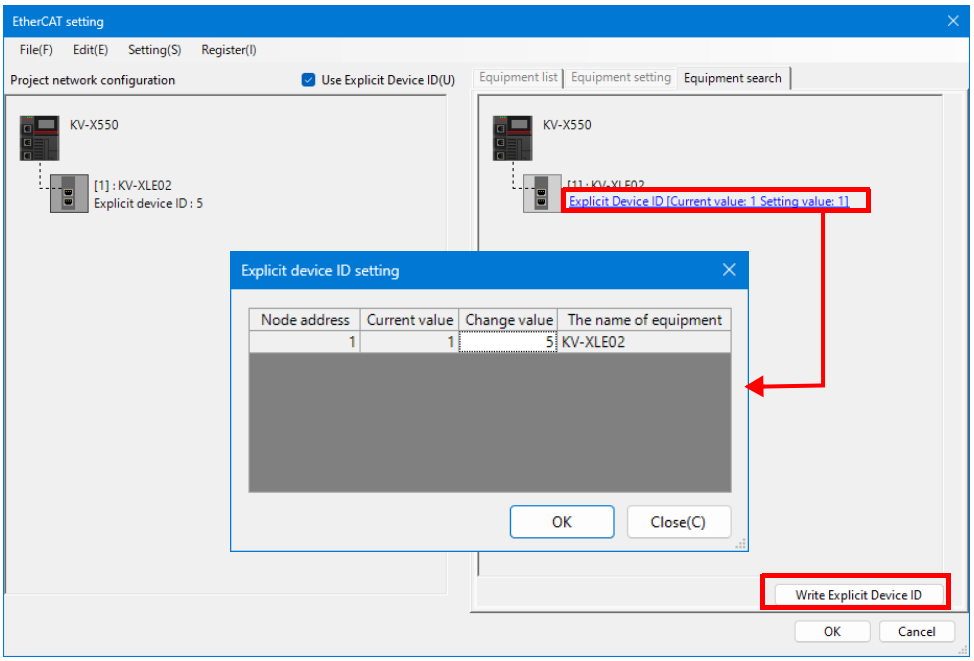


Checking and writing Explicit Device IDs

Searching for devices with “Use Explicit Device ID” enabled makes it possible to check the Explicit Device ID set on the project’s network and the Explicit Device ID actually set on the connected device. Checking both the Explicit Device ID set for the project and the Explicit Device ID actually set for the connected device makes it possible to identify EtherCAT devices added later.



Clicking the Explicit Device ID in the device search field will display the [Explicit device ID setting] dialog box. Enter the new Explicit Device ID and click “Write Explicit Device ID” to rewrite the Explicit Device ID of the connected device.



Point

- **When the rotary switch on the node device is set to a value other than 0**
  The value of register 0x0134 of the node device is read as the current value of the Explicit Device ID.
  In this situation, you cannot rewrite the Explicit Device ID value from “KV STUDIO” .
- **When the rotary switch on the node device is set to 0**
  The value of register 0x0012 of the node device is read as the current value of the Explicit Device ID.
  In this situation, you can rewrite the Explicit Device ID value from “KV STUDIO” .
  However, the node device must be turned off and on again for the Explicit Device ID to be applied.

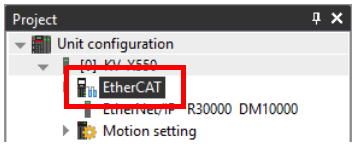
Manual setting

Registration of an ESI file

The ESI (EtherCAT Slave Information) file, which is offered by each EtherCAT node, contains communication settings. You can download an ESI file from, for instance, the web site of each manufacturer.
The following shows the procedure for registering an ESI file with the “KV STUDIO”.

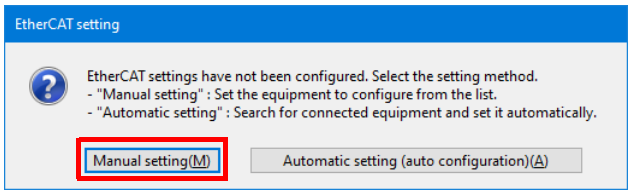
Important **An ESI file exists for each node model. To use multiple node models, register ESI files for all models to be used.**

1 Double-click “EtherCAT” in the workspace.



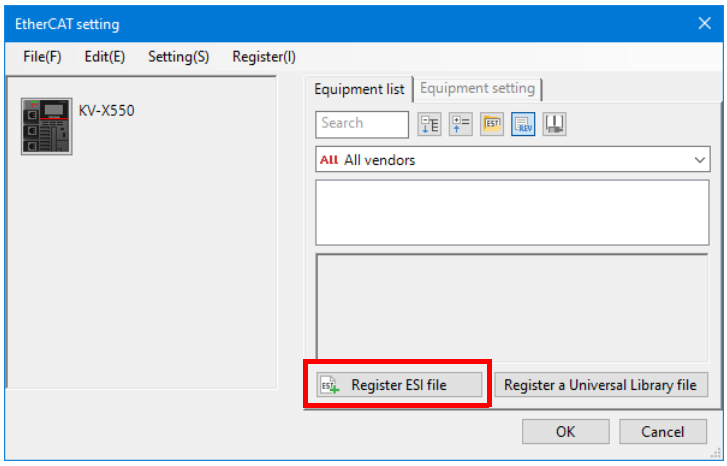
The [EtherCAT setting] dialog box appears.

2 Click [Manual setting] in the [EtherCAT setting] dialog box.



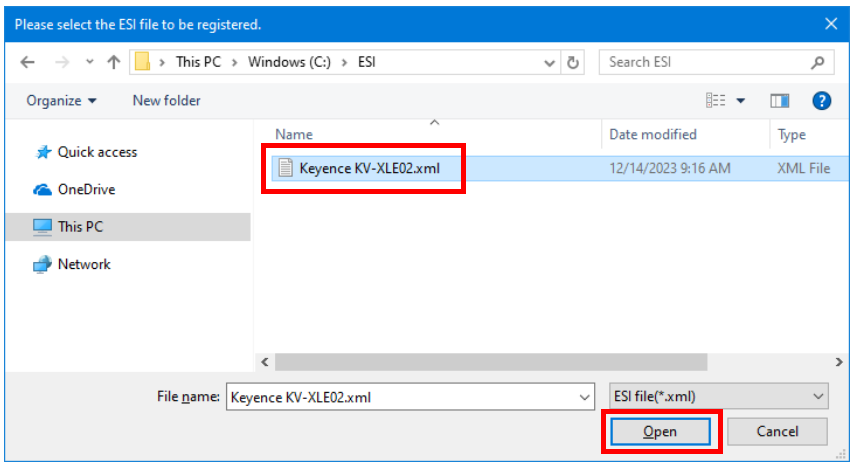
Reference When the node configuration is already set up, the [EtherCAT setting] dialog automatically opens.

3 Click [Register ESI file] in the [EtherCAT setting] dialog box.

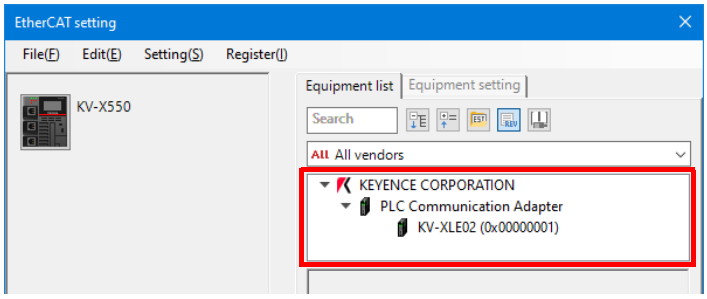


The dialog box “Please select the ESI file to be registered.” appears.

4 Select an ESI file for the node to register, and click [Open].



The selected ESI file is registered, and the node is displayed in the “Equipment list” in the [EtherCAT setting] dialog box.

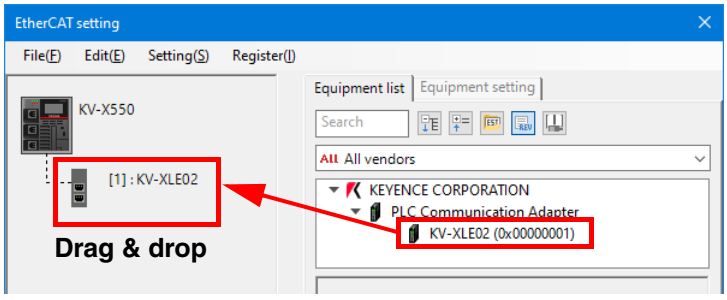


Configuration setting of node

Set the configuration of the node (setting of the topology). Select a node from the components with ESI files registered.

1 Select a node to add to the configuration from the “Equipment list” in the [EtherCAT setting] dialog box.

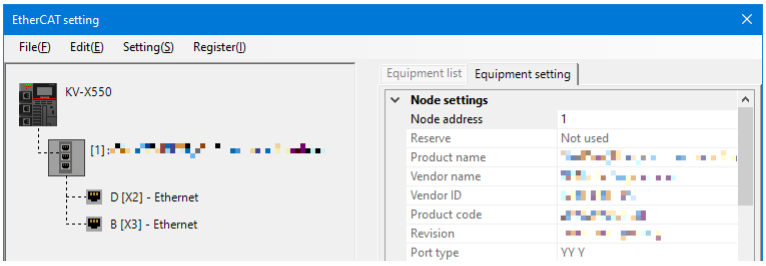
2 Drag and drop the selected node to the left area in the [EtherCAT setting] dialog box.



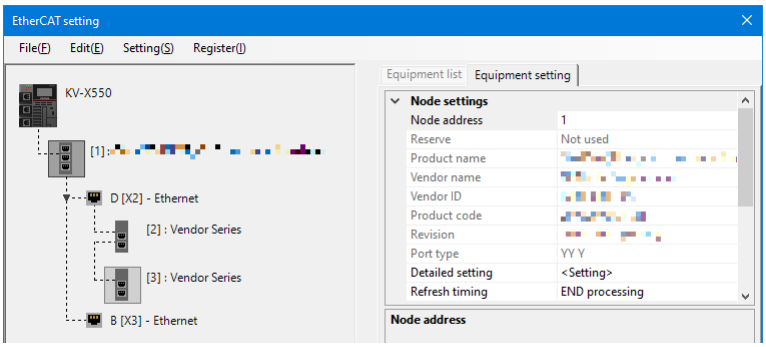
3 Repeat the same procedure to add all nodes.

Reference To configure the ring, make settings as follows.

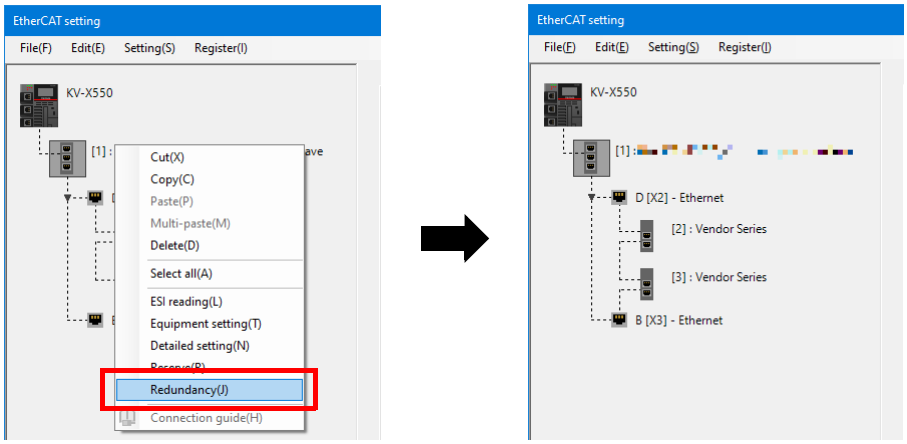
1. Add a branching slave for the ring.



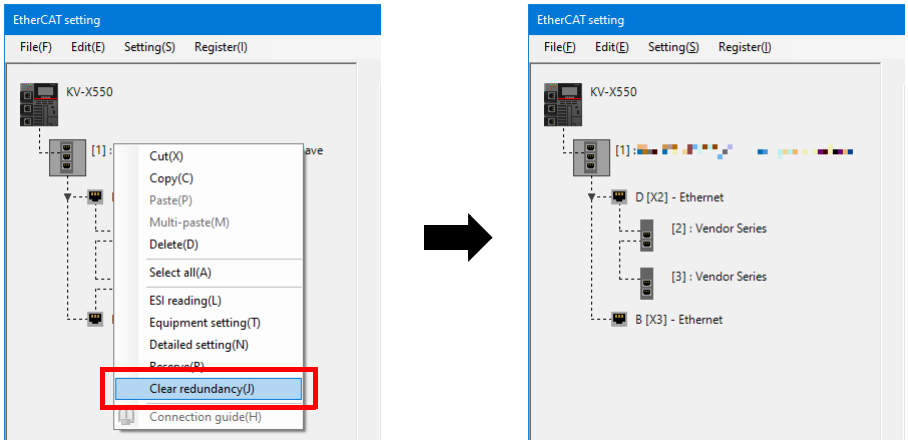
2. The [EtherCAT setting] dialog box shows “D” as the start port and “B” as the end port. Add a node that makes up the ring to a branch destination of the D port. “Ring configuration”



3. Select a branching slave and click [Redundancy] from the right-click menu.
   A ring is formed.



To release the ring, select a branching slave and click [Clear redundancy] from the right-click menu.



Node setting

Set basic parameters for a node.

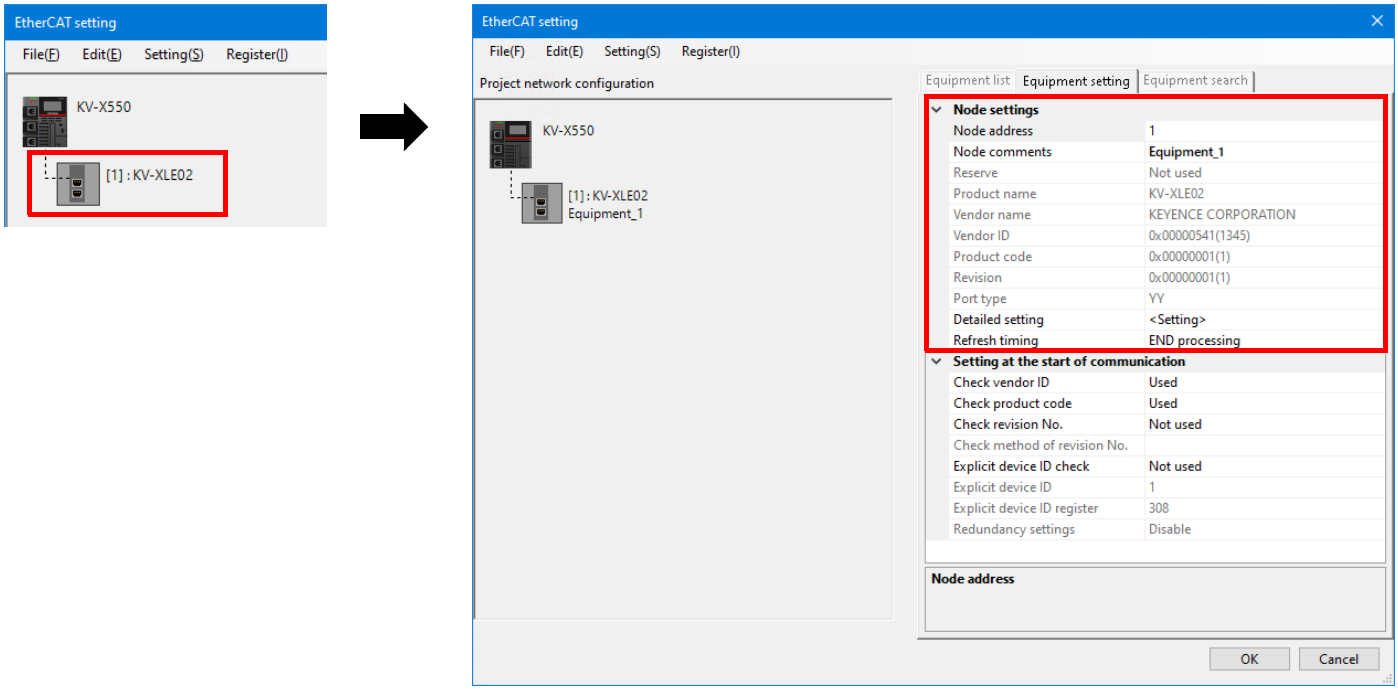
Setting items list

Item	Description
Node address	Set the node address*1.
Node comments	Set a comment to distinguish nodes on the EtherCAT setting screen (up to 64 characters, either full-width or half-width). The comment you set will be displayed in the EtherCAT setting dialog and in the workspace.
Reserve	Set this item when reserving the node. "Reservation of nodes"
Product name	Displays the product name of the node based on the information of the ESI file.
Vendor name	Displays the vendor name of the node based on the information of the ESI file.
Vendor ID	Displays the vendor ID of the node based on the information of the ESI file.
Product code	Displays the product code of the node based on the information of the ESI file.
Revision	Displays the revision of the node based on the information of the ESI file.
Port type	Displays the port type of the node based on the information of the ESI file.
Detailed setting	Make detailed settings for the node. "Detailed node setting"
Refresh timing	Set the refresh timing of the variable allocated to the sending and receiving buffer in the PDO mapping, by selecting from the "Batch (END processing)" and "Inter-unit synchronization". "PDO mapping"

*1: You cannot set a duplicated node address between different nodes.

How to set

Select a node in the [EtherCAT setting] dialog box, and set the items outlined in red in the "Equipment setting".



Check items when starting communication

Set items to check between the master and nodes when starting communication.
Use these items to check whether the node configuration set in the “KV STUDIO” is consistent with the configuration of the actual machine.

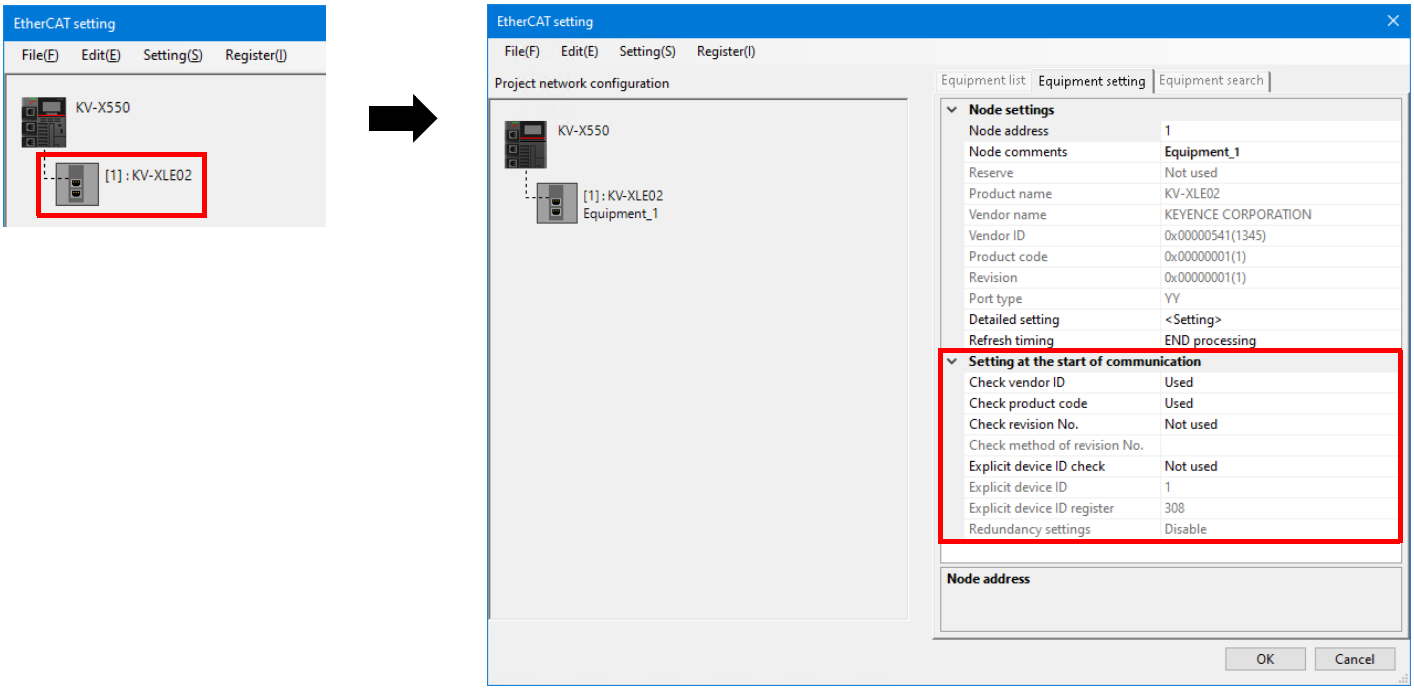
Setting items list

Item	Description
Check vendor ID	To check the vendor ID when staring communication, set this item to “Used”.
Check product code	To check the product code when staring communication, set this item to “Used”. When the ring is going to be configured, set the Check product code for all the ring component devices to “Used”.
Check revision No.	To check the revision No. when staring communication, set this item to “Used”.
Check method of revision No.	Set this item when the Check revision No. is set to “Used”.
Explicit device ID check	To check the Explicit device ID when staring communication, set this item to “Used”. Using the Explicit device ID check enables you to check multiple node devices individually even if they have the same vendor ID, product code, and revision No. To reserve a node, be sure to set the Explicit device ID check to “Used”. “Reservation of nodes” When configuring a ring, set the Explicit Device ID check to "Yes" for all devices in the ring. “Notes and restrictions on the ring topology”
Explicit device ID	When the Explicit device ID check is set to “Used”, specify the Explicit device ID set for the node. Check the value set here against the value of Explicit device ID set for the node device. For instructions on how to set the Explicit Device ID on the node device, refer to the node device manual. The settings can be set using the rotary switches on the main unit or by writing them using the node device's configuration software. Depending on the node device configuration, it may be possible to set the Explicit Device ID from “KV STUDIO” via the following procedure. “Checking and writing Explicit Device IDs” “Station alias settings”
Explicit device ID register	When the Explicit device ID check is set to “Used”, set this item. A value to set differs depending on the setting method of the Explicit device ID. For Configured Station Alias Set the Explicit device ID by writing the ROM of the node. You can use [Station alias settings] to set from the “KV STUDIO”. To configure the settings from “KV STUDIO”, use device search or station alias settings. “Checking and writing Explicit Device IDs” “Station alias settings” Set the Explicit device ID register to 18 (0x12). For ID-Selector Set the Explicit device ID with hardware, such as a rotary switch. This item cannot be changed by communication. Set the Explicit device ID register to 308 (0x134) or a desired value. *For details about the settings, see the manual of the node.
Redundancy settings	To form a ring topology, set this item to “Enable”. Redundancy can be set only when the components include a branching slave that supports the ring topology. “Ring”

Reference If the ESI file includes descriptions about setting items, initial values are set according to the information in the ESI file.

How to set

Select a node in the [EtherCAT setting] dialog box, and set the items outlined in red in the “Equipment setting”.

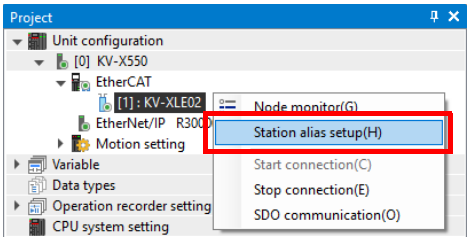


Reference

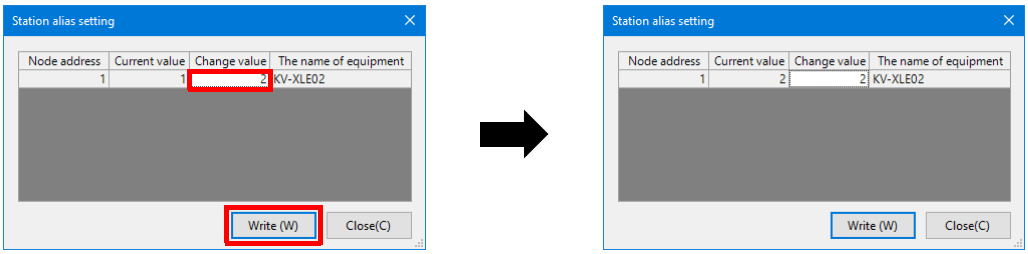
Station alias settings

The Explicit device ID needs to match the settings in both the master KV-X500/X300 series’ side and the node side. When matching the node side’s settings to the Explicit device ID which is set in the KV-X500/X300 series’ side, the node device’s Explicit device ID can be rewritten from “KV STUDIO” via the following the steps. To rewrite the Explicit Device ID of a node device from “KV STUDIO”, change to Monitor mode or Online Edit mode and perform the following steps.

Select a node in the workspace, and from the right-click menu, click [Station alias setup].



The [Station alias setting] dialog box appears. Set the Explicit device ID for the “Change value” and click [Write].



- When the rotary switch on the node device is set to a value other than 0

  The value of register 0x0134 of the node device is read as the current value of the Explicit Device ID.
  In this situation, you cannot rewrite the Explicit Device ID value from “KV STUDIO”.

- When the rotary switch on the node device is set to 0

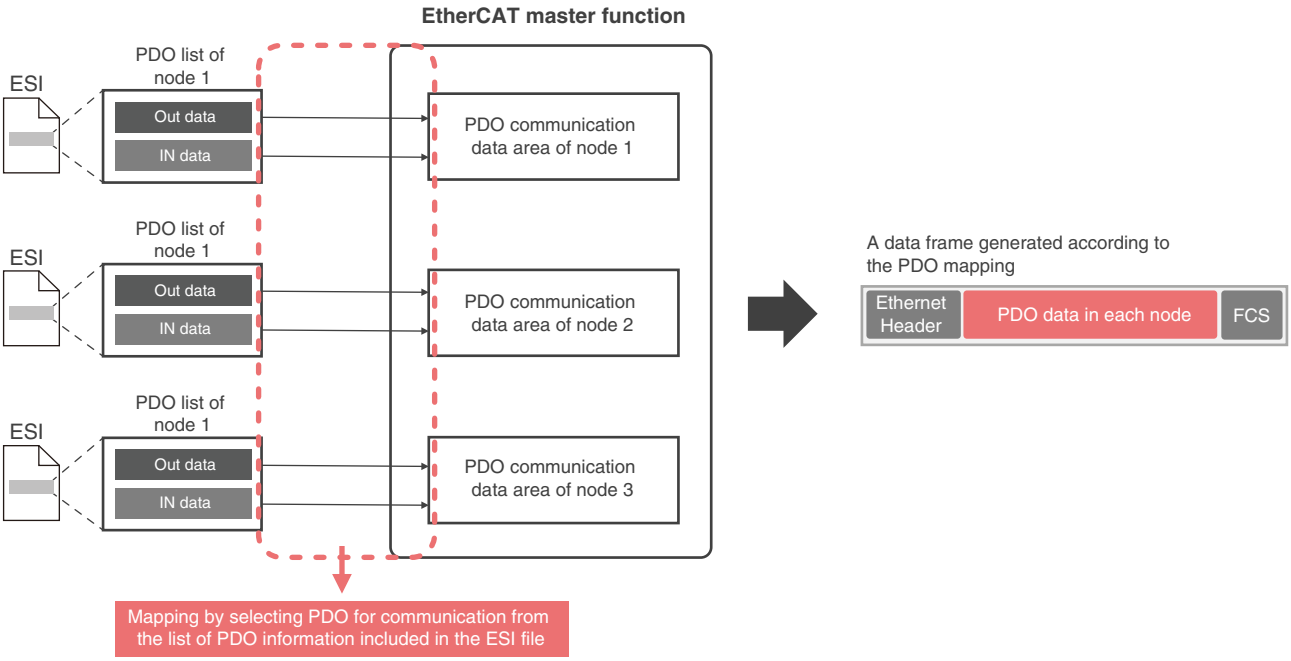
  The value of register 0x0012 of the node device is read as the current value of the Explicit Device ID.
  In this situation, you can rewrite the Explicit Device ID value from “KV STUDIO”.
  However, the node device must be turned off and on again for the Explicit Device ID to be applied.

Detailed node setting

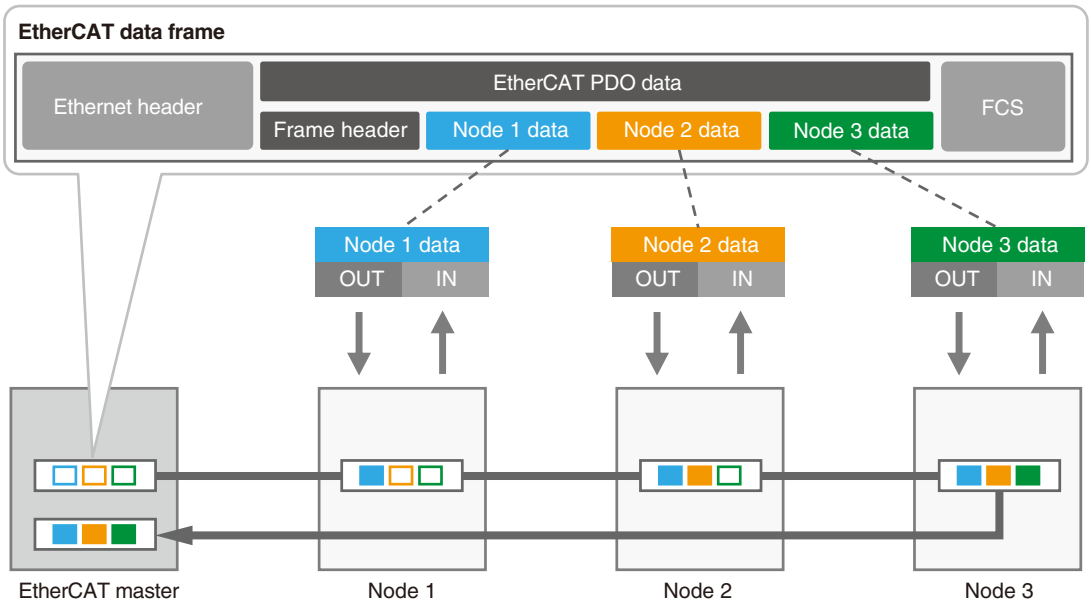
The following describes detailed settings for nodes.
Set the mapping of data to be exchanged in the PDO communication of EtherCAT, and set the configuration of a module to be attached to the coupler unit of a remote I/O device.

PDO mapping

This function maps data sent and received between the master of EtherCAT and the nodes within the network.
The EtherCAT allows data to be communicated between all the nodes in one data frame. Determining data to be sent and received between the master and each node is called PDO mapping, which needs to be set for each node. Data available for mapping depend on the content of ESI file.



When data are sent from the EtherCAT master, each node reads or writes data in own data area according to the settings of the PDO mapping. At this time, data are exchanged on-the-fly.

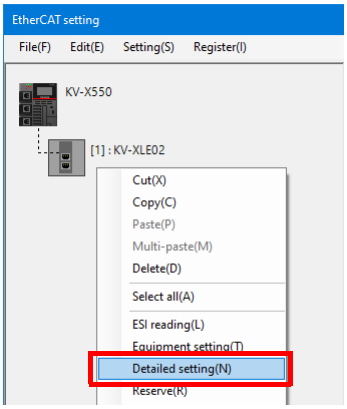


List of PDO mapping items

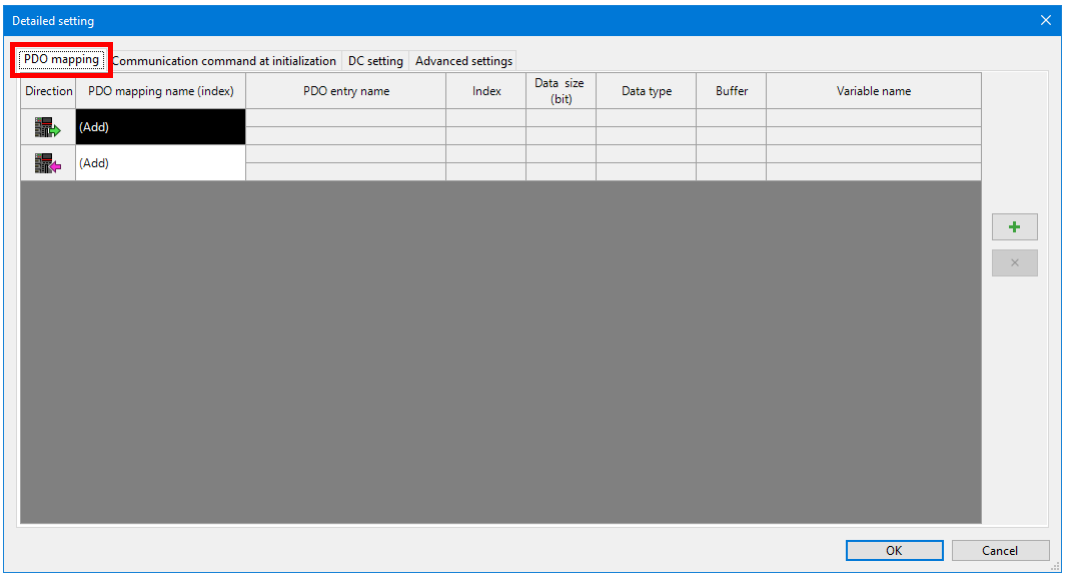
Item	Description
PDO mapping name	Set the PDO mapping name. When setting multiple PDO mappings, be careful that each PDO mapping name is not duplicated.
PDO entry name	Set the PDO entry name.
Index	Set the index number and sub-index number of the PDO. Basically, there is no need changing the default value.
Data size	Set the data size of PDO in 1-bit units. This item is automatically set based on the information in the ESI file, and thus cannot be changed.
Data type	Displays the data type of PDO. This item is automatically set based on the information in the ESI file, and thus cannot be changed.
Buffer	Displays the address of each element on the send buffer and receive buffer. The offset is automatically calculated. Variables set in the PDO mapping are allocated to the transmit/receive buffers. "Transmit/receive buffer"
Variable name	Set variables to be allocated to the transmit/receive buffer. The node devices that are used via the Motion Control Function, and which are registered as the axes do not need to be allocated variables because they use the motion control system variables and CPU devices to exchange data. For details on how to set the motion axes, see the "KV-X500/X300 Series Motion Control Function User's Manual".

How to set

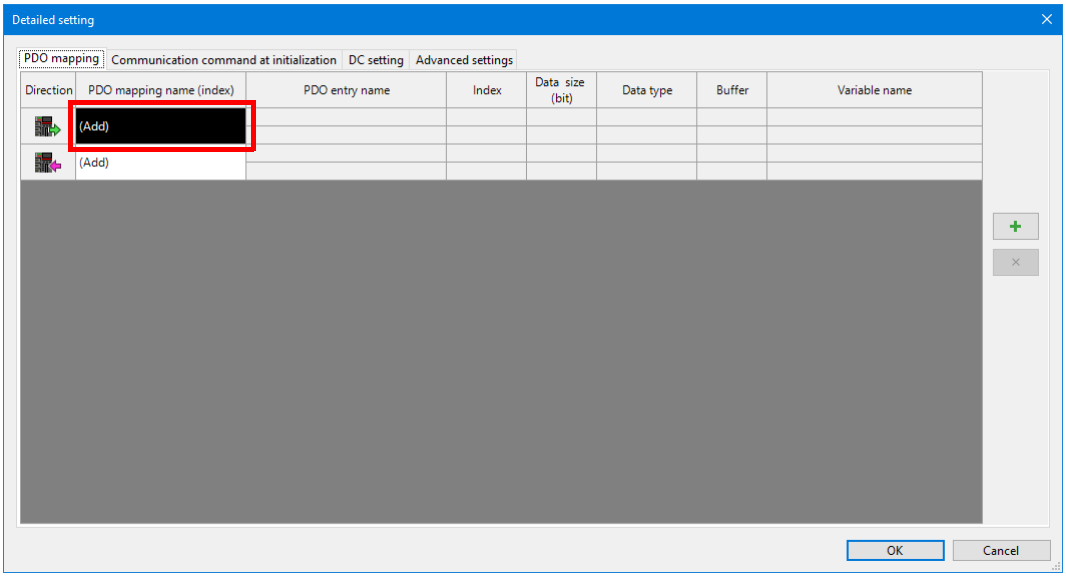
Select a node in the [EtherCAT setting] dialog box, and click [Detailed setting] from the right-click menu.



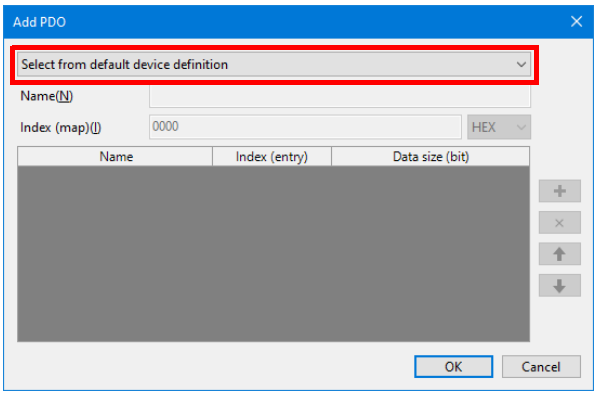
The [Detailed setting] dialog box appears. Select [PDO mapping].



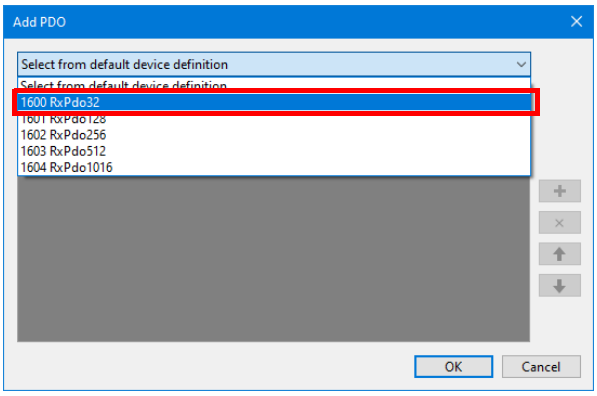
Map data to be sent and received in each direction: from the master to the node, and from the node to the master.
The following example sets the PDO mapping in the direction from the master to the node. Double-click [Add] outlined in red.



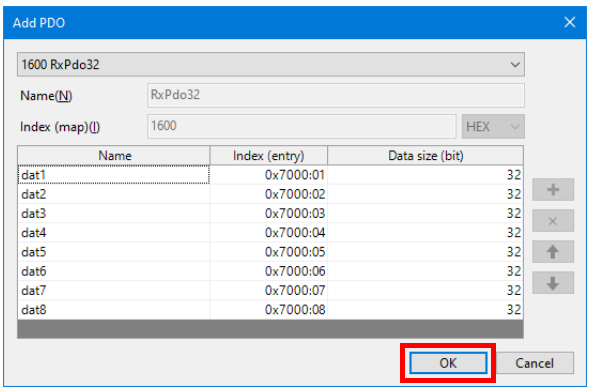
The [Add PDO] dialog box appears. Click “Select from default device definition”.



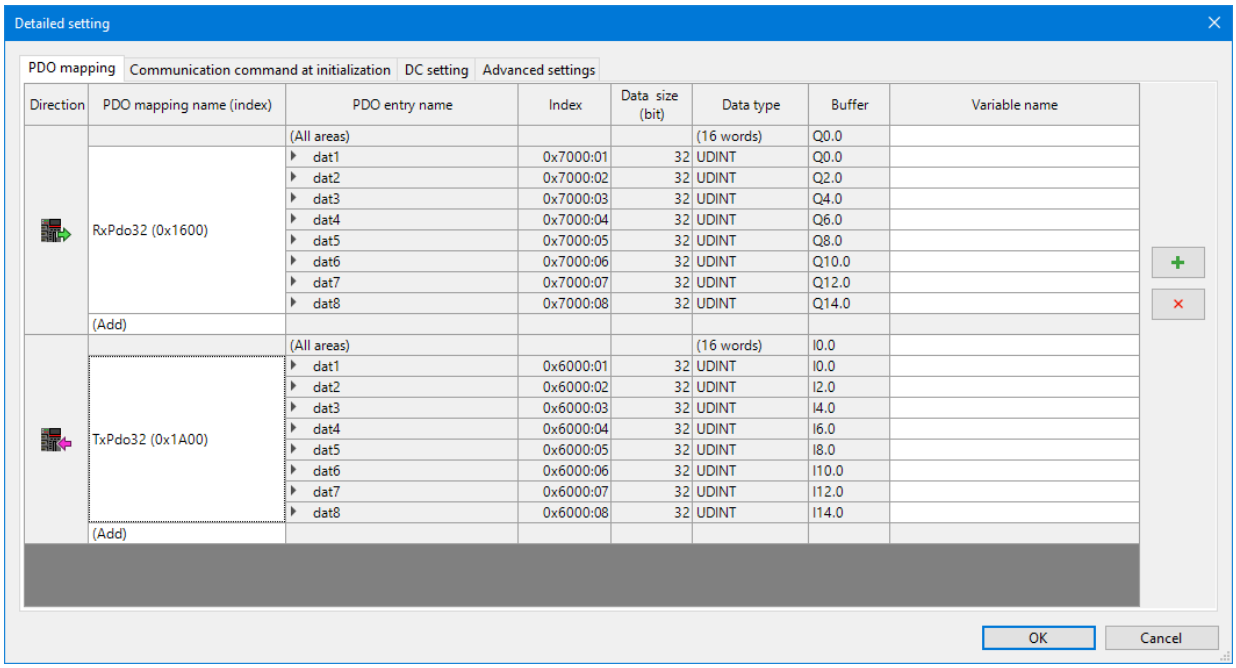
Pre-set settings are displayed according to the information defined in the ESI file. Select an appropriate setting.



The PDO mapping name, index of the object dictionary, and data size are added. Click [OK].



Repeat the same procedure to map all PDO for data to be sent and received in each direction: from the master to the node, and from the node to the master.

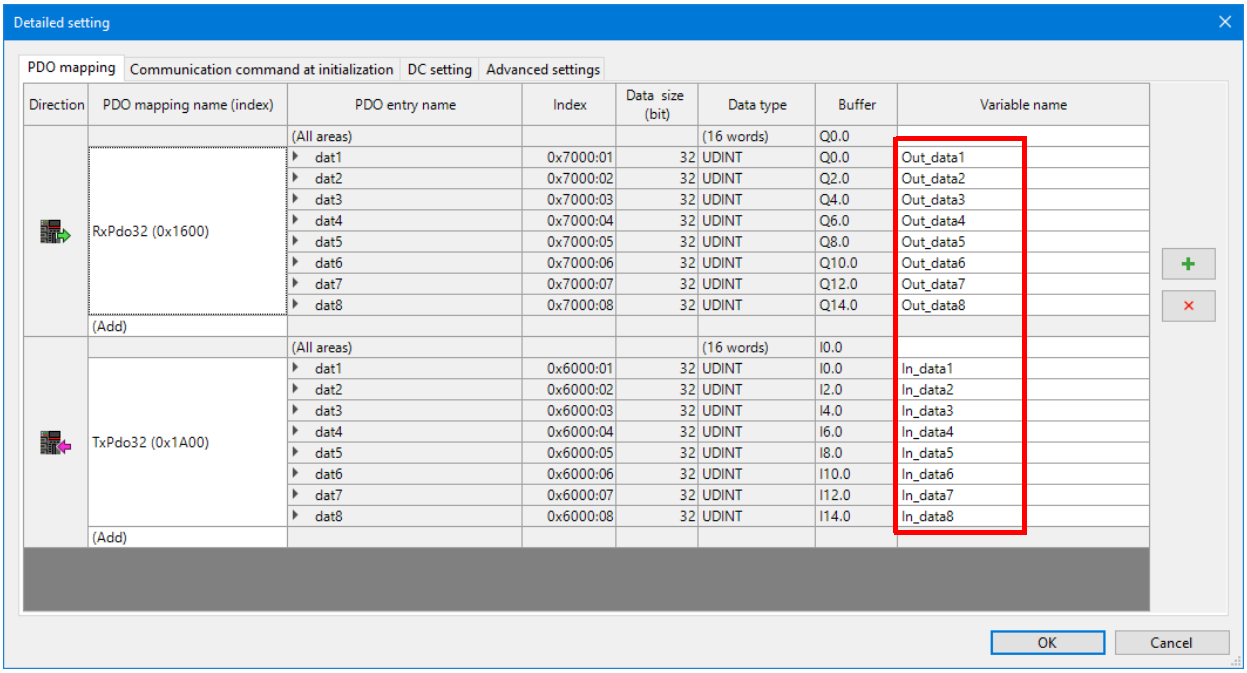


Allocate variables to store data sent and received in the PDO communication to the transmit/receive buffers. Data are refreshed between the variables allocated here and the PDO communication area. "Refresh of EtherCAT"
Fill the "Variable name" fields with variable names and Click "OK".

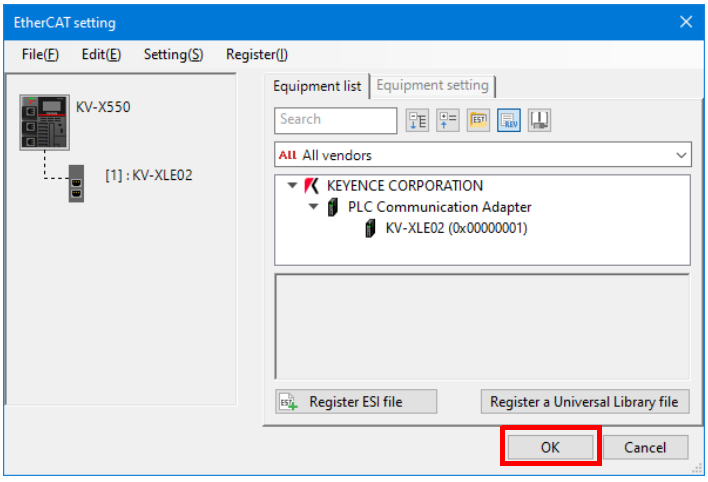
Point

- **Devices cannot be assigned to the send/receive buffer.**
- **The transmit/receive buffer cannot be directly read or written using commands such as UnitBufMemRead.**
- **To read and write the send and receive buffers, you must use the allocated global variables.**
- **Transmission/receiving buffers are allocated in word units. As such, if a PDO is allocated for a data type that is less than one word, such as 8-bit or 1-bit, communication data will not be refreshed in the remaining area within the word. Do not refer to data in areas not allocated to PDOs. Data in these areas may contain indeterminate values.**
  For example, in the following allocation, PDOs are not allocated to bits 8 to 15 of I1.0 and bits 1 to 15 of I3.0, so these areas should not be referred to in the program.

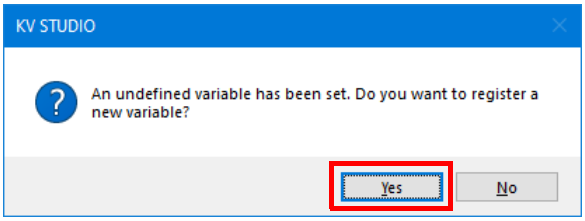
Data size (bit)	Data type	Buffer
16	UINT	I0.0
8	SINT	I1.0
16	UINT	I2.0
1	BOOL	I3.0
16	UINT	I4.0



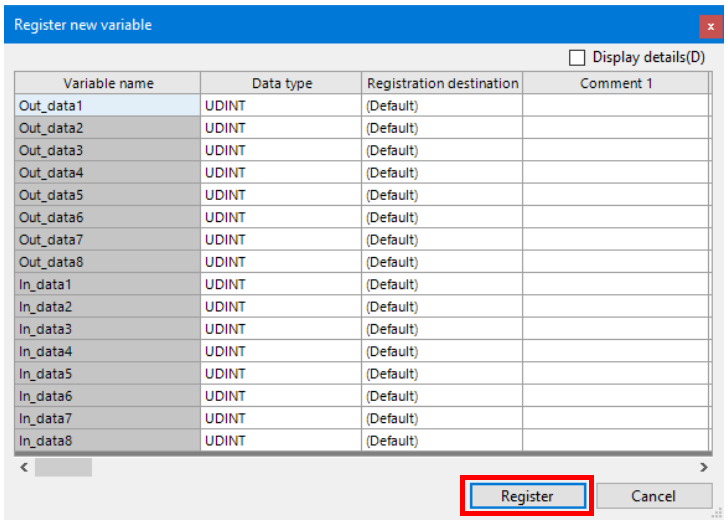
In the [EtherCAT setting] dialog box, click [OK].



If any of the set variable names is not registered, the following dialog box appears. Click [Yes].

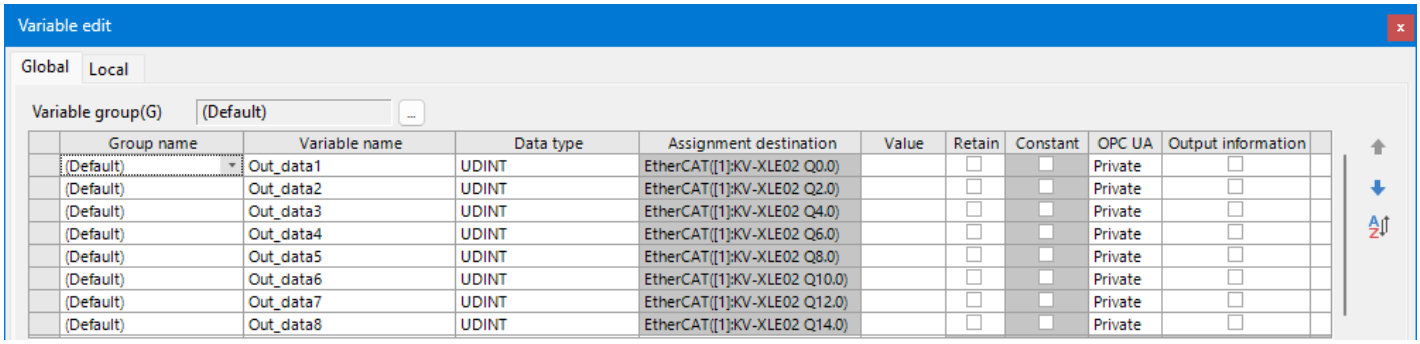


The [Register new variable] dialog box appears. The data type is automatically set according to the data type of PDO. Click [Register] to register the variables.



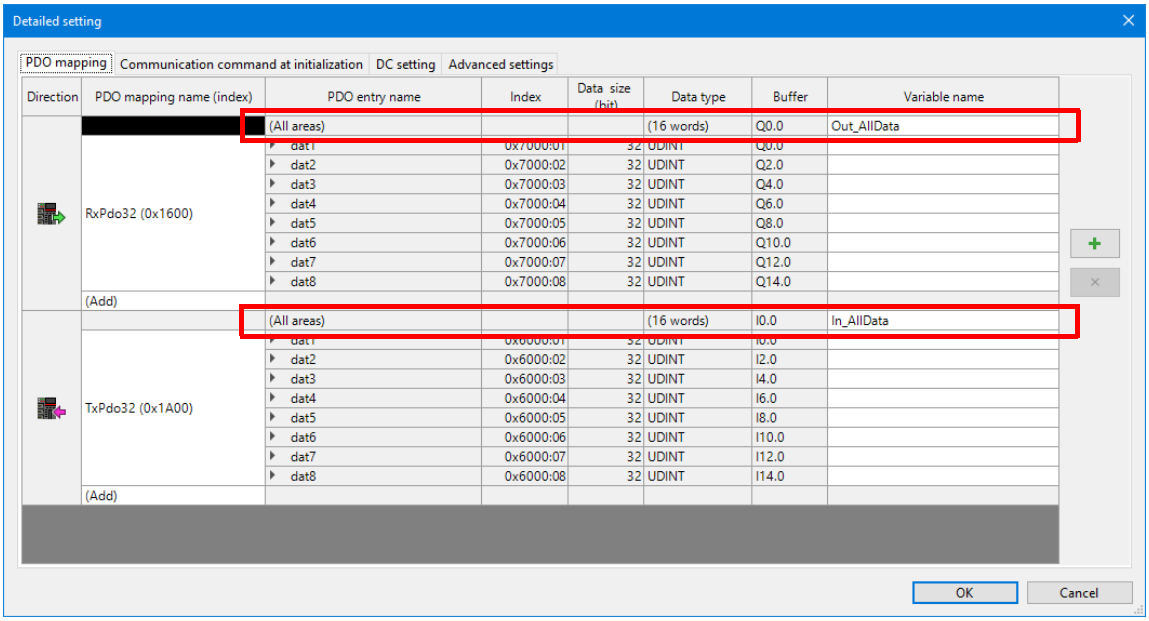
Variable name	Data type	Registration destination	Comment 1
Out_data1	UDINT	(Default)	
Out_data2	UDINT	(Default)	
Out_data3	UDINT	(Default)	
Out_data4	UDINT	(Default)	
Out_data5	UDINT	(Default)	
Out_data6	UDINT	(Default)	
Out_data7	UDINT	(Default)	
Out_data8	UDINT	(Default)	
In_data1	UDINT	(Default)	
In_data2	UDINT	(Default)	
In_data3	UDINT	(Default)	
In_data4	UDINT	(Default)	
In_data5	UDINT	(Default)	
In_data6	UDINT	(Default)	
In_data7	UDINT	(Default)	
In_data8	UDINT	(Default)	

The allocation destination of the registered variables are registered with the EtherCAT ([Node number: model name send and receive buffer address]) as shown below.



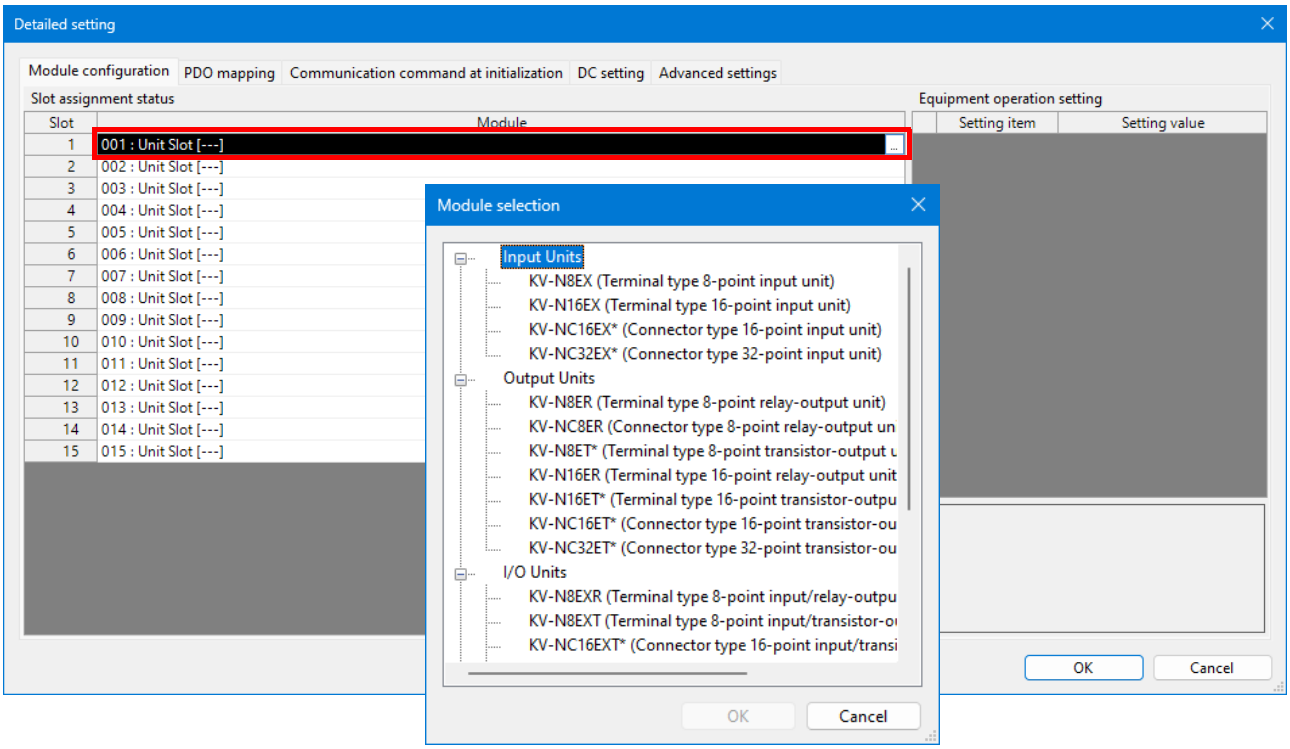
Group name	Variable name	Data type	Assignment destination	Value	Retain	Constant	OPC UA	Output information
(Default)	Out_data1	UDINT	EtherCAT([1]:KV-XLE02 Q0.0)				Private	
(Default)	Out_data2	UDINT	EtherCAT([1]:KV-XLE02 Q2.0)				Private	
(Default)	Out_data3	UDINT	EtherCAT([1]:KV-XLE02 Q4.0)				Private	
(Default)	Out_data4	UDINT	EtherCAT([1]:KV-XLE02 Q6.0)				Private	
(Default)	Out_data5	UDINT	EtherCAT([1]:KV-XLE02 Q8.0)				Private	
(Default)	Out_data6	UDINT	EtherCAT([1]:KV-XLE02 Q10.0)				Private	
(Default)	Out_data7	UDINT	EtherCAT([1]:KV-XLE02 Q12.0)				Private	
(Default)	Out_data8	UDINT	EtherCAT([1]:KV-XLE02 Q14.0)				Private	

Reference You can also allocate array type variables or structures suitable for the data structure of PDO, instead of allocating variables to individual PDOs. In such a case, allocate variables to “All areas”. If the set variables are unregistered, these are registered as array type variables of UINT appropriate for the data size.



Module configuration

Set the module configuration when installing multiple modules to one node, such as a remote I/O coupler unit. A tab appears only for a node for which module configuration is required.
Double-clicking shows a dialog to select the module to be attached to the coupler unit.



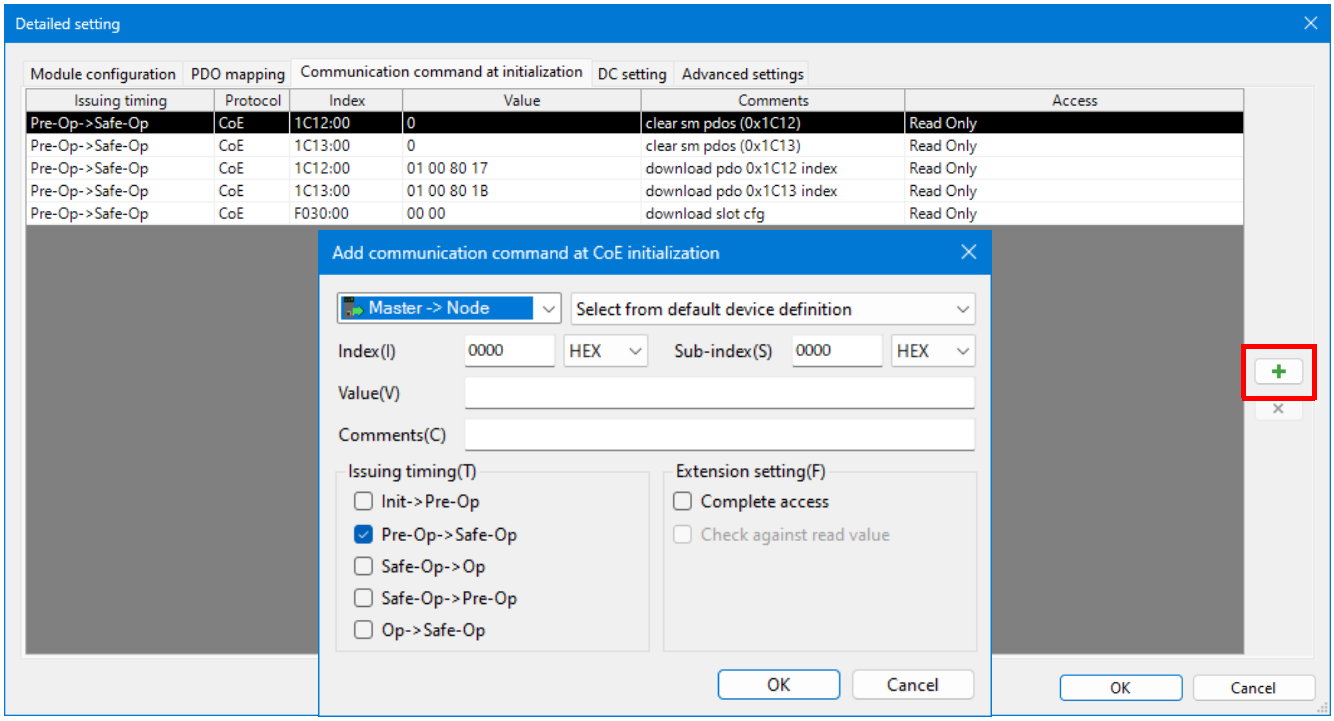
PDO is automatically added to the PDO mapping according to the added module. Allocate variables to the data to be read and written in the program.

“PDO mapping”

When changing the module settings, use the communication command at initialization.

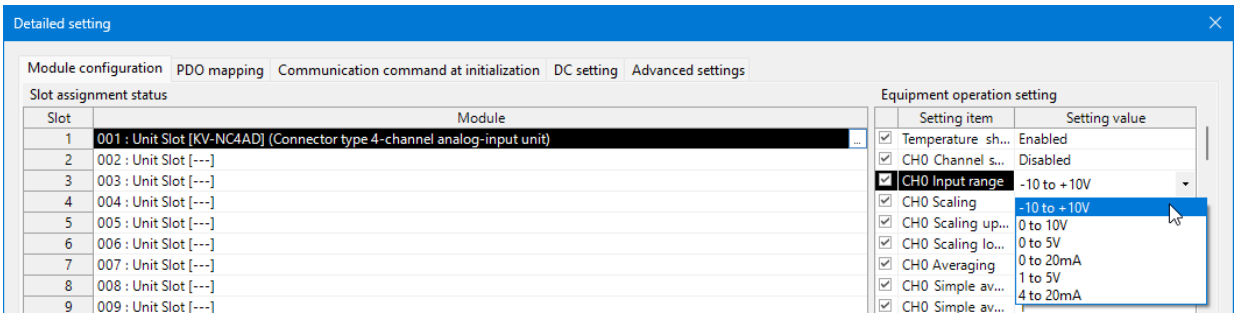
Communication command at initialization

This function writes values in specified objects according to the EtherCAT state of the node. When changing the settings for the module attached to the coupler unit, set this item.
Click [+], and set the content of the command. See the manual of the connected EtherCAT node for the indexes and value ranges.



Reference

When KV-EC01 is used for the EtherCAT node, the following special screen appears on the module configuration screen. You can also set the modules to be connected on the module configuration screen. Therefore, the settings of communication command at initialization is not required.



DC setting

These are settings for DC (Distributed Clock) synchronization. Basically, there is no need of setting this item.

Advanced settings

These are advanced settings for the EtherCAT communication. Basically, there is no need of setting this item.

Master setting

Set parameters for the master.

Setting items list

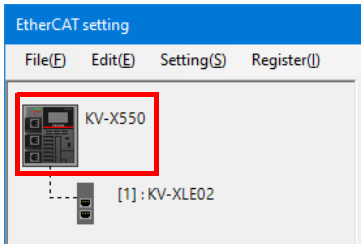
Item	Description
Communication cycle (µs)	Set the communication cycle of the EtherCAT communication. Setting range: 62.5 (KV-X500 series only) , 125, 250, 500, 750, 1000, 1500, 2000, 3000, 4000, 5000, 6000, 7000, 8000, 9000, 10000
Number of detected consecutive PDO communication timeouts	A time-out error occurs if the measured communication cycle exceeds the set communication cycle consecutively for the specified number of times. Setting range: 0 to 16 Default: 3
Start wait time (s)	An error is issued if no communication is established with the node after the specified time has passed since the start of PDO communication. Set this item when the power-on start-up time differs between the master and the node or when it takes time to complete initialization communication due to a number of connected units.
Automatic PDO communication start	Select whether to automatically start the PDO communication when the power is turned on. When not set to automatic start, the communication starts when the operation is permitted.

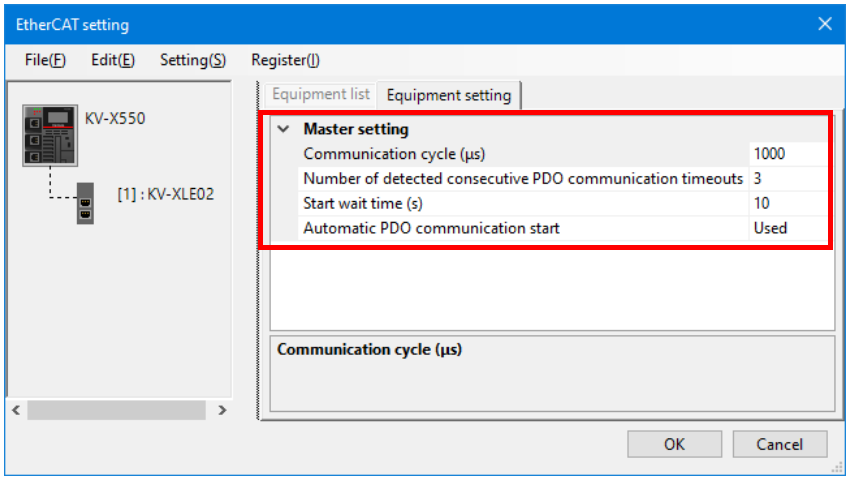
Point

- **The configurable communication cycle varies depending on the number of nodes or the size of PDO per node.**
- **When selecting 62.5µs on the KV-X310, an error occurs during conversion.**

How to set

Select the master (KV-X500/X300 series) in the [EtherCAT setting] dialog box, and set the items outlined in red in the [Equipment setting] tab.





Reference

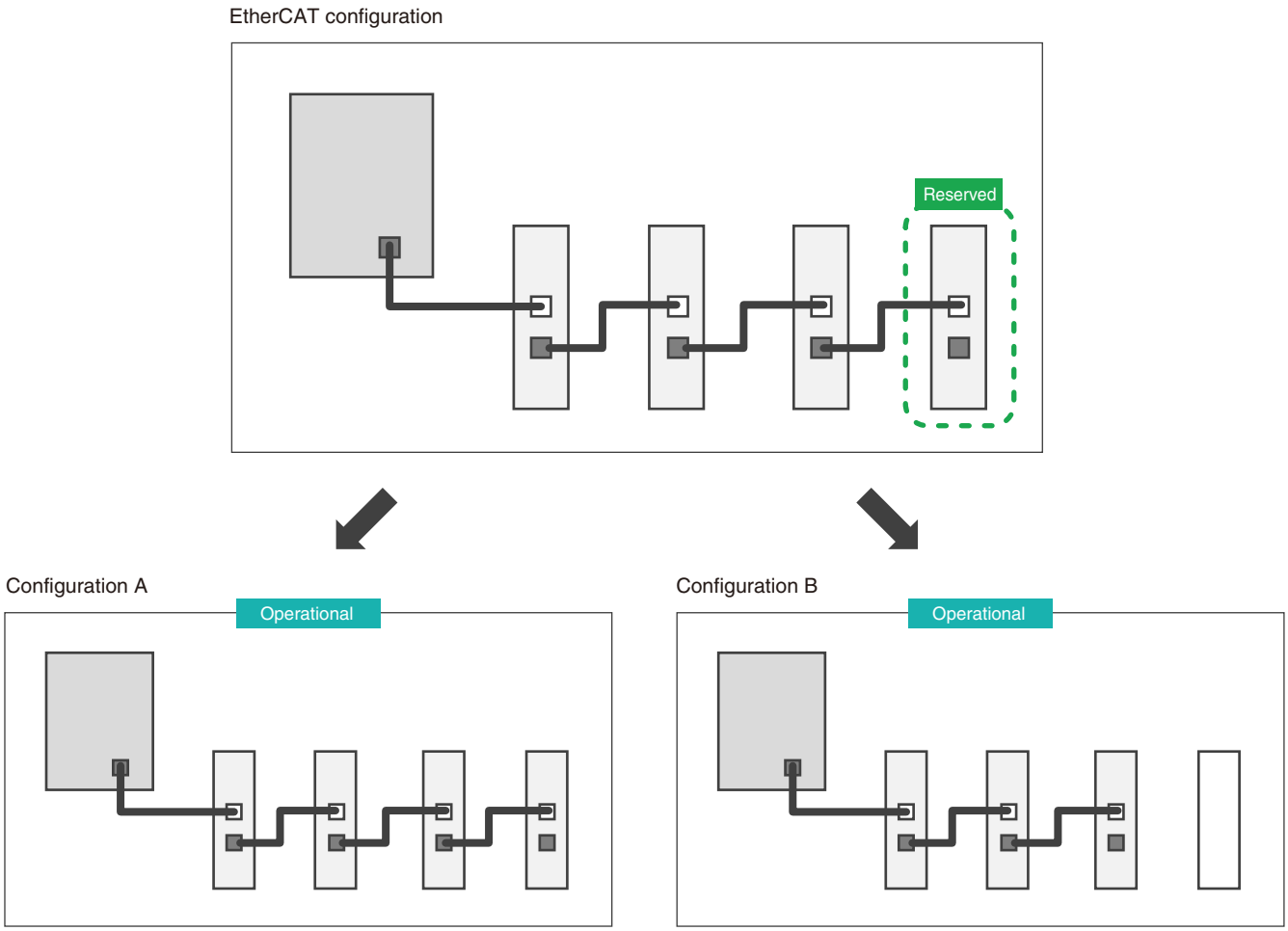
When the “Automatic PDO communication start” is set to “Not used”, the EtherCAT state of all nodes will change to Pre-Operational. You can start the communication by turning on the operation permission.

“EtherCAT state machine”

Reservation of nodes

This function is used when the configuration of actually connected nodes changes depending on whether the device has optional functions or other factors. The reserved node operates normally when connected. When not connected, no error occurs. Even with different configuration, the node can be managed in the same project.
The master usually recognizes the node devices in the order they are connected to it, so when the same device is successively connected to it, the master cannot determine which node is reserved.
Therefore, when using the reservation function, be sure to set the Explicit device ID. By checking the Explicit device ID, the master can determine the reserved node.

"Check items when starting communication"

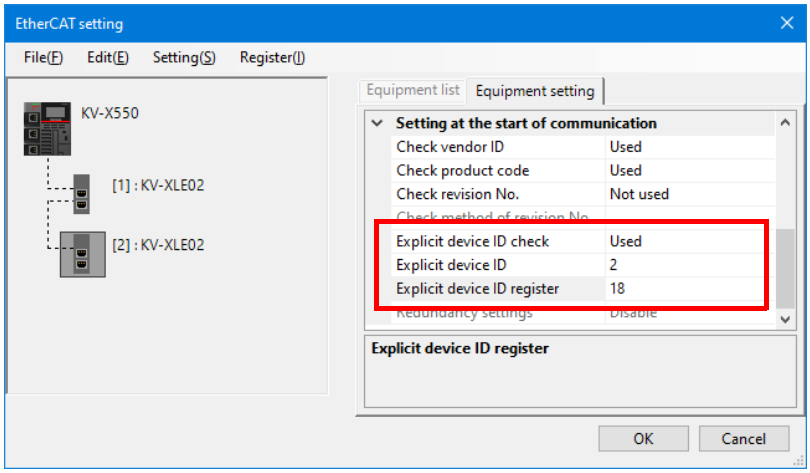


Reference After participating in the PDO communication, if the reserved node has a communication abnormality, an error occurs. To remove the node after it participates in the PDO communication, use Stop/Restart.

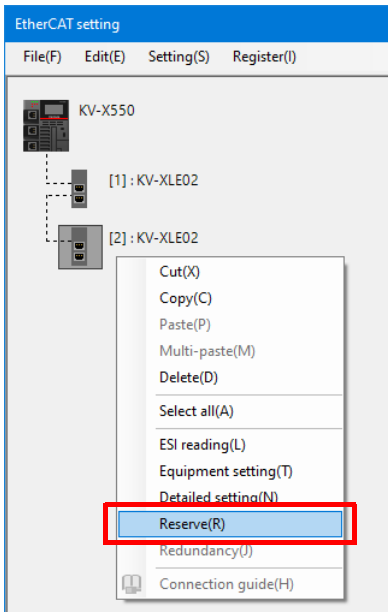
"Stop/resume"

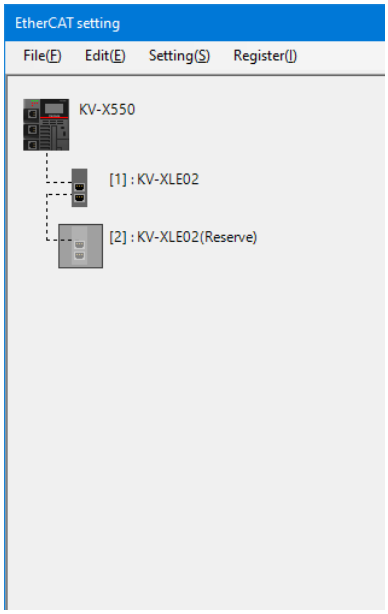
How to set

Open the [EtherCAT setting] dialog box. Set the “Explicit device ID check” of the node to “Used”, and then set the “Explicit device ID” and “Explicit device ID register”.



Select the target node and click [Reserve] from the right-click menu.
“(Reserve)” is shown next to the node name.





Notes and restrictions on reservation

- All the downstream components from the node will be reserved.
- When cancelling reservation, be sure to select the reserved leading node. Also, when reservation is cancelled, the reservation of all the downstream nodes will be cancelled.
- When a branching slave is reserved, all the components connected to each OUT port will be reserved.
- Of the components of the ring, only a branching slave, which is the start point of the ring, can be reserved; the other nodes cannot be reserved.
- If all nodes are preconfigured as reserved but no actual devices are connected, an EtherCAT link-off error will occur.

Batch setting

The following describes how to set in a batch node addresses, Explicit device ID, and PDO mapping variables allocation.

Batch setting of node addresses

Set node addresses in a batch.

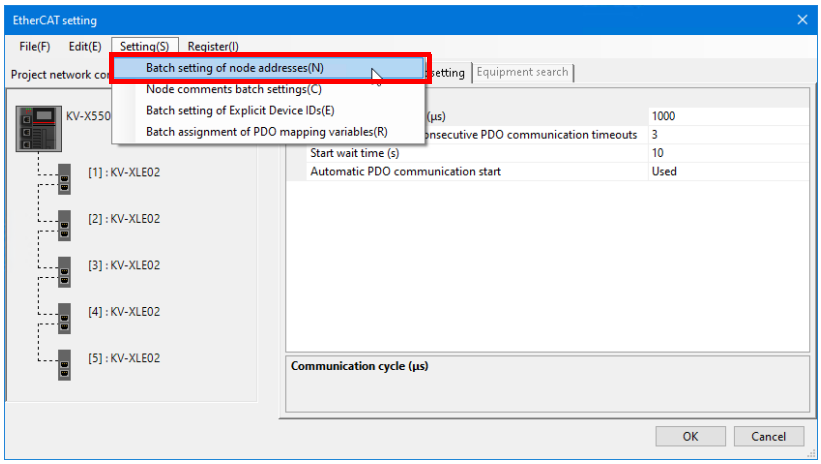
When changing the order of nodes or adding/deleting a node after the setting is complete, you can correctively change the node addresses.

Setting items list

Item	Description
Current value	The current value of the node address
Change value	The changed value of the node address. Set a desired value.
The name of equipment	The component name of the node.

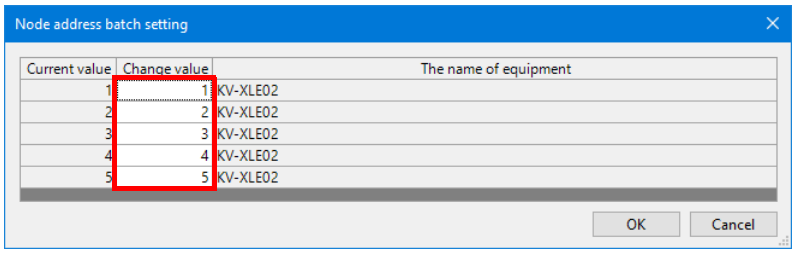
How to set

From the [Setting] menu in the [EtherCAT setting] dialog box, select [Batch setting of node addresses].



The [Node address batch setting] dialog box appears.

Set the “Change value” to node addresses to be allocated to the nodes, and click [OK].



Batch setting of node comment

Batch setting of node comment will be performed.

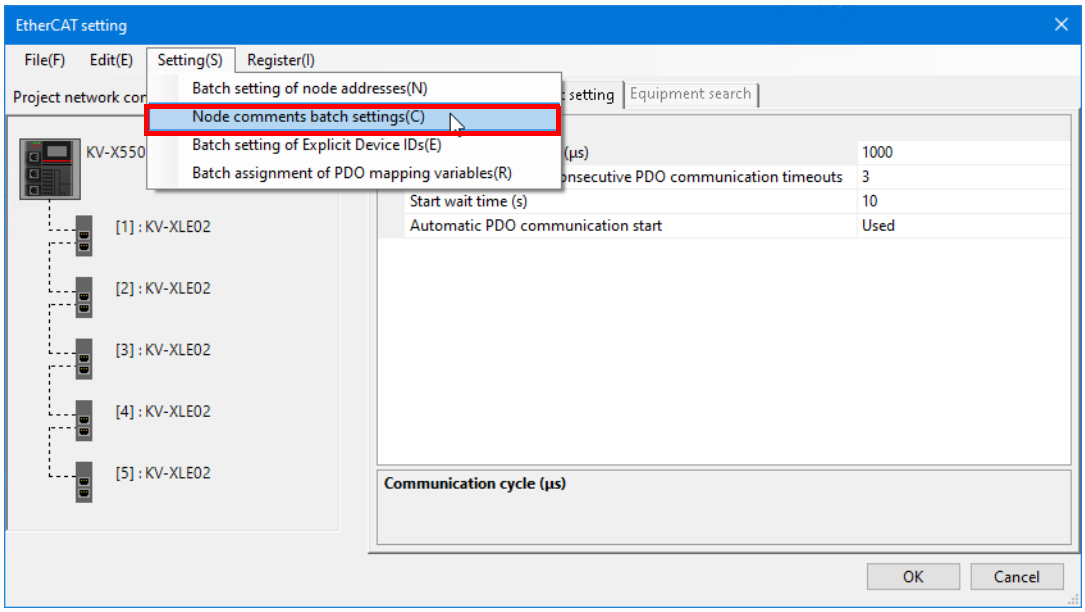
If there are many node devices, setting devices in bulk rather than individually will reduce the configuration workload.

Setting items list

Item	Description
Node comment	It is a node comment.
Device name	The component name of the node.

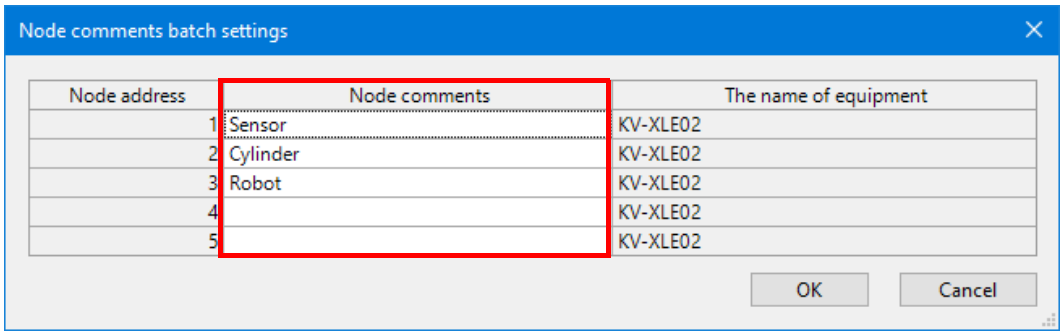
How to set

From the [Setting] menu in the [EtherCAT setting] dialog box, select [Node comments batch settings].



The [Node comments batch setting] dialog box appears.

Set the Node Comment to be assigned to each node in the “Node Comment” field, then click “OK”.



Batch setting of Explicit device ID

Set the Explicit device ID in a batch.

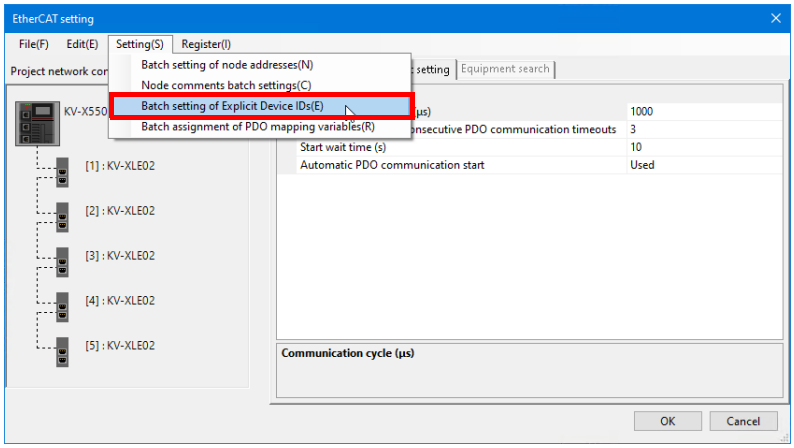
When setting the Explicit device ID of multiple nodes, changing the order of nodes, or adding/deleting a node, you can collectively change the values of Explicit device ID.

Setting items list

Item	Description
Check	To check the Explicit device ID, set this item to “Used”.
ID	When the Check is set to “Used”, set the Explicit device ID.
Register	Set the Explicit device ID register. “Check items when starting communication”
Component name	The component name of the node.

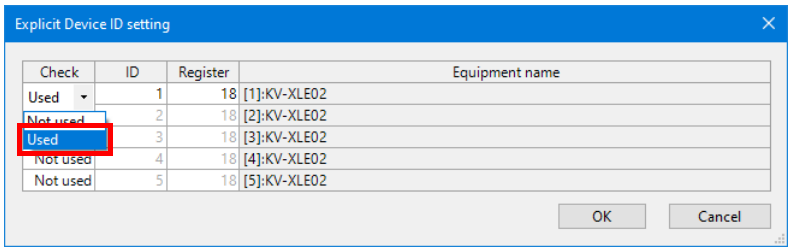
How to set

From the [Setting] menu in the [EtherCAT setting] dialog box, select [Batch setting of Explicit Device IDs].

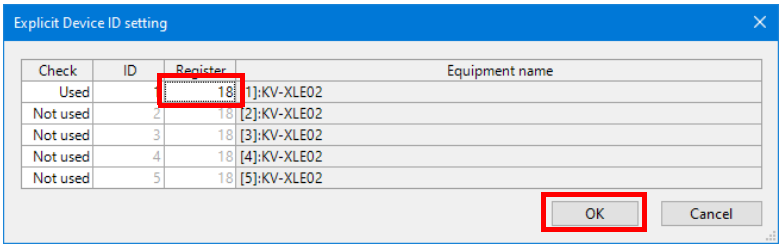


The [Explicit Device ID setting] dialog box appears.

Set “Used” for the “Check” of the node for which Explicit device ID is to be checked.



Set the Explicit device ID and the Explicit device ID register, and click [OK].



Batch setting of variable allocation of PDO mapping

Set the PDO mapping variables allocation in a batch.

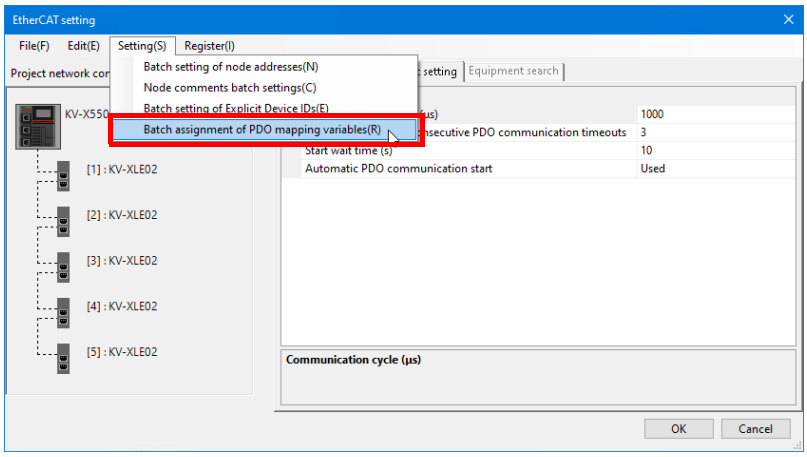
Variable allocation can be collectively set without separately opening each PDO mapping setting screen.

Setting items list

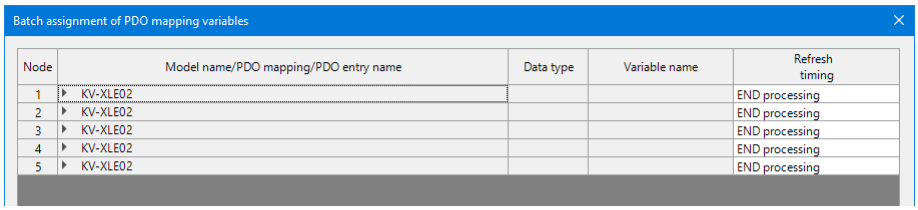
Item	Description
Node	Displays the node address.
Model name/PDO mapping/PDO entry name	Displays the model name, PDO mapping name, and PDO entry name.
Data type	Displays the data size and data type of PDO.
Variable name	Set the names of variables to be allocated to the PDO.
Refresh timing	Select the refresh timing from the “END processing” and “Inter-unit synchronization”.

How to set

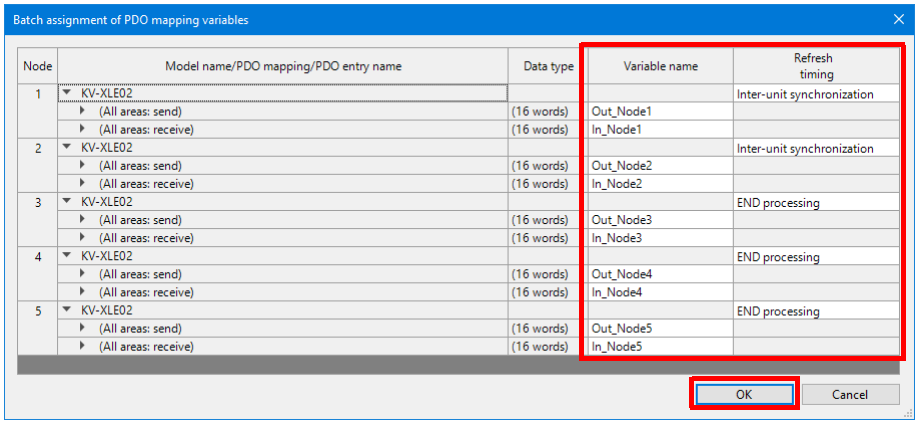
From the [Setting] menu in the [EtherCAT setting] dialog box, select [Batch assignment of PDO mapping variables].



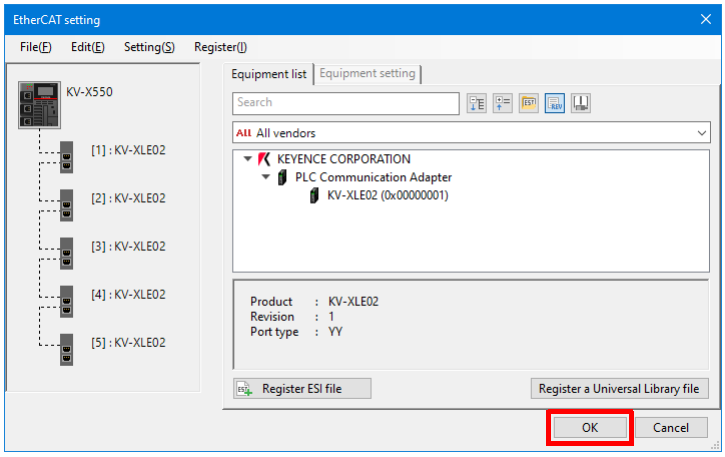
The [Batch assignment of PDO mapping variables] dialog box appears.



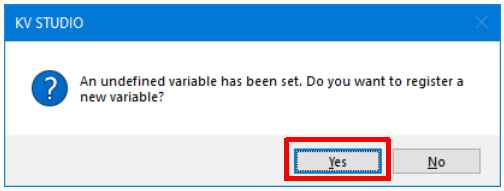
Set the “Variable” to be allocated to the PDO and “Refresh timing”, and click [OK].



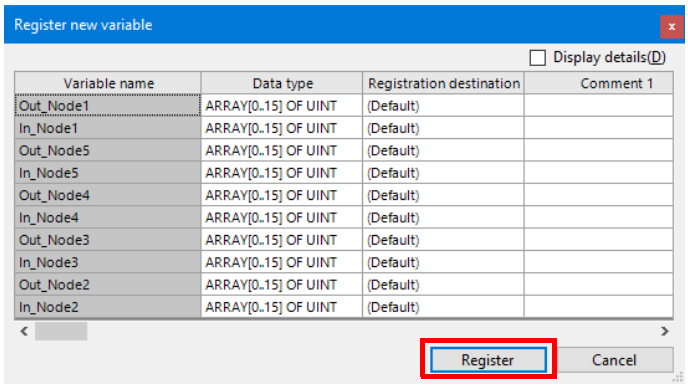
In the [EtherCAT setting] dialog box, click [OK].



If any of the set variable names is not registered, the following dialog box appears. Click [Yes].



The [Register new variable] dialog box appears. The data type is automatically set according to the data type of PDO. Click [Register] to register the variables.



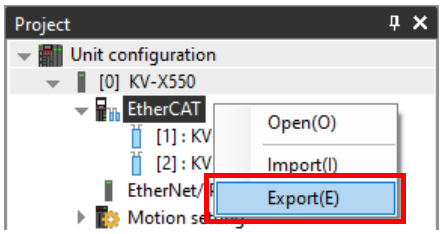
Variable name	Data type	Registration destination	Comment 1
Out_Node1	ARRAY[0..15] OF UINT	(Default)	
In_Node1	ARRAY[0..15] OF UINT	(Default)	
Out_Node5	ARRAY[0..15] OF UINT	(Default)	
In_Node5	ARRAY[0..15] OF UINT	(Default)	
Out_Node4	ARRAY[0..15] OF UINT	(Default)	
In_Node4	ARRAY[0..15] OF UINT	(Default)	
Out_Node3	ARRAY[0..15] OF UINT	(Default)	
In_Node3	ARRAY[0..15] OF UINT	(Default)	
Out_Node2	ARRAY[0..15] OF UINT	(Default)	
In_Node2	ARRAY[0..15] OF UINT	(Default)	

3-3 Import/export of EtherCAT setting

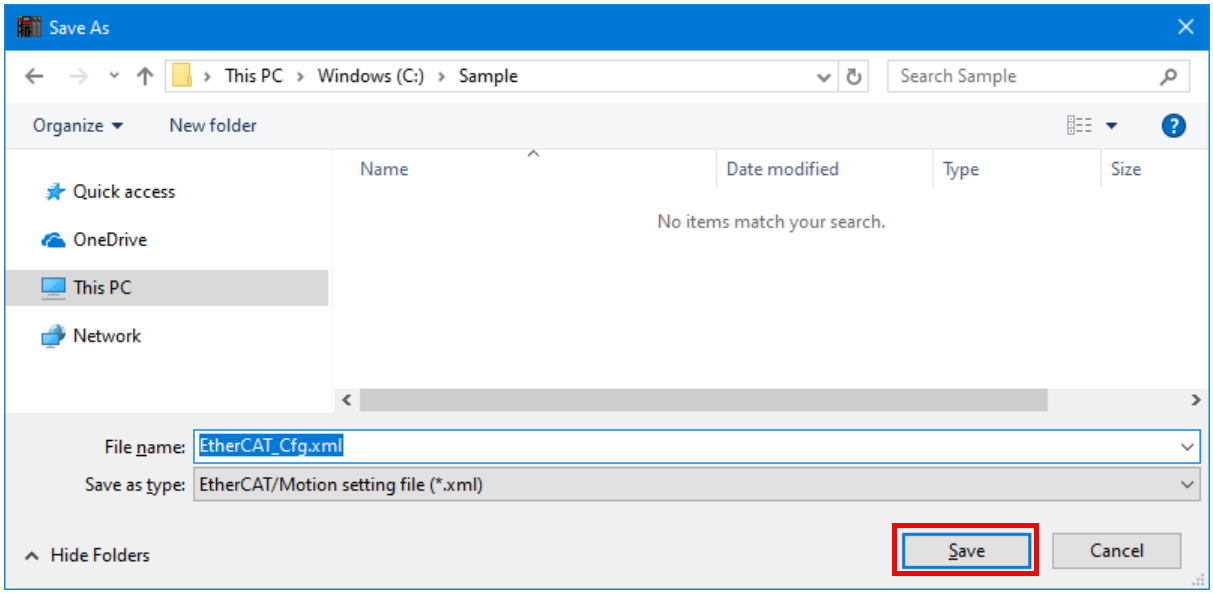
The following describes how to import/export the settings made in [EtherCAT setting].

Export

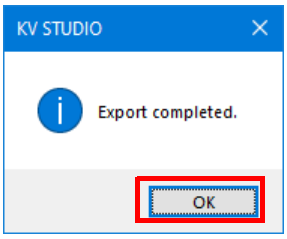
Select “EtherCAT” in the workspace and click “Export” from the right-click menu.



Save the file with a new name in the XML format.

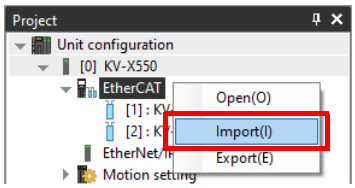


The following dialog appears. Click [OK].

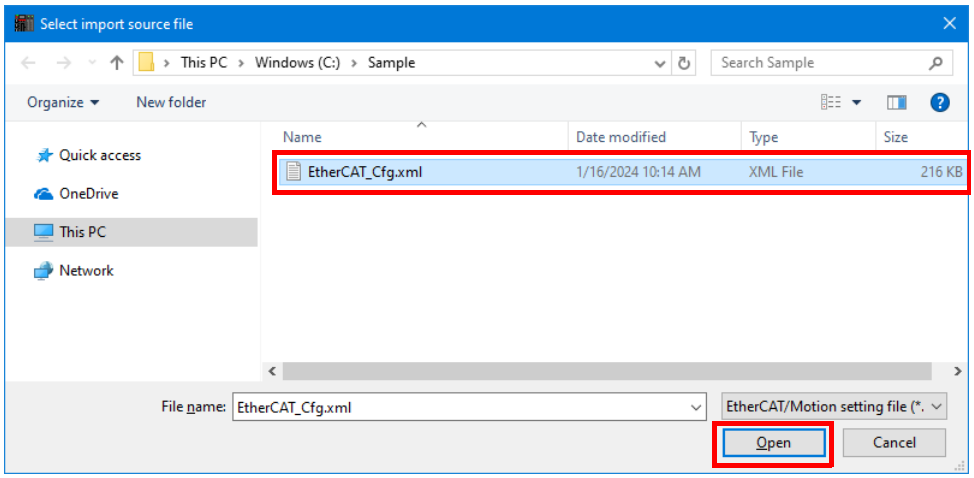


Import

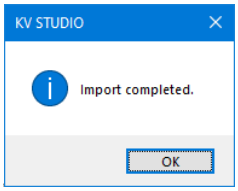
Select “EtherCAT” in the workspace and click “Import” from the right-click menu.



The following dialog appears. Select a file to import and click [Open].



Once the import is completed, the following dialog box appears.



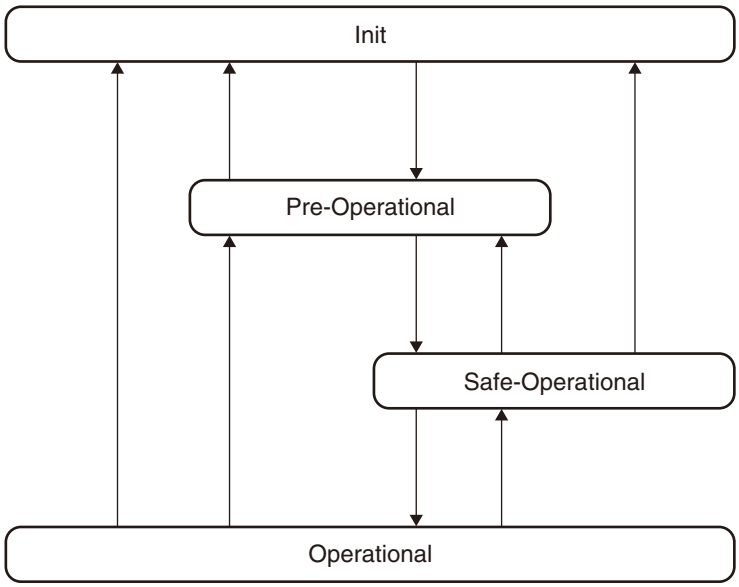
Reference You can also import/export the settings of EtherCAT setting from the following.

- Export
  [Tool] > [EtherCAT/motion setting] > [Export]
- Import
  [Tool] > [EtherCAT/motion setting] > [Import]

Chapter 4 Communication Function of EtherCAT

4-1 EtherCAT state machine

Each node connected to the EtherCAT master has four states that the node transitions to control communication. Available communication functions depend on the state that the node has transitioned to. State transition is automatically performed by instructions from the master.



The following shows the availability of PDO/SDO communication at each state.

EtherCAT state	PDO: Output (M -> N)	PDO: Input (N -> M)	SDO
Init	×	×	×
Pre-Operational	×	×	√
Safe-Operational	×	√	√
Operational	√	√	√

*M: KV-X500/X300 series N: Connected component

Reference

- To check the state of each node, use the system variable “_EC_MASTER.EcStateTbl[n]” (n: node address).
  0: Not connected
  1: Init
  2: Pre-Operational
  3: Safe-Operational
  4: Operational
- If "Auto start of PDO communication" is set to "No" in the master settings of the EtherCAT settings, the EtherCAT state of all nodes will transition to Pre-Operational.
  When operation permission is turned ON, PDO communication will start.
  “Master setting”

Communication start sequence when the power is turned on or operation is permitted

The EtherCAT master function of the “KV-X500/X300 series” performs the following communication start sequence when the power is turned on or the operation is permitted after the EtherCAT settings are changed.

(1) The function checks the settings of the current EtherCAT communication against the settings after change, and if any inconsistency is found,
the function stops the EtherCAT communication and performs the sequence from (2) to (4). When these settings of the EtherCAT communication are the same,
the function stops the communication start sequence and continues the EtherCAT communication.

(2) The function checks the set network configuration against the configuration of the node actually connected. For a node with the same settings, the sequence from (3) to (4) is performed. For a node with different settings, the function cannot start the EtherCAT communication, and thus a node error or an EtherCAT master error occurs.

(3) The function starts the EtherCAT communication and moves the EtherCAT state of the “KV-X500/X300 series” and node to the Safe-Operational state.

(4) The function moves the EtherCAT state of the “KV-X500/X300 series” and node to the Operational state.
After the communication sequence above ends, when the Operation permission is turned from OFF to ON, the Operational turns ON. It may take several seconds after the operation permission is turned ON.

Reference

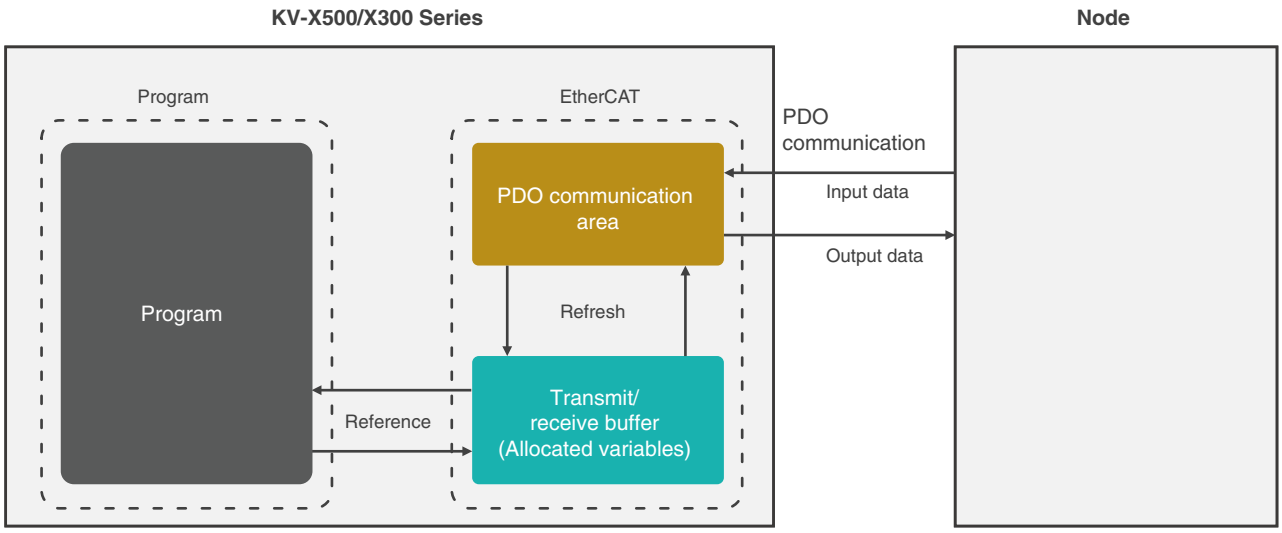
- If the operation permission fails, the EtherCAT communication is not established. After resolving the cause of the error, turn again the operation permission from OFF to ON.
  The following are the methods for turning the operation permission from OFF to ON, depending on the settings of the “instruction method” of the unit editor.
  - When “function blocks” are used for “instruction method”
    Turn Enable in MC_OperationEnable from OFF to ON.
  - When “relays” are used for “instruction method”
    Turn the operation permission relays from OFF to ON.
- When the “Automatic PDO communication start” is set to “Not used” in the EtherCAT master setting, the EtherCAT state of all nodes transitions to Pre-Operational when the power is turned on. To start the communication, turn the operation permission to ON.
  “Master setting”
- The settings of the EtherCAT communication are reflected when the operation permission is turned from OFF to ON. The EtherCAT communication stops when the settings are reflected, and automatically restarts after reflecting the settings.

4-2 Refresh of EtherCAT

The following describes the refresh of EtherCAT communication.

Overview

The PDO communication of EtherCAT performs cyclic data exchange between the master and nodes. The transmitted and received data of PDO communication is refreshed between the transmit and receive buffers inside the CPU. The “KV-X500/X300 series” allocates variables to the transmit/receive buffer to refer to the transmit/receive buffer from the program. By using in the program the variables allocated to the transmit/receive buffer, the function can read and write data to be exchanged through the PDO communication in the program.

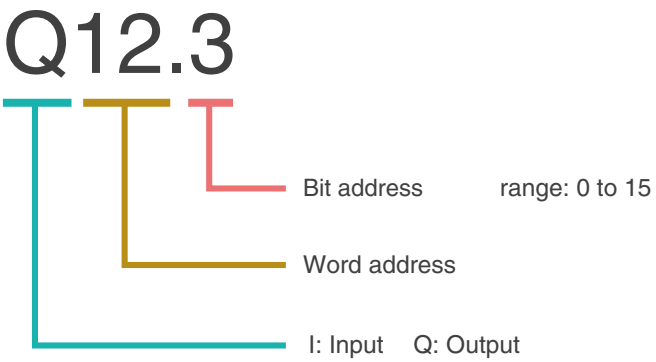


Reference The node devices that are used via the Motion Control Function, and which are registered as the axes do not need to be allocated variables because they use the motion control system variables and CPU devices to exchange data.
For details on how to set the motion axes, see the “KV-X500/X300 Series Motion Control Function User’s Manual”.

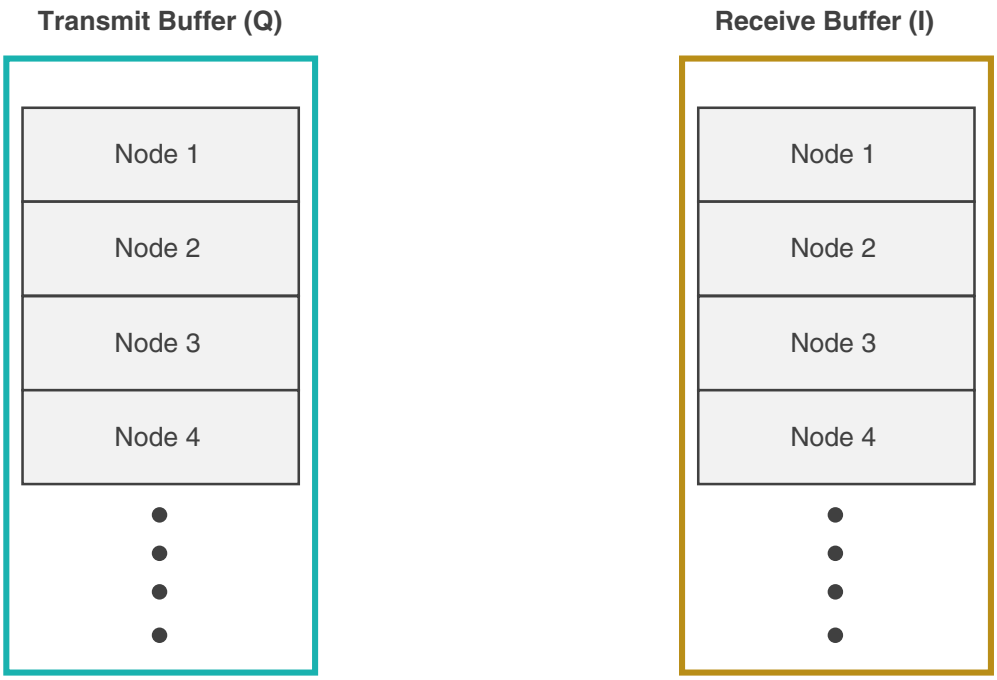
Transmit/receive buffer

The transmit and receive buffers are divided into a transmit buffer (Q) and a receive buffer (I). The buffer address is indicated according to the following rule.

Indication of the buffer address



The offset can be found out from the buffer address at the time when a structure/array is allocated to the whole PDO. You cannot set the buffer address of the node. For communication with multiple nodes, the data of each node are stored in the order that the nodes are placed in the EtherCAT setting.



Reference

Variables do not need to be assigned to the entire send/receive buffer. Variables can be allocated to only a part of the send/receive buffer.
If a send buffer does not have a variable assigned to it, 0 will be sent.
Data received from the node is stored in the receive buffer without assigned variables; however, it cannot be accessed by programs or other processes.

Point

- **Ring Usage Notes and Restrictions·Devices cannot be assigned to the send/receive buffer.**
- **The transmit/receive buffer cannot be directly read or written using commands such as UnitBufMemRead.**

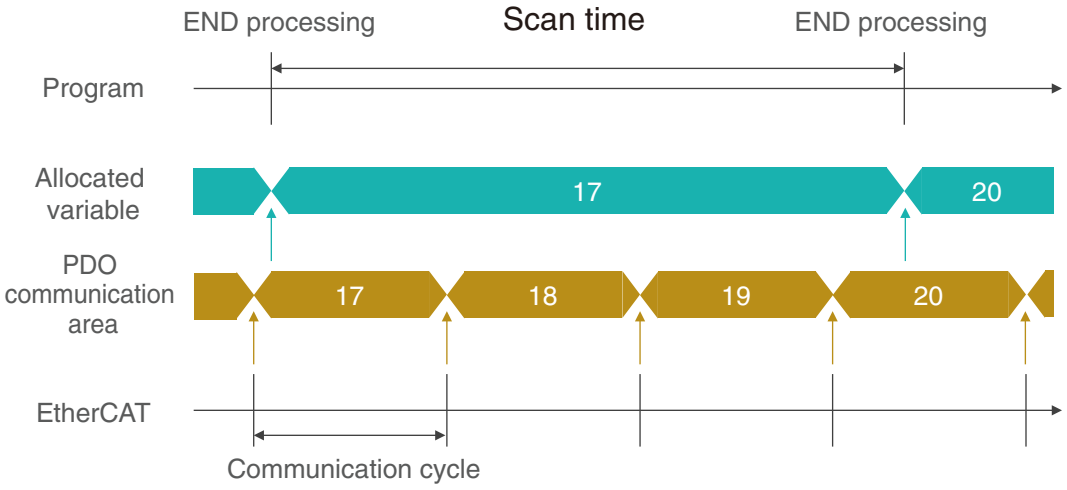
Timing of refresh

Select the timing of data refresh from the “END processing” and “Inter-unit synchronization” for each node.

“How to set refresh”

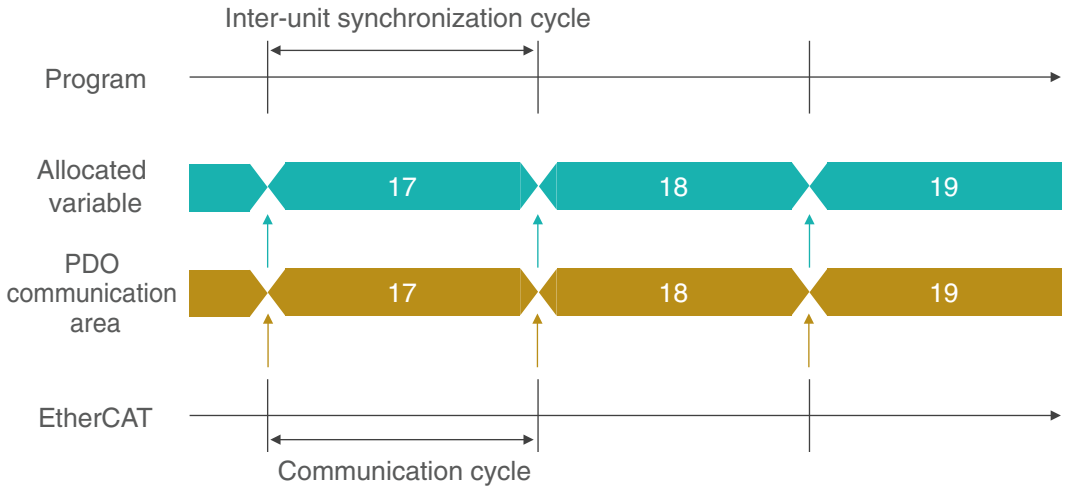
For END processing

The variables allocated to the node with its refresh timing set to “END processing” are refreshed with the PDO communication data of the immediately preceding END processing. The variables can be referenced simultaneously from every-scan module, standby module, and initialization module.



For inter-unit synchronization

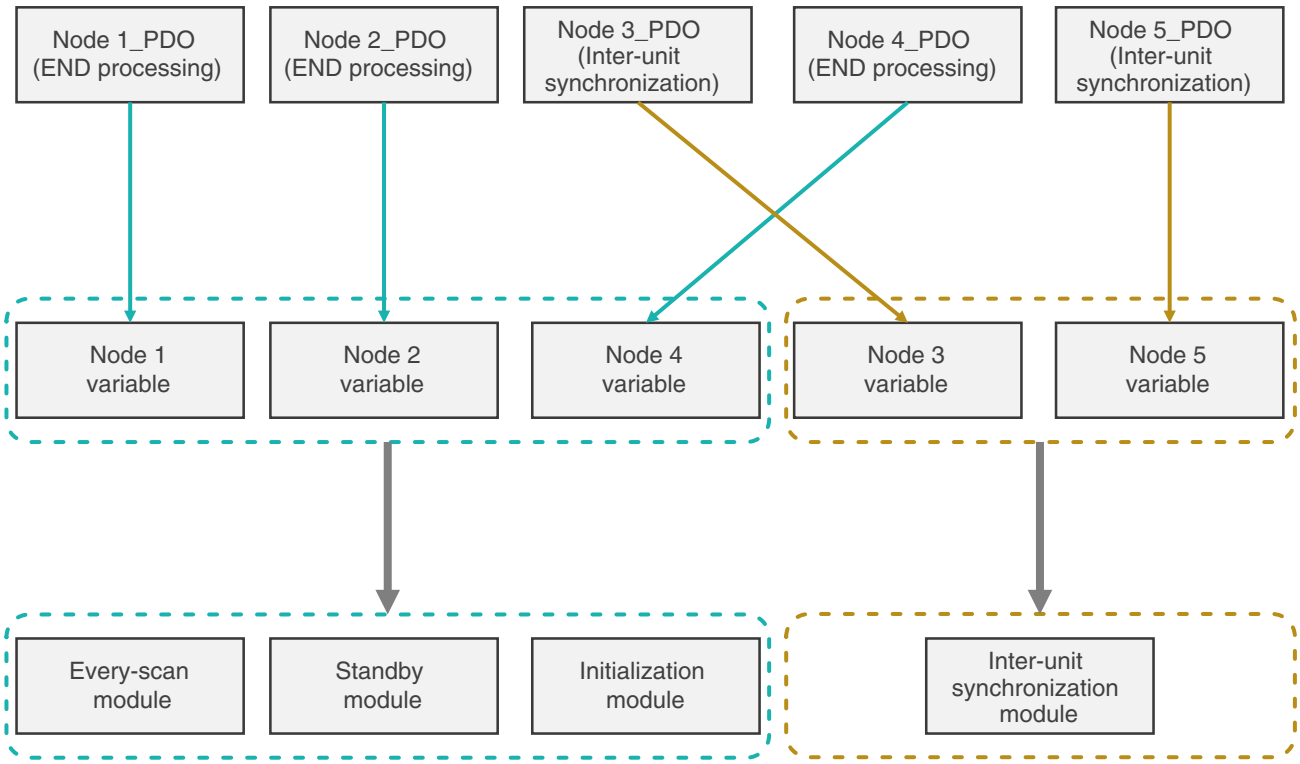
The variables allocated to the node with its refresh timing set to “Inter-unit synchronization” are refreshed with the PDO communication data at the time when the inter-unit synchronization has started. The variables can be referenced simultaneously from the inter-unit synchronization module.



*This is an operation example when the communication cycle and the inter-unit synchronization cycle are the same.

Operation when the refresh timing differs for each node

When there are different refresh timings within the same network, the synchronism of data is guaranteed between nodes where the same timing is set.



Reference

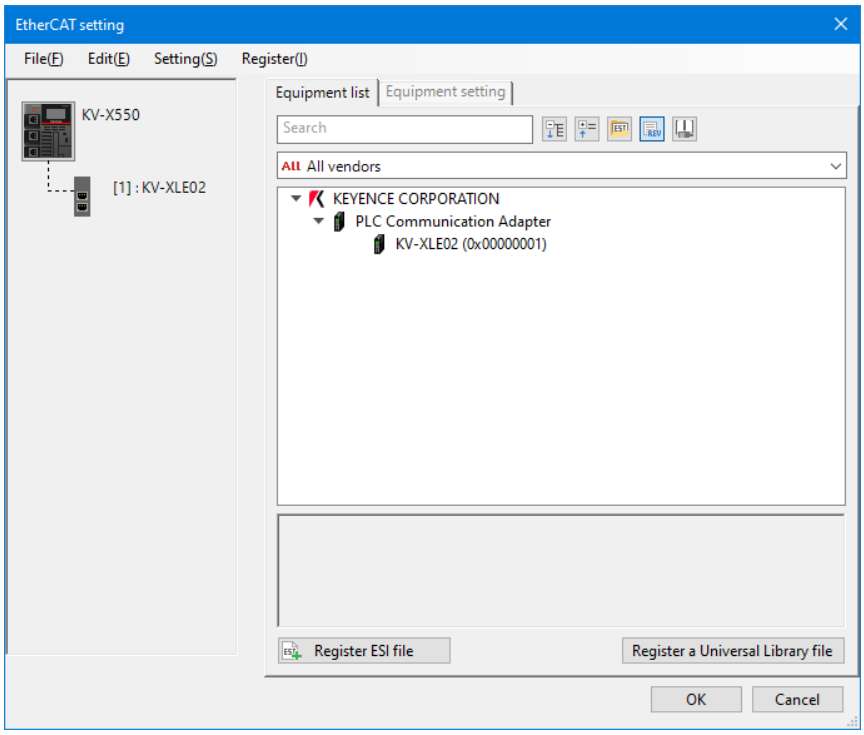
- If variables are read or write from a module that does not synchronize with the refresh timing, there is no data synchronization between variables (when, for instance, the refresh timing uses the every-scan module to refer to the variable allocated to the node with “Inter-unit synchronization” set). The value of the variable may be overwritten during the scan, or the values of variables for different EtherCAT communication cycles may coexist.
- During module interruption, the variable allocated to the transmit/receive buffer is not refreshed.

4-3 How to set refresh

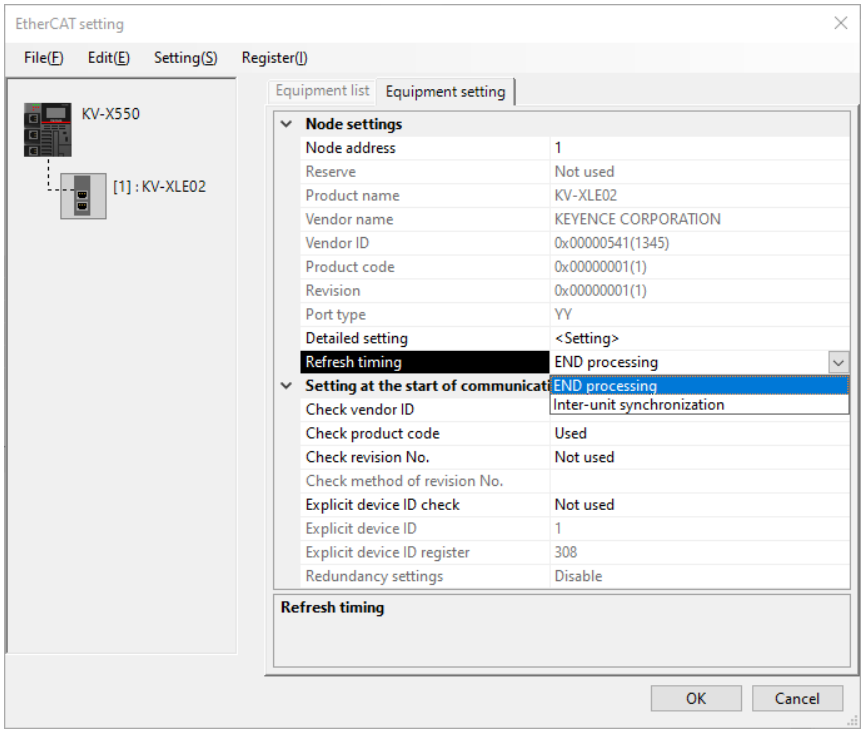
The following describes how to set refresh.

Setting of the refresh timing

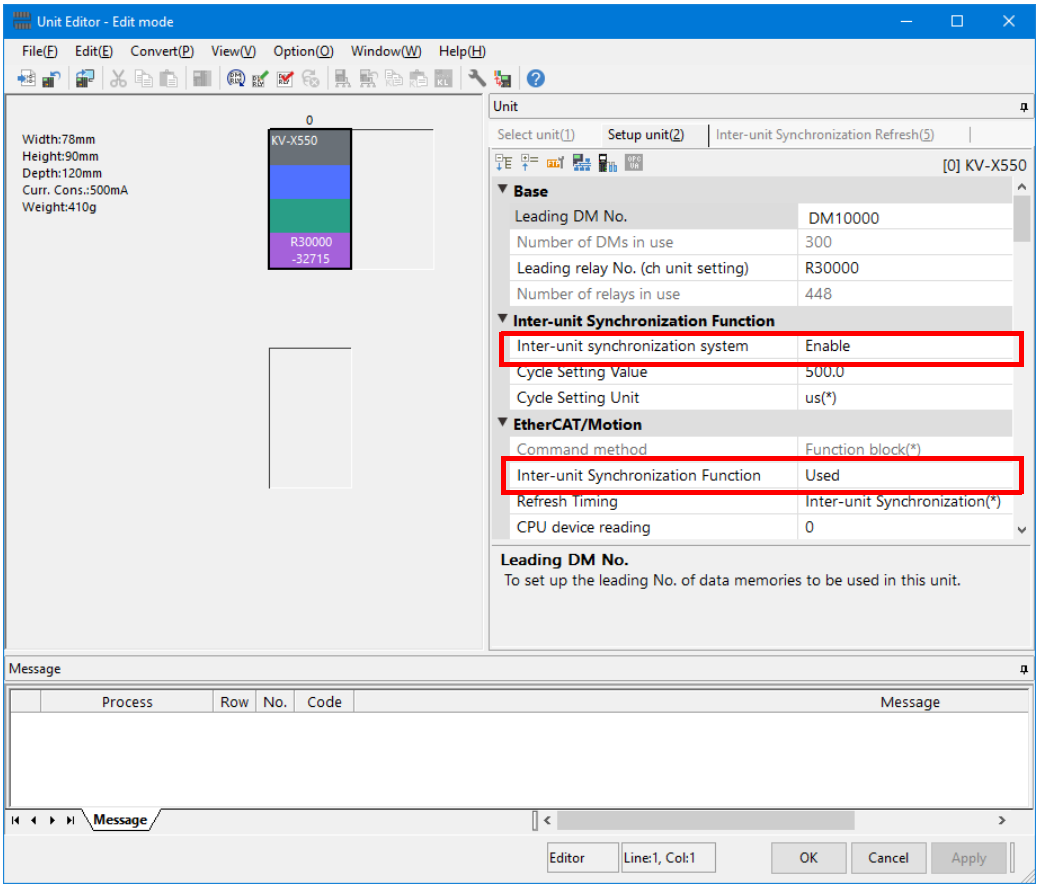
Double-click “EtherCAT” in the workspace, and open the EtherCAT setting dialog box.



Select the node and then select the “END processing” or “Inter-unit synchronization” for the “Refresh timing” in the node setting.



Reference To set the “Refresh timing” in the [EtherCAT setting] to “Inter-unit synchronization”, you need to set the “Inter-unit synchronization system” in the [Unit Editor] and then set the “Inter-unit synchronization function” in the EtherCAT motion to “Used”.



Setting of variable allocation

Set variables to be allocated to the transmit/receive buffers.

“PDO mapping”

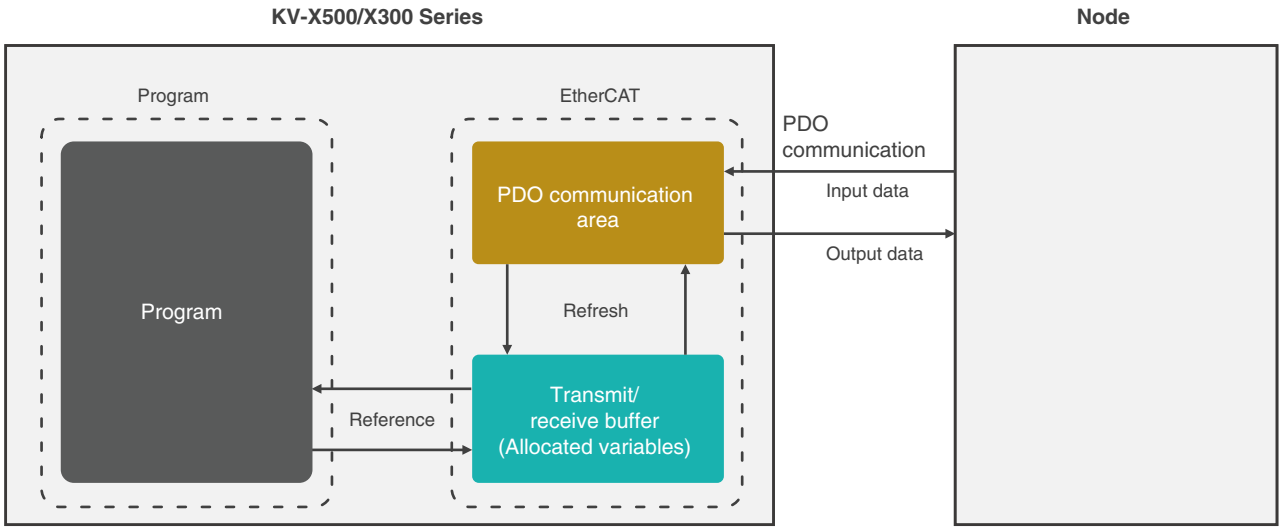
4-4 PDO communication function

The following describes the PDO communication function of the EtherCAT.

Overview

The PDO (Process Data Objects) communication of the EtherCAT is a communication function for cyclic data exchange between the master and nodes. This function exchanges data in a certain cycle with all node connected to the master. The content of data to be exchanged between the master and nodes depends on the setting of the PDO mapping.

"PDO mapping"



Notes and restrictions

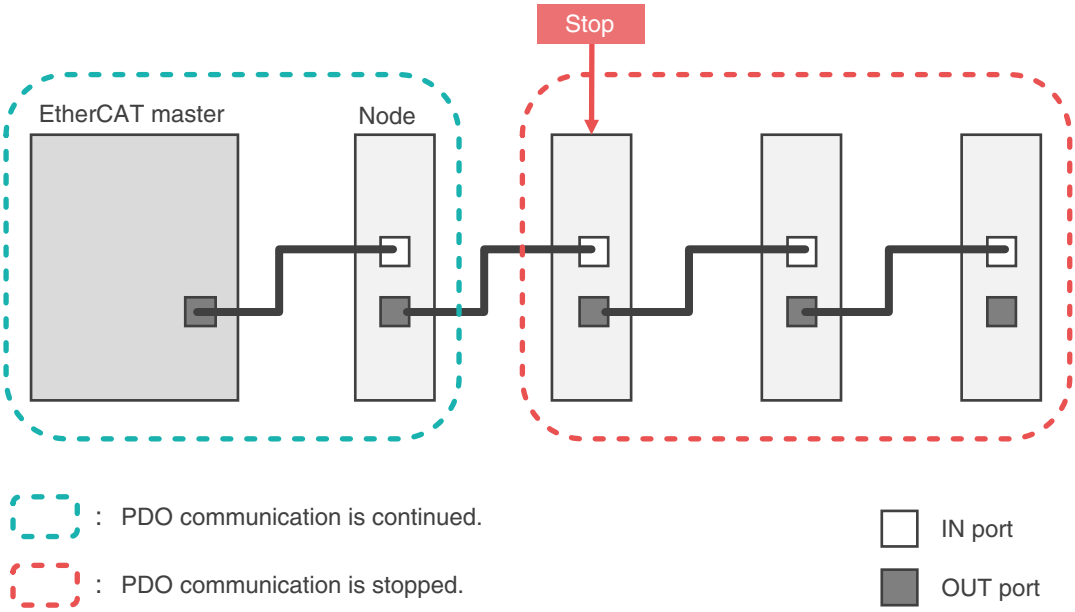
- When the following operation is performed, the data temporarily stop being updated (the previous values are retained) in the node -> master direction in all node components.
  - Connection/disconnection of a node component
  - Cable disconnection/reconnection in ring operation
  - Stop/resume of a node

Point **During the temporary suspension of data update in the node -> master direction in all node components, the system variable_EC_Master.IndataInvalid is set as TRUE.**

- Not supported for the PDO communication that uses multiple FMMU in the same direction (Tx/Rx). NU-EC1, a Keyence network communication unit, cannot be connected. Use NU-EC1A.
- At the start of the PDO communication, the transmit/receive buffers are initialized to 0 (zero). With no variable allocated to the PDO in the direction from the master to the slave, 0 is sent.

Stop/resume

This function temporarily stops the PDO communication when maintenance work is carried out on the device or a node is replaced. The PDO communication is stopped in the node that has requested deactivation and the downstream nodes as shown below.

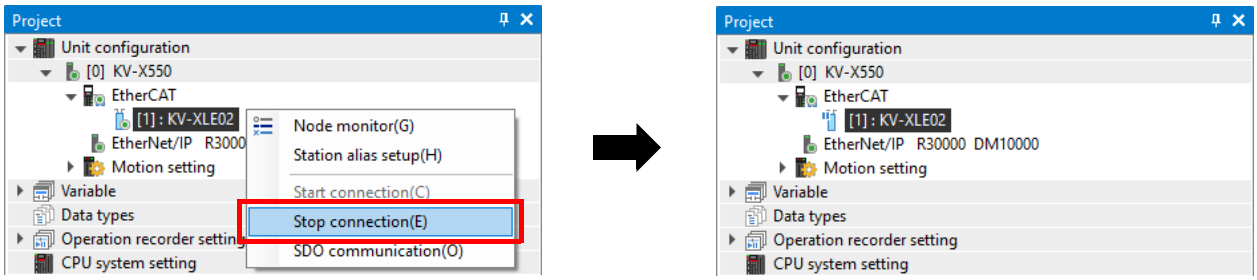


Reference In the node where the PDO communication is stopped, the EtherCAT state will transition to Pre-Operational.

How to use

When running form “KV STUDIO”

Select a node in the workspace, and select “Stop connection” or “Start connection” from the right-click menu.



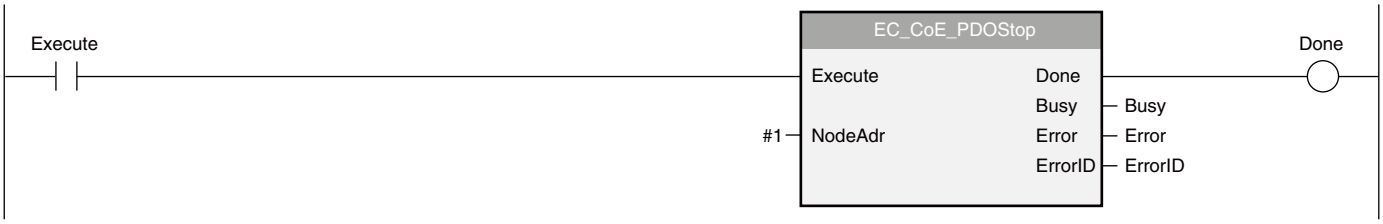
When running from the function block

To stop or resume the PDO communication from the program, use EC_CoE_PDOStop or EC_CoE_PDORestart, respectively.

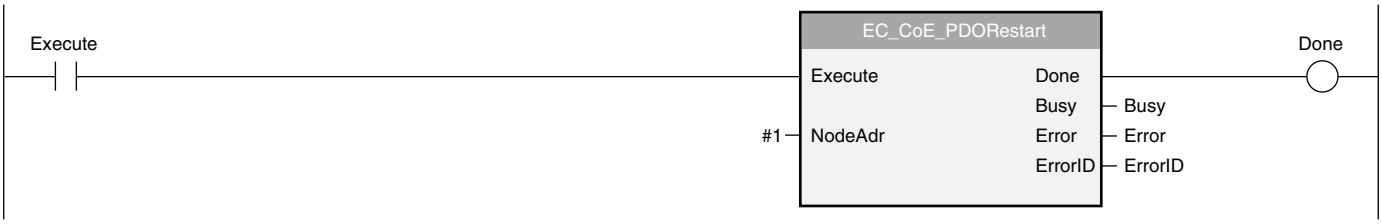
Reference program

The following shows sample programs to stop/resume the communication from the function block. Run the function by specifying the node address of the target node.

Stop



Resume



For details about the EC_CoE_PDOStop and EC_CoE_PDORestart, see “KV-X500/X300 Series Command Reference Manual”.

Procedure for partially stopping devices and replacing components

The following describes how to replace a node.

1. **Run the “Stop connection” from “KV STUDIO” or run EC_CoE_PDOStop from the program to stop the PDO communication with the node. The PDO communication stops in the node that has requested deactivation and the downstream nodes.**
2. **Turn OFF the node and replace the node.**
3. **Turn ON the node.**
4. **Run the “Start connection” from “KV STUDIO” or run EC_CoE_PDORestart from the program to resume the PDO communication.**

Reference

If the Explicit device ID check is set to “Used” in the node to be replaced, set the Explicit device ID for a new node before performing replacement work. If the Explicit device ID differs between the old and new nodes, an error occurs.

“Check items when starting communication”

Notes and restrictions

- You can run the stop/resume also for a reserved node and a node for which connection is being processed.
- Ring-configured devices cannot be stopped or restarted individually.
- The stop/resume is unavailable in the following cases.
  - During the initialization of the EtherCAT communication*1
  - When initialization of the EtherCAT communication fails
  - During the stop/resume processing*2
- The stop status is cleared when the power is turned on again or the operation permission is turned ON*3.
- The _EC_Node[n].NodeWarning status is cleared by stop/resume. Check the situation and resolve the cause of the warning before running the stop/resume.

*1: From when the Operation permission is turned on until when the Operational turns ON

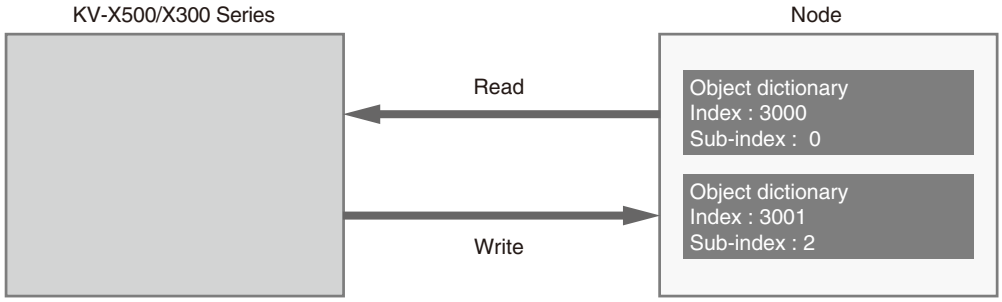
*2: You cannot simultaneously run the stop/resume in the “KV STUDIO” and the EC_CoE_PDOStop/EC_CoE_PDORestart.

*3: If no setting change is made to the EtherCAT setting, the deactivation status is not be cleared when the operation permission is turned ON.

4-5 SDO communication function

Overview

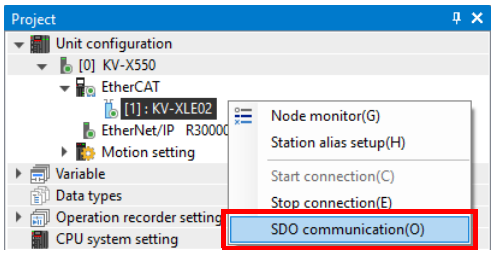
The SDO (Service Data Objects) communication of the EtherCAT is a communication function for the master to read and write data at any time in the nodes. Specify the index, sub-index and data size of the object dictionary of the target node, and run the function. By using the complete access, you can also read and write data of multiple objects having different sub-index numbers. "Object dictionary"



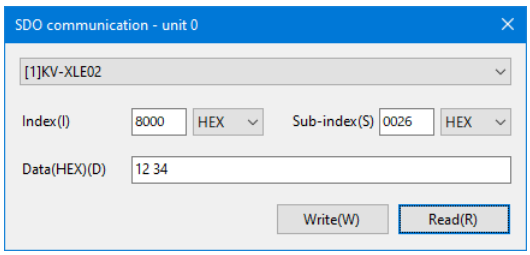
How to use

When running form "KV STUDIO"

Select a node in the workspace, and click [SDO communication] from the right-click menu.



Set the index, sub-index, and data, and click [Write] or [Read].



When running from the function block

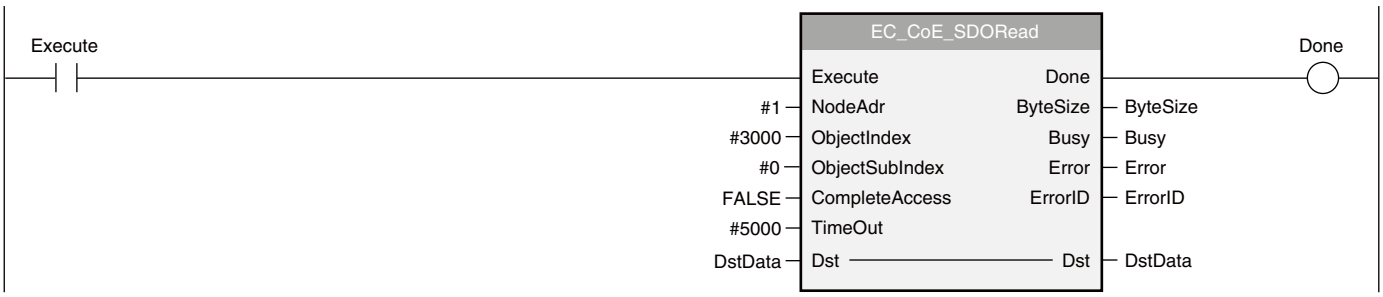
To read or write parameters of the node from the program in the SDO communication, use EC_CoE_SDORead or EC_CoE_SDOWrite.

Reference program

The following shows sample programs to perform the SDO communication. Use the function block for the SDO communication.

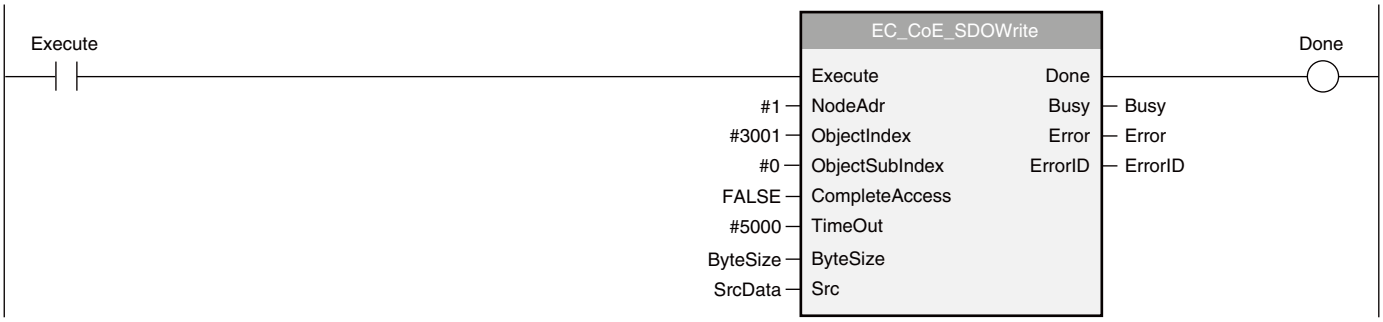
When reading data from the node in the SDO communication

Use EC_CoE_SDORead.



When writing data to the node in the SDO communication

Use EC_CoE_SDOWrite.



Reference

Complete access (CompleteAccess)

Use the CompleteAccess to read or write values at the same time for multiple objects having the same index and different sub-index numbers. Specify only the indexes of objects where data are read/written. Whether to support the CompleteAccess depends on the specifications of the node.

For details about the EC_CoE_SDORead and EC_CoE_SDOWrite, see “KV-X500/X300 Series Command Reference Manual”.

4-6 How to check the communication status

The following describes how to check the communication status.

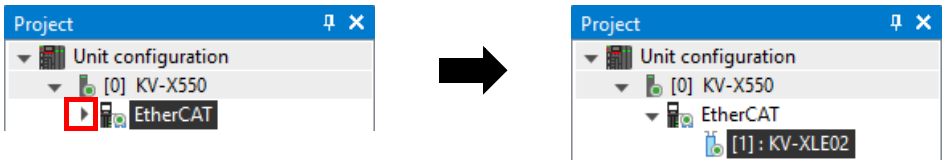
Workspace

Develop the EtherCAT in the workspace to check the communication status of each node registered with the EtherCAT setting.

Point **The communication status can be checked in the monitoring mode and online edit mode.**

How to check

Click the red outlined part for the “EtherCAT” in the workspace to display the connected node.



Communication status list of the node

The following is the communication status list of the node. The status to be displayed preferentially is the one with the smallest number for the “Priority”.

Icon	Status	Description	Priority
	Normal	The communication status is normal.	6
	Reserved	This is a reserved node that is unconnected.	5
	Transitioning	This node is transitioning to the status for participating in the PDO communication.	2
	Stopping	This is a node where the PDO communication is being stopped by the stop operation.	4
	Error	This is a node with an error.	1
	Unregistered	This is a node unregistered with the EtherCAT setting.	3

Reference When a disconnection has occurred in the ring, the node is shown in yellow.

Node monitor

Node monitor enables you to start the registration monitor with the variable for a selected node registered, where you can monitor the value of the variable.

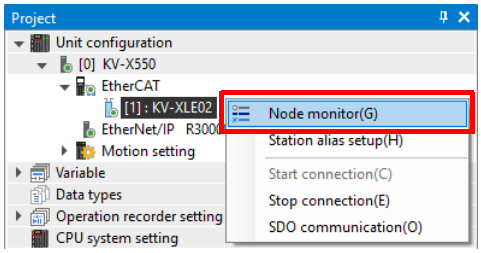
Monitoring items list

Item	Description
_EC_MASTER.EcStateTbl[n]	This enables you to check the transition status of the EtherCAT. 0: Not connected 1: Init 2: Pre-Operational 3: Safe-Operational 4: Operational
_EC_Node[n]	This enables you to check the information on the node. "System variable list"
Variable allocated to the transmit/receive buffer	The variables allocated to the transmit/receive buffer can be checked in the settings of PDO mapping. "PDO mapping"

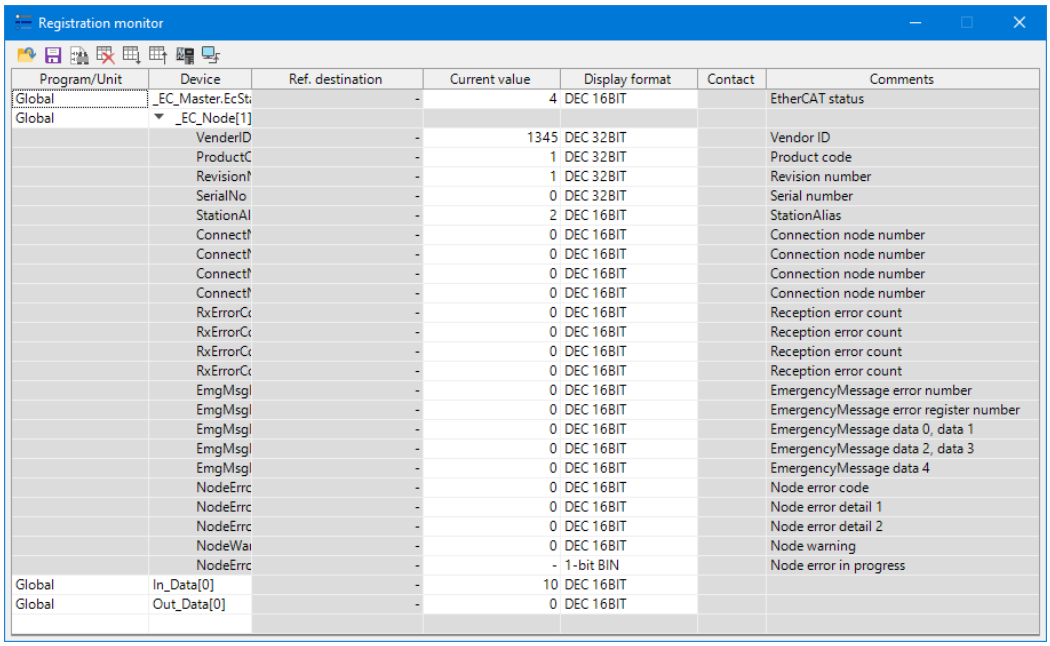
*n represents the node address.

How to start

Select a node in the workspace, and click [Node monitor] from the right-click menu.



The [Registration monitor] is displayed with the node-related system variables registered.



System variables

You can check for an error with the EtherCAT master and the communication status of each node connected to the master by referring to the system variable "_EC_MASTER".

"System variable list"

LED

You can check the communication status by the LED light on the CPU unit.

"Names and functions of parts"

Communication in the event of an error

In EtherCAT communication, the behavior in the event of an error varies depending on the error location and the nature of the error, as described below. When an error occurs, an automatic retry process is performed, and EtherCAT communication automatically resumes once the cause of the error is resolved.

Communication in the event of a CPU unit error

If not a severe error, the EtherCAT communication continues.

If a severe error, the EtherCAT state of all nodes is moved to the Safe-Operational or lower state.

Communication when switching the CPU unit from RUN to PROG

The EtherCAT communication continues.

Communication in the event of an alarm or warning of the node

The EtherCAT communication continues.

Note that in the case of an alarm or warning of the node related to the EtherCAT communication, follow the description in the "Communication in the event of an error in the EtherCAT master" or "Communication in the event of an error in the EtherCAT communication with a specific node" shown below.

Communication in the event of an error in the EtherCAT master

In the event of an error in the EtherCAT master, the operation differs depending on the error as follows.

In either case, if an error occurs when the communication is started, you need to turn the operation permission from OFF to ON and restart communication.

"Communication start sequence when the power is turned on or operation is permitted"

When the Link of the EtherCAT communication is off

In the event of this error, communication with all nodes stops, and an error is issued.

Automatic retry is performed during the error. When the cause of the error is removed, the communication resumes automatically.

This error will occur when the cable is not correctly plugged into the EtherCAT port.

In the event of an EtherCAT communication error

In the event of this error, communication with all nodes stops. Automatic retry is performed. When the cause of the error is removed, the communication resumes automatically. An error is issued to all the connected nodes.

In the event of matching discrepancy between nodes

Communication continues with nodes other than the unmatched node.

When the unmatched node is disconnected, the error is automatically cleared. This error occurs when the node set in the EtherCAT setting differs from the actually connected component.

When the refresh is not working for the EtherCAT communication

The EtherCAT communication continues with only the refresh operation stopped.

When the inconsistency of the setting is resolved between the EtherCAT master and the node, the error is automatically cleared.

This error occurs when the registered variable information differs from the settings of refresh for the EtherCAT communication.

Reference When a variable is allocated to the PDO in the PDO mapping setting, transfer both the variable setting and the EtherCAT setting to the PLC. Transferring either of these settings stops the refresh of the EtherCAT communication, causing an EtherCAT master error to occur.

Communication in the event of an error in the EtherCAT communication with a specific node

The PDO communication with the node having this error stops.

Automatic retry is performed during the error. When the cause of the error is removed, the communication resumes automatically.

Appendix

A-1 Error list

This is a list of error codes that appear when an error occurs in EtherCAT communication. For information on communication and retries in the event of an error, please see below.

"Communication in the event of an error"

How to see the table

Error code

Codes are stored into the following system variables according to the error details.

Function	Category	Description	Storage destination
EtherCAT	Error	Error code	_EC_Master.MasterErrorCode _EC_Node[N].NodeErrorCode
		Detailed information 1 to 2	_EC_Master.MasterErrorDetail1 _EC_Master.MasterErrorDetail2 _EC_Node[N].NodeErrorDetail1 _EC_Node[N].NodeErrorDetail2

Detailed information

In the event of an error in the EtherCAT communication, these variables store information including the node address where the error has occurred.

EtherCAT communication error code list

The error code is stored to the ErrorID function block for EtherCAT communication.

Error code	Detailed information 1	Detailed information 2/3	Error message	Error details and measures	Deactivated node
0	0	0	No error	There is no error.	None
40006	Node number where the error has occurred	0	EtherCAT PDO object error	The set object was not correctly read. Check if the Controlword object is set for the control word of the motion function map in the axis setting. For a multi-axis component, check if the offset of the object index correctly shifts for each axis.	Node with the error
40008	0	0	EtherCAT LINK off	The link of EtherCAT communication is disconnected. Check the cable connection. * See the LED light of LINK/ACT to check the connection status.	All nodes
40009	Node number where the error has occurred	0	EtherCAT connection failed	An attempt to connect to a EtherCAT node component failed. See the manual of the node component to check if the connection is supported for the EtherCAT settings (node detailed setting), communication cycle, and other communication settings. A component not supported for the EtherCAT may be connected. Check the connected components. The start wait time of the master setting may be short. Check the start wait time.	Node with the error

Error code	Detailed information 1	Detailed information 2/3	Error message	Error details and measures	Deactivated node
40010	0	0	EtherCAT communication error	A receive error has occurred consecutively for the specified number of times or more in the EtherCAT master. Noises may be applied to the communication line. Take measures against noises. A component not supported for the EtherCAT may be connected. Check the connected components. If the communication cycle is fast, data may not be received due to time-out. Review the usage conditions to take measures, such as increasing of the communication cycle or decreasing of the PDO size.	All nodes
40011	Node number where the error has occurred	0	Matching discrepancy between EtherCAT node components 1	The EtherCAT configuration settings do not match the connected node components. Check the settings and the configuration of the actual machine. Check if the cables are connected in reverse to the IN/OUT ports of the node component.	Node with the error
40012	Node number where the error has occurred	Status code of the node with the error	Status information of the EtherCAT node component error	The EtherCAT status of the node component has changed to Error. Check the status code from the error details and remove the cause from the node component. This error occurs also when the set communication cycle or the synchronization cycle of DC does not match the node specifications. Check the node specifications.	Node with the error
40013	Node number where the error has occurred	0	Invalid state of the EtherCAT node component	The EtherCAT state of the node component has changed to state other than the OP. Remove the cause from the node component.	Node with the error
40014	Node number where the error has occurred	0	Wrong connection to EtherCAT ports	The cables are connected in reverse to the IN/OUT ports of the EtherCAT node component. Check the ports to which the cables are connected.	Node with the error
40016	Node number before verification failure node	0	Matching discrepancy between EtherCAT node components 2	A node that is not included in the EtherCAT configuration settings is connected. Or the wiring may be wrong. Check the settings and the configuration of the actual machine.	All nodes
40017	Node number where the error has occurred	0	Node not exist	Not found the node device specified in the EtherCAT configuration settings. Check the settings and the configuration of the actual machine.	Node with the error
40018	Node number where the error has occurred	Type of error	Communication error at the time of EtherCAT node initialization 1	An error occurred with the communication command at EtherCAT initialization. The settings of the PDO mapping may violate the specifications of the node component. Check the specifications of the object in the manual of the node component. Or the communication command at initialization may violate the specifications of the object dictionary of the node component. Check the specifications of the object dictionary in the manual of the node component. Or the set communication cycle or the synchronization cycle of DC may not match the node specifications. Check the specifications of the node component.	Node with the error
40019	Node number where the error has occurred	0	Communication error at the time of EtherCAT node initialization 2	The initialization command for the EtherCAT mailbox timed out. The mailbox communication specifications (e.g. buffer size) of the node component may not match the settings of the master or node.	Node with the error
40020	Node number where the error has occurred	0	Failure of read communication for setting up EtherCAT	An attempt to establish communication for reading failed when setting up the EtherCAT. Check the specifications of the object dictionary in the operation mode of the node component.	Node with the error
40021	Node number where the error has occurred	0	Failure of write communication for setting up EtherCAT	An attempt to establish communication for writing failed when setting up the EtherCAT. Check the specifications of the object dictionary in the operation mode of the node component.	Node with the error
40023	0	0	EtherCAT master error	An error occurred in the EtherCAT master. Check the error information of the master that stores the system variable (_EC_Master.MasterErrorCode), and remove the cause of the error from the EtherCAT master.	Master
40024	0	0	EtherCAT refresh deactivated	The EtherCAT refresh is deactivated due to either of the following inconsistencies: • Between the setting of the global variable and the setting of the variable allocated to the PDO mapping in the EtherCAT settings • Between the setting of the inter-unit synchronization in the unit editor and the setting of the refresh timing in the EtherCAT settings Transfer the whole project. Or if a node exists where the refresh timing is set to inter-unit synchronization, check whether interruption is enabled.	-
40025	0	0	EtherCAT send resource depletion	The send resource has been depleted. Increase the communication cycle.	All nodes

Error code	Detailed information 1	Detailed information 2/3	Error message	Error details and measures	Deactivated node
40026	Node number where the error has occurred	0	SV3 connection error (parameter read failure)	Failed to read parameters during the connection with SV3. Please check if the connection with SV3 is correct.	Node with the error
40027	Node number where the error has occurred	0	SV3 connection error (parameter write failure)	Failed to write parameters during the connection with SV3. Please check if the connection with SV3 is correct.	Node with the error
40028	Node number where the error has occurred	0	EtherCAT device type mis-match error	The node device of the axis with the device type set to SV3 is not actually of the SV3 type. Please verify that the node device setting and connection are correct.	Node with the error

EtherCAT communication warning code list

The warning code is stored to the ErrorID function block for EtherCAT communication.

Error code	Detailed information 1	Detailed information 2/3	Error message	Error details and measures	Deactivated node
41001	0	0	EtherCAT SDO communication parameter warning	There is an abnormal parameter (e.g.: Index, Subindex, Data size) on the master for performing the EtherCAT SDO communication. Review the parameter value.	None
41002	0	0	Invalid destination node of EtherCAT SDO communication	The destination node for EtherCAT SDO communication does not exist. Check the parameter value (node address).	None
41003	0	0	EtherCAT SDO communication service unavailable	The SDO communication cannot be performed on the EtherCAT master side. Perform the operation after the STATUS LED lights up green.	None
41004	0	0	EtherCAT SDO communication protocol warning	A protocol error occurred in the EtherCAT SDO communication. Try again.	None
41005	0	0	EtherCAT SDO communication time-out	The EtherCAT SDO communication timed out. Try again.	None
41006	0	0	EtherCAT SDO communication data type warning	The data type does not match the specifications of the node component in the EtherCAT SDO communication. Make the data type of the parameter match the specifications.	None
41007	0	0	EtherCAT SDO communication write access warning	An attempt to perform write access failed in the EtherCAT SDO communication. Non-writable data may have been accessed, or the write range may be exceeded. Check and correct the specifications of read and write parameters of the node component.	None
41008	0	0	EtherCAT SDO communication object access warning	An attempt to perform object access failed in the EtherCAT SDO communication. The object number (Index, Subindex) may be incorrect or the access range may be exceeded. Check and correct the parameter number range of the node component.	None
41009	0	0	EtherCAT SDO communication application warning	An application error occurred in a node component in the EtherCAT SDO communication. The node device may have been disabled for communication in a case where, for instance, a parameter is being saved. Wait for a while, and run the SDO communication. Or check the CoE specifications of the node component.	None
41010	0	0	Other warning in the EtherCAT communication	An error occurred in a node component in the EtherCAT SDO communication. Check the CoE specifications of the node component and take measures.	None
41021	0	0	Invalid node address	The specified node address cannot be used. Check if the specified node address can be used in the function block.	None
41022	0	0	Operational warning	The processing could not be executed due to non operational status. Make the status operational.	None
41024	0	0	EtherCAT communication status warning	The EtherCAT communication is not started. Check the EtherCAT configuration settings.	None
41025	0	0	Specify argument warning	The IN argument, OUT argument or local variable for the FUN instruction has been specified by the IN-OUT argument. Check the specified argument.	None
41026	0	0	PDO transmissions resumption warning	A node which does not exist in the node configuration settings is connected. Check the settings and the actual device configurations.	None

A-2 System variable list

System variables for EtherCAT

The following describes system variables for the EtherCAT.

The following shows when to update and read/write attributes of the variables.

- When to update the variable
  Update the value according to the settings of the refresh timing (END processing/inter-unit synchronization).
- Attributes
  R: Read-only
  R/W: Readable and writable

Other notes

- About MN (Maximum number of nodes)
  The “MN” shown in the table means the maximum number of nodes.
  For example, in the case of KV-X550, the notation ARRAY[1..MN] OF _EC_NODE becomes ARRAY[1..512] OF _EC_NODE.
  Please refer to the following for the maximum number of nodes for each CPU unit.
  “EtherCAT communication specifications”

_EC_MASTER

Use this variable when referring to the information held by the master of EtherCAT.

Update the variables according to the settings of the refresh timing (END processing/inter-unit synchronization).

Variable name / Member name	Data type	Name	Description	Attri-bute	UG
_EC_MASTER	_EC_MASTER	EtherCAT master structure		R	900000
MasterError	BOOL	Master error	An EtherCAT communication error in the master	R	900000
PDOActive	BOOL	All nodes PDO communication under way*1	TRUE: The PDO communication is under way in all nodes. FALSE: The PDO communication has been deactivated in any of the nodes.	R	900000.01
PDOInactive	BOOL	All nodes PDO deactivated	TRUE: The PDO has been deactivated in all nodes. FALSE: The PDO communication is under way in any of the nodes.	R	900000.02
PDOCommError	BOOL	PDO communication error	TRUE: A communication error has occurred in any of the nodes participating in PDO. FALSE: No communication error has occurred in all nodes participating in PDO.	R	900000.03
RingError	BOOL	Ring disconnected*2	TRUE: A disconnection has been detected in any of the ring configurations. FALSE: No disconnection has been detected in all ring configurations.	R	900000.04
InDataInvalid	BOOL	Invalid PDO input data*5	TRUE: The values in the node -> master direction have stopped being updated in the PDO communication. FALSE: The values in the node -> master direction continue being updated in the PDO communication.	R	900000.05
PDOParticipationTbl	ARRAY[1..512] OF BOOL	Table of nodes participating in PDO	TRUE: Participating in PDO FALSE: Not participating in PDO	R	900001.00 to 900032.15
PDOActiveTbl	ARRAY[1..512] OF BOOL	Table of nodes in PDO communication	TRUE: In PDO communication (EtherCAT state = Operational) FALSE: PDO deactivated (EtherCAT state ≠ Operational)	R	900033 to 900064.15
PDOCommErrorTbl	ARRAY[1..512] OF BOOL	Table of nodes in abnormal PDO communication	TRUE: Abnormal PDO communication FALSE: Normal PDO communication	R	900065 to 900096.15
SDOActiveTbl	ARRAY[1..512] OF BOOL	Table of nodes available for SDO communication*3	TRUE: SDO communication is available. FALSE: SDO communication is unavailable.	R	900097 to 900128.15
RegTbl	ARRAY[1..512] OF BOOL	Table of registered nodes	TRUE: Register FALSE: Unregistered	R	900129 to 900160.15
ReserveTbl	ARRAY[1..512] OF BOOL	Table of reserved nodes	TRUE: Reservation set FALSE: No reservation set	R	900161 to 900192.15
PDOStopReqTbl	ARRAY[1..512] OF BOOL	Table of nodes requesting deactivation	TRUE: With deactivation request FALSE: With no deactivation request	R	900193 to 900224.15
RingErrorTbl	ARRAY[1..512] OF BOOL	Table of nodes with the ring disconnected	TRUE: A disconnection has been detected in the ring configuration. FALSE: No disconnection has been detected.	R	900225 to 900256.15
EcStateTbl	ARRAY[1..512] OF UINT	EtherCAT status	0: Not connected 1: Init 2: Pre-Operational 3: Safe-Operational 4: Operational	R	900257 to 900768
MasterErrorCode	UINT	Master error code*4		R	900769
MasterErrorDetail1	UINT	Master error details 1		R	900770
MasterErrorDetail2	UINT	Master error details 2		R	900771

*1: All nodes do not include the reserved nodes that are not participating in the PDO communication and the nodes requesting deactivation.

*2: The reserved nodes that are not participating in the PDO communication are not applicable.

*3: When the EtherCAT state is Pre-Operational or higher, the status becomes TRUE.

*4: If multiple errors occur at the same time, the error code of the first error will be retained.

*5: In the case of, for instance, suspension/resume of the PDO communication, plugging and unplugging of a cable, and disconnection/recovery during ring operation, the value in the node -> master direction may not be updated (the previous value is retained) during the number communication cycle.
In such a case, use this variable as interlock to prevent values not updated from being referred to.

_EC_NODE

Use this variable when referring to the information held by a node of EtherCAT.
The variable is updated immediately.

Variable name / Member name	Data type	Name	Description	Attri-bute	UG
_EC_NODE	ARRAY[1..MN] OF _EC_NODE	EtherCAT node structure		R	900772
VenderID	UDINT	Vendor ID	Vendor ID read from the actual machine	R	900772
ProductCode	UDINT	Product code	Product code read from the actual machine	R	900774
RevisionNo	UDINT	Revision number	Revision number read from the actual machine	R	900776
SerialNo	UDINT	Serial number	Serial number read from the actual machine	R	900778
StationAlias	UINT	StationAlias	StationAlias read from the actual machine (Value of the ESC 0x12 register)	R	900780
ConnectNodeNo	ARRAY[0..3] OF UINT	Connected node number	Node number of the component connected to each port (Port A to D*1) [0]: Port A, [1]: Port B, [2]: Port C, [3]: Port D	R	900782 to 900785
RxErrorCount	ARRAY[0..3] OF UINT	Receive error count*2	Integrated value of the number of received error frames detected at each port (Port A to D) (Ring operation with 0 to 65535) [0]: Port A, [1]: Port B, [2]: Port C, [3]: Port D	R	900786 to 900789
EmgMsgErrorCode	UINT	EmergencyMessage Error code	EmergencyMessage error code	R	900790
EmgMsgErrorReg	UINT	EmergencyMessage Error register number	Stores the EmergencyMessage error register number to the last 8 bits. The first 8 bits are filled with 0.	R	900791
EmgMsgData0_1	UINT	EmergencyMessage Data 0, Data 1	Stores EmergencyMessageData[0] and [1]. EmergencyMessageData[0]: The last 8 bits EmergencyMessageData[1]: The first 8 bits	R	900792
EmgMsgData2_3	UINT	EmergencyMessage Data 2, Data 3	Stores EmergencyMessageData[2] and [3]. EmergencyMessageData[2]: The last 8 bits EmergencyMessageData[3]: The first 8 bits	R	900793
EmgMsgData4	UINT	EmergencyMessage Data 4	Stores EmergencyMessageData[4]. EmergencyMessageData[4]: The last 8 bits The first 8 bits are filled with 0.	R	900794
NodeErrorCode	UINT	Node error code*3		R	900795
NodeErrorDetail1	UINT	Node error details 1		R	900796
NodeErrorDetail2	UINT	Node error details 2		R	900797
NodeWarning	UINT	Node warning*4	Bit00: PDI watchdog warning Bit01: EtherCAT frame lost	R	900798
NodeError	BOOL	Node error		R	900799.00

*1: The IN port of the node is allocated to Port A. The OUT port of the node is allocated to Port B to D.
For a node having one IN port and one OUT port, Port A and Port B are used.

*2: The integrated value is cleared when the parameter (e.g. EtherCAT related settings) is reflected when the “KV-X500/X300 series” is turned on or by operation permission.

*3: If multiple errors occur at the same time, the error code of the first error will be retained.

*4: This item is zero-cleared when the EtherCAT state moves to Operational (when the communication starts, when the error is resolved) or the node is deactivated.

Revision History

Printing Date	Version	Details of Revision
Jun 2024	Official release	
Sep 2024	Revised 1st edition	
April 2025	2nd Revision 1st edition	
Sep 2025	3rd Revision 1st edition	

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A4WW1-MAN-2033

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